C. Chloroplasts and vacuole D. Chloroplasts and mitochondria

Explanation

- **Chloroplasts** are the site of photosynthesis where transformation of light energy in to chemical energy takes place.
- **Mitochondria** is an organelle of a cell which is concerned with the release of energy.

Answer: D

5. The chlorophyll pigment that can transform excited electron to acceptor is:

A. Chlorophyll "b" C. Chlorophyll "a"

B. Xanthophylls D. Carotene

Explanation: accessory pigments absorb light energy and transfer it to chlorophyll 'a', then Chlorophyll 'a' transfer the electron to primary electron acceptor.

Answer: C

6. Based on the given chemical equation. $12H_2O + 6CO_2 \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$. Such type of metabolic process:

A. Photosynthesis C. Transpiration

B. Respiration D. Osmosis

Explanation: the metabolic process that is indicated above is photosynthesis.

Answer: A

7. Light is composed of particles of energy called

A. Carotenoids C. Photo systems

B. Antenna complexes D. Photons

Answer: D

8. In addition to chlorophyll, most plants contain accessory photosynthetic pigments such as

A. PEP C. $NADP^+$

B. Carotenoids D. G_3P

Explanation: Accessory pigments includes carotenoid, chlorophyll b, and c.

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Answer: B

Photosystem

Cluster of pigment molecules and the proteins that the pigment molecules are embedded in are referred to collectively as a photo system. There are two types of photo systems: Photo system II (PsI) and Photo system I (PsII) They contain similar kinds of pigments, but they have different roles in the light reactions.

- **Photo system I** consists of a pair of chlorophyll a molecules with an absorption peak at 700 nm and is referred to as P700.
- **Photo system II** is made up of a pair of chlorophyll a molecules with an absorption peak of about 680 nm and is referred to as P680.

9.1.2 Light Dependent and Light Independent Reactions

Photosynthesis occurs in two stages. Theses are: Light dependent reactions, and Light - independent reactions.

i. Light dependent reactions

- They are the steps of photosynthesis that convert solar energy to chemical energy.
- Water molecule is splitting (photolysis) to produce electrons and protons (H^+) and giving of O_2 as a by - product.

The light energy is used to derive the synthesis of two molecules:

i. ATP that provides energy for the dark reaction

ii. Reduced NADP/NADPH2/ that provide hydrogen for a reduction.

Photo phosphorylation is the light deriving formation of ATP. There are two types of photophosphorylation in light dependent reactions. These are Non - cyclical, and Cyclical photo phosphorylation.

A. Non - cyclical photo phosphorylation

- It involves both photo system II and photo system I
- It involves one way of electron flow from water to $NADP^+$
- The electrons lost from the chlorophyll are not recycled in any way
- First electron donor is **water** and final electron acceptor is $NADP^+$
- Produces energy rich molecules **ATP**, and **NADPH**

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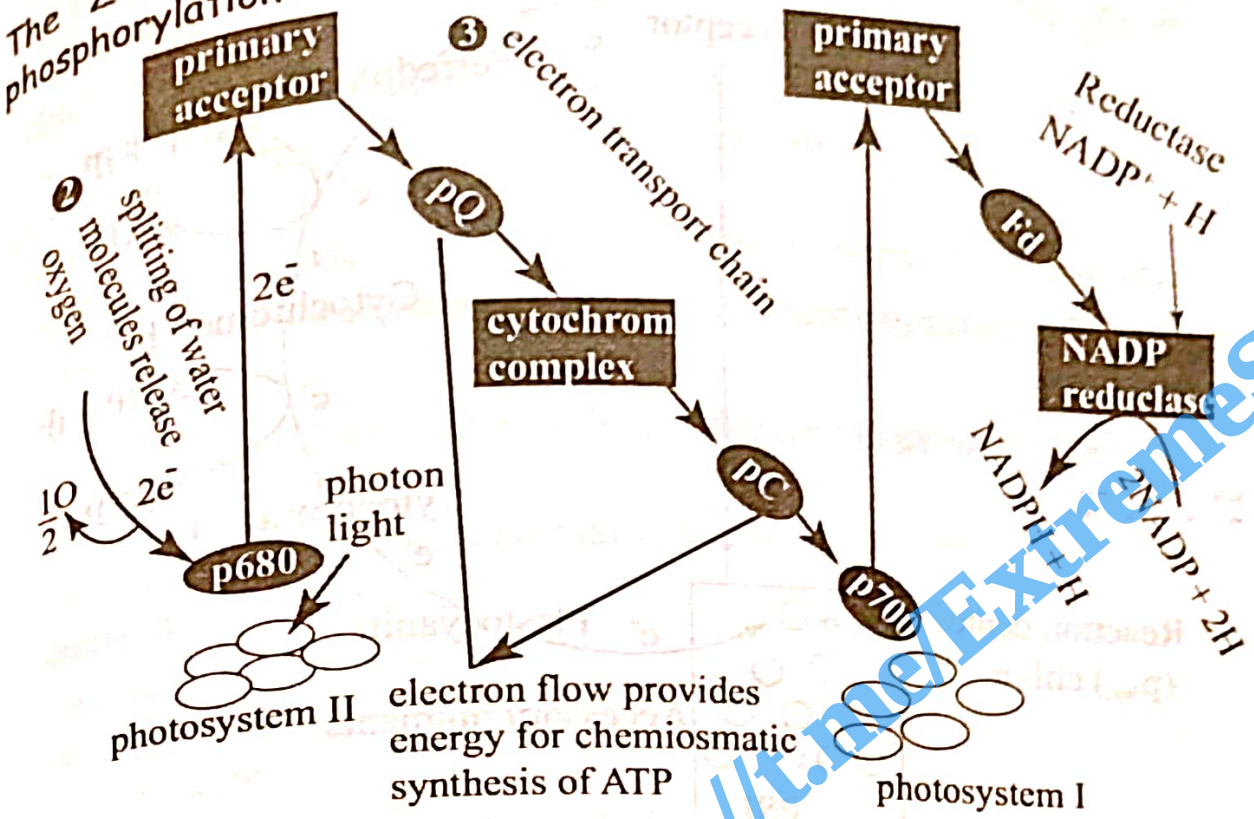
The Z - scheme diagram below shows non - cyclic photo phosphorylation:



B. Cyclical photo phosphorylation

It involves only photo system I/PSI/

The pathway is cyclic

- Because energized electrons that originate from P700 eventually return to P700

Initial donor and final electron acceptor is the same chlorophyll molecule.

The net product is only ATP molecule.

The significance of cyclic electron transport to photosynthesis in plants is unclear.

- Cyclic electron transport may occur plant cells when there is too little $NADP^+$ to accept electrons from ferredoxin.
- There is evidence that cyclic electron flow may help maintain the optimal ratio of ATP to NADPH required for carbon fixation, as well as provide extra ATP to power other ATP to power other ATP requiring processes in chloroplasts.
- It may be adaptation that occurs when there is shortage of carbon

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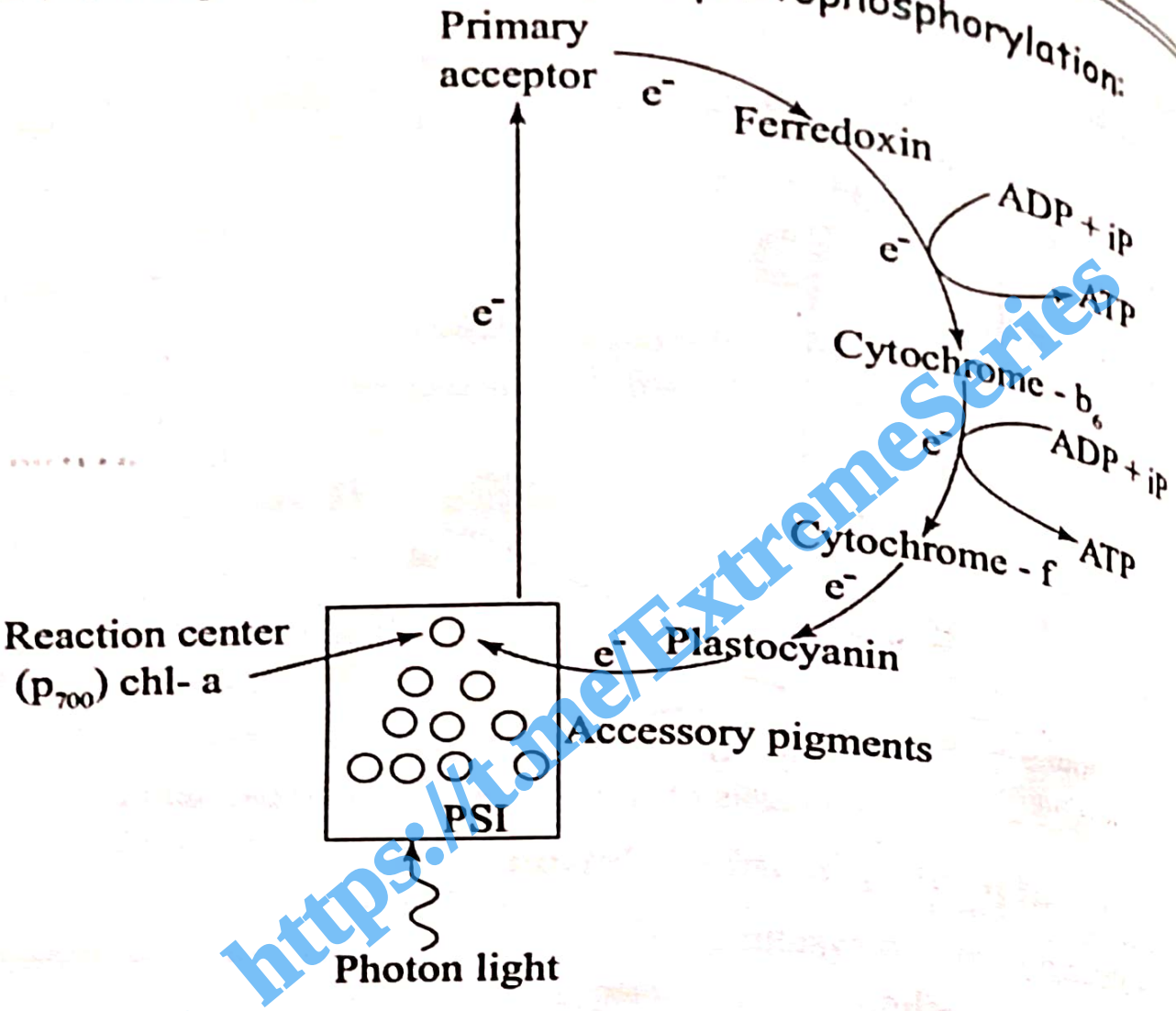
dioxide for dark reaction.

The following diagram shows cyclic photophosphorylation:



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Comparison between non - cyclic and photophosphorylation

Feature	Noncyclic Electron Transport	Cyclic Electron Transport
Electron source	H_2O	None - electrons cycle through the system
Oxygen released	Yes (from H_2O)	No
Terminal electron	NADP	None - electrons cycle through the system
Form in which energy is temporarily captured	ATP (by chemiosmosis):	ATP (by chemiosmosis)
Photo system (s) required	PSI (P700) and PS II (P680)	PSI (P700) only

The light reactions begin when accessory pigment molecules in both photo systems absorb light.

By absorbing light, those molecules acquire some of the energy carried by light.

In each photo system, the acquired energy is passed quickly to the other pigment molecules until it reaches to specific pair of chlorophyll a can also absorb light

The following major reactions take place inside PsII and PsI:

1. Electrons in **Photo system II** excited and more energetic by the energy in photons
   - These excited electrons from chlorophyll pass to **primary electron acceptor.**
2. Photolysis of water molecule takes place to produce **hydrogen ions, electrons** and **oxygen**
   - $2H_2O \rightarrow 4H^+ + 4e^- + O_2$
   - These electrons replace the electrons that lost from chlorophyll molecule
3. Primary electron acceptor passes electrons to the next electron acceptors sequentially:
   - Plastoquinone (Pq) → Cytochrome → Plastocyanin (Pc)
   - The electrons lose energy as they pass from one carrier to the next, this empower the proton pump to pump protons from outer stroma to inner space of thylakoid.

Some energy from energized electrons is used to phosphorylate (ADP), forming (ATP).

4. In the Cytochrome complex of electron transport chain there is a proton pump between stroma and inner space of thylakoid.

☑ This creates protons (H^+) accumulation inside the theylakoid, which drives the chemiosmosis synthesis of ATP.

5. Electrons in **photo system I** are excited as it absorbs photons and escape from chlorophyll a molecule.

This happens at the same time that light is absorbed by photo system II.

Energized electrons then pass along a **second electron transport chain**

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that involving ferredoxin (Fd) and NADP reductases.

- Some energy of the energized electrons is used to photophorylate (ADP), forming (ATP).

The electrons lost by theses chlorophyll a molecules are replaced by the electrons that have passed through the electron transport chain from photo system II.

Finally, electron reacts with protons (H^+) and $NADP^+$ in the stroma of the chloroplast to form reduced NADP/NADPH/

- NADPH is a hydrogen carrier, which has the ability to provide high - energy electrons to power certain reactions in anabolic pathways, such as the carbon fixation reactions of photosynthesis.

Thus, the products of the light dependent reactions, ATP and NADPH, are both needed in the energy - requiring carbon fixation reactions (during light independent reaction)

Summary: The light - dependent reactions are summarized as follows:

$$12H_2O + 12NADP^+ + 18ADP + 18P_i \xrightarrow[\text{Chlorophyll}]{\text{Light}} 6\,O_2 + 12NADPH + 18ATP$$

Illustrative Example

9. In cyclic photophorylation what is the sources of the recycled electron?

A. Reduced NADP
B. Chlorophyll molecule
C. Adenosine triphosphate
D. Photolysis of water molecule

Explanation: An excited electron from chlorophyll PsI is used in as a source of electron in cyclic photophosphorylation. No oxygen and no NADH are formed. Only ATP is produced.

Answer: B

10. Which of the following is not is not true about photosystem II?

A. It is reaction center molecule is P680
B. It passes its excited electron to photosystem I
C. The energy lost from its exited electrons reduces NADH
D. It replenishes its lost electrons from photolysis of water

Explanation: In photosystem II, the energy from excited electrons is used to reduce NADP in to NADPH.

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Answer: C

11. What is the importance of chemiosmosis in photosynthesis?

A. Splitting water molecule
B. Producing glucose
C. Combining hydrogen and carbon
D. Synthesizing ATP

Explanation: Chemiosmosis is the diffusion of hydrogen ions or proton pumps through ATP synthase to produce ATP

Answer: D

12. Where does the light dependent reaction of photosythesis occur in the chloroplast?

A. In the theylakoid membrane
B. In the fluid of the stroma
C. In all parts of the chloroplast
D. In the stomatal opening

Explanation: The light dependent reaction of photosynthesis is the first stage of photosynthesis, it takes place in the stacks of granna in the thylakoid membrane.

Answer: A

13. Which of the following event does not happen during light phase of photosynthesis?

A. Reduced NADP
B. ATP generation
C. Photolysis of water
D. Reduction of CO_2

Explanation: During the light dependent state of photosynthesis

- ATP is generated
- Water molecules split (photolysis).
- NADP is reduced and $NADPH^+$ is formed.

Answer: D

14. The ultimate source of electron for light dependent reaction is:

A. Water B. Glucose C. P700 D. Light

Explanation: The electrons and protons formed from the hydrolysis of water.

Answer: A

15. Cyclic phosphorylation produces:

A. Oxygen and ATP
B. Reduced NADP and ATP
C. Oxygen
D. ATP only

Explanation: Cyclic phosphorylation involves only PSI, which

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produces only ATP.

ii. Light - independent reaction Answer: D

- The second step in the mechanism of photosythesis
- It takes places in the Stroma of the chloroplast
- It is also called dark reaction of Calvin cycle
- It is purely enzymatic and is slower than the light reaction
- Sugars are synthesized from CO_2.

Carbon fixation occurs in the stroma through a three sequence reactions known as the Calvin cycle. These are:

- **CO_2 uptake, carbon reduction, and RuBP regeneration**

1. Carbon fixation (CO_2 uptake)

A molecule of CO2 reacts with a phosphorylated five - carbon compound, ribulose disphosphate (RuBP).

- This reaction is catalysed by the enzyme ribulose bisphospate carboxylase/oxygenase, also know as rubisco

The product of this reaction is an unstable six - carbon intermediate, which immediately breaks down into two molecules of phosphoglycerate (PGA) with three carbons each.

- The carbon that was originally part of a CO2 molecule is now part of a carbon skeleton; the carbon has been "fixed".
- The Calvin cycle is also known as the **C3 pathway** because the product of the initial carbon fixation reaction is a three carbon compound.
- Plants that initially fix carbon in this way are called C3 plants.
- About 95% of plants on the earth are C3 plants. For example: wheat, rye, oat, orchard grasses

2. Carbon Reduction

The second phase of the Calvin cycle

- Consists of two steps in which the energy and reducing power from ATP and **NADH** are used to convert the **PGA** molecules to glyceraldehyed - 3 - phosphate (G3P).
- For every six carbons that enter the cycle as CO2, two molecules of G3P, to be carbohydrate synthesis. Each of these three - carbon molecules of

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- G3P is essentially half a hexose (glucose) molecule.
- The name is derived from the fact that **NADPH** donates electrons to a three - carbon intermediate to form G3P.

3. Regeneration

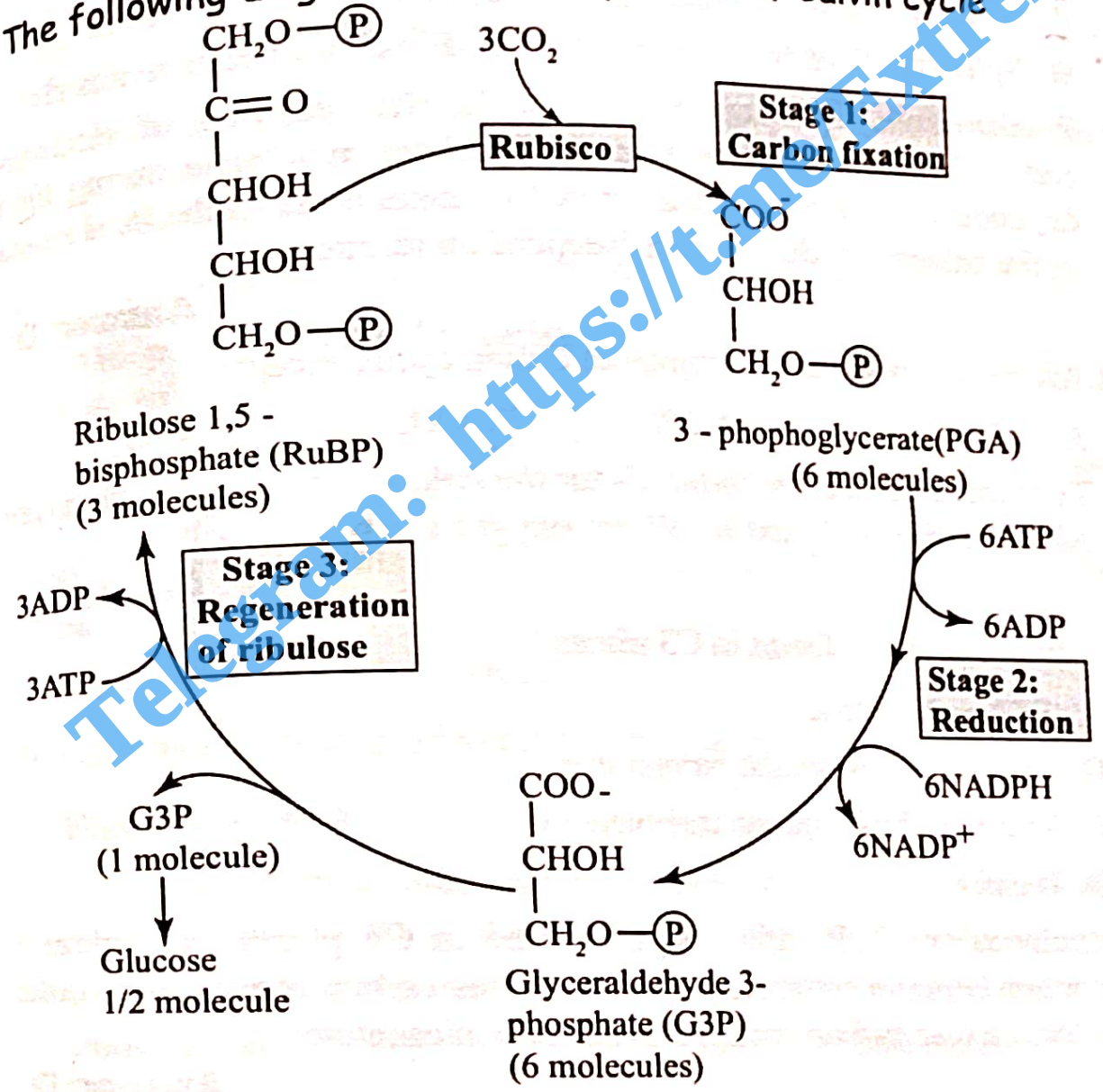
Two G3P molecules form glucose, while the other 10 G3P need to be recycled so that they can regenerate the 6 **RuBP** acceptor.

- Each of which becomes phosphorylated by **ATP** to produce **RuBP**, the five - carbon compound with which the cycle started.

In the Calvin cycle, 12NADPH2 and 18 ATPS are required to fix $6CO_2$ molecules into one hexose sugar molecule

- $12NADPH + 18\ ATP + 6CO2 \rightarrow C6H_{12}O_6 + 12\ NADP^+ + 18\ ADP + 18Pi + 6H_2O$

The following diagram shows the process of Calvin cycle



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Illustrative Examples

16. What is the name of enzyme that catalyse the combination of RuBP with CO2 molecule in the Calvin of photosynthesis?

A. RuBP carboxylase
B. PEP carboxylase
C. RuBP oxyginase
D. ATP synthase

Explanation: In light independent reaction (Calvin cycle), Ribulose which is found in the stroma bind with carbon dioxide by an enzyme RuBP carboxylase (Rubisco).

Answer: A

17. Which of the following process occurs during the light independent reactions of photosynthesis?

A. ATP formation
B. Splitting of water
C. Excitation of electrons
D. Fixation of carbon dioxide

Explanation: Splitting of water molecules, exciting of electrons and ATP formation takes place in thylakoid membrane during light dependent reaction. The fixation of Co2 takes place in the fluid stroma of the chloroplast during light independent reaction.

Answer: D

18. All the following are the inputs of Calvin cycle, except:

A. ATP B. NADPH C. CO_2 D. O_2

Explanation: The raw materials for the light independent reactions are ATP, NADPH, CO_2 and RuBP, but oxygen is a by product.

Answer: D

19. Which enzyme is found in C3 plants?

A. PEP carboxylase
B. Rubilose biphosphate oxygenase
C. Rubilose biphosphate carboxylase
D. B and C

Explanation: PEP carboxylase is used in C4 plants to catalyze a reaction between carbondioxide and a three carbon compound in order to form a four carbon compound called oxaloacetate.

Answer: D

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20. In light independent reaction reduced NADP is used to:

A. Oxidize GP to TP
B. Oxidize TP to GP
C. Reduce GP to GP
D. Reduce 3PG to G3P

Explanation: Reduced NADP which is produced during light dependent reaction is used to reduce 3PG to G3P is light independent reaction.

Answer: D

21. The three steps of light independent reaction are _____

Explanation: Carbon fixation, carbon reduction, and regeneration of RuBP.

9.1.3 C_4 Plant

Tropical and sub - tropical regions contain plants such as maize, crabgrass, sorghum and sugarcane utilize C_4 path way or undergo C_4 photosynthesis.

Many plant species living in hot, dry environments have adaptations that facilitate carbon fixation. The stomata close to reduce the loss of water vapor, greatly diminishing the supply of CO2.

- C4 plants first fix CO2 into a four - carbon compound, oxaloacetate
- CAM plants initially fix carbon at night through the formation of oxaloacetate.

The first product of carbondioxide fixation is the four carbon compound known as **oxaloacetic acid.**

- Plant that utilize this pathway are commonly called the C4 or four carbon plants.
- It has spatial or place separation steps.

C4 path way consists of four basic stages. These are:

i. Fixation of CO_2

- CO_2 reacts with a three carbons compound is called Phosphoenol pyruvate /PEP/ to form four carbons compound is called oxaloacetate
- The reaction is catalyzed by the enzyme PEP carboxylase (Pepco).

$$\text{Phosphoenolpyruvate} + CO_2 \xrightarrow{\text{PEP carboxylase}} \text{Oxaloacetate}$$

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- Takes place in **mesophyll cell.**

ii. Transport of the malate to the bundle sheath cells

- Oxaloacetate is converted in to another four carbons compound called malate by malate dehydrogenas enzyme in the mesophyll cell
- Malate is passed from the palisade mesophyll cell to the bundle sheath cell
- The chloroplast in the bundle sheath cell lacks thylakoid as the result light dependent reactions do not take place.

iii. Decarboxylation of the malate

- In the bundle sheath cell malate (4C) is converted in to pyruvate (3C) with the releases of carbon dioxide (1C) and the reaction is catalyzed by malice enzyme.
- The generated CO_2 enters the Calvin cycle and binds with Ribulose biphosphate and the reaction is catalyzed by Rubisco.
- Calvin cycle is takes place in the bundle sheath cell.

iv. Transport of the pyruvate

The pytuvate that is formed by the decarboxylation step moves to the mesophyll cell and regenerates the CO_2 acceptor is called phosphoenolpvruvate.

Note: Photorespiration is negligible in C4 plants such as crabgrass because the concentration of CO_2 in bundle sheath cells (where rubisco is present) is always high, which out compete oxygen for rubisco.

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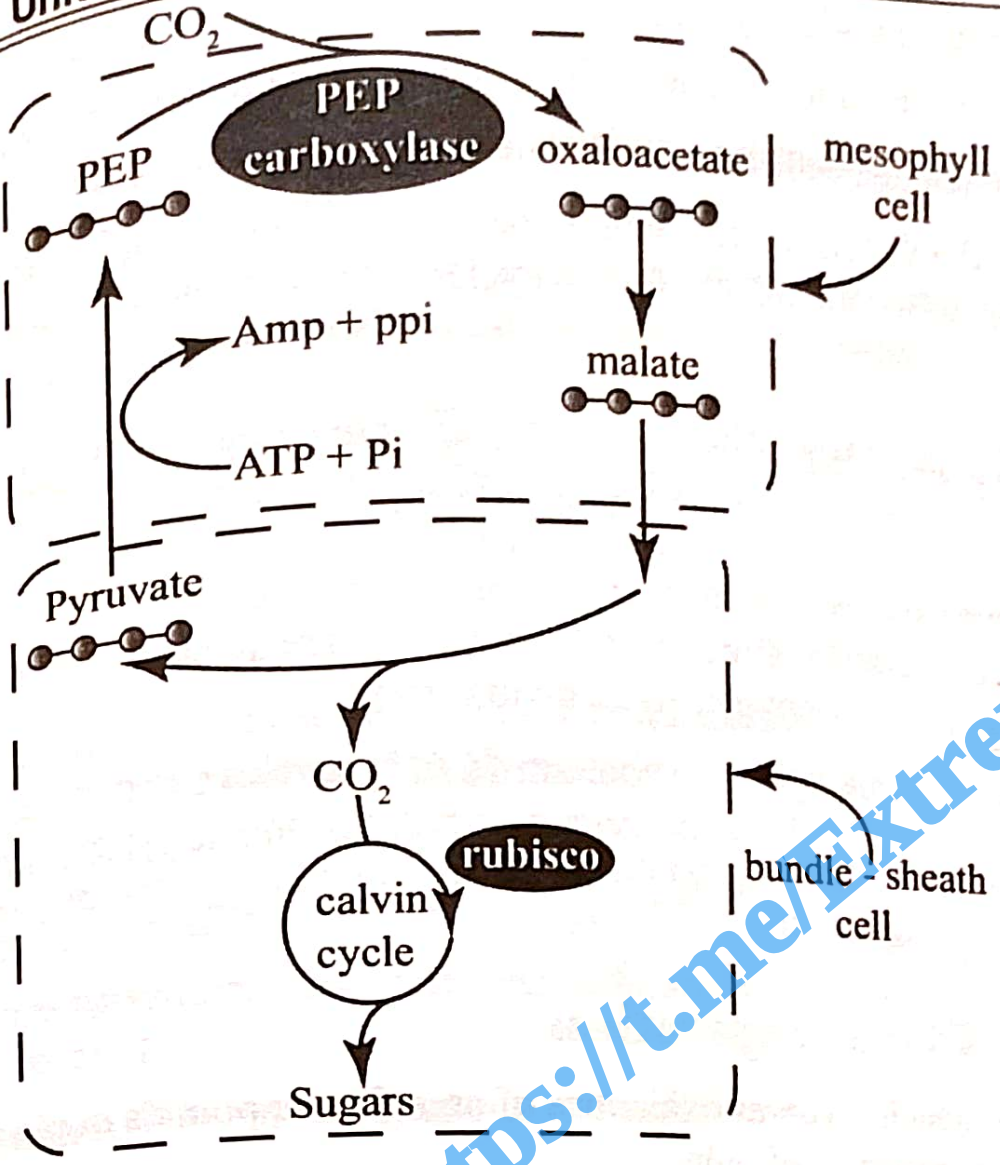

9.1.4 CAM photosynthesis

- It is also called crassulacean acid metabolism (CAM).
- Plants that are growing in the arid condition such as pineapple, cactus etc. uses CAM photosynthesis.
- It enables plants to improve water use efficiency.
- CAM is much like the metabolism of C4 plants in that CO_2 is initially fixed into a four - carbon compound.

However, In CAM plants the processes of initial CO_2 fixation and the Calvin cycle are separated in time, rather than in space.

A. At night time the plants open their stomata to allow entry of carbon dioxide

- Phosphoenolpvruvate + CO_2 $\xrightarrow[\text{carboxylase}]{\text{Phosphoenolpvruvate}}$ oxaloacetate
- Oxaloacetate is converted in to malate and stored in the vacuole of

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mesophyll cell.

B. During the day time the plants close their stomata to conserve water

➥ But the light dependent reaction generates ATP and NADPH

➥ So that the Calvin cycle continue because decarboxylation of malate releases carbon dioxide for Calvin cycle.

Note

In the C_4 pathway, the enzyme PEP carboxylase binds CO_2 effectively, even when CO_2 is at a low concentration. C_4 reactions takes place within mesophyll cells. The CO_2 is fixed in to oxaloacetate, which is then converted to malate. The malate moves into a bundle sheath cell, and CO_2 removed from it. The released CO_2 then enters the Calvin cycle.

The crassulacean acid metabolism (CAM) pathway is similar to the C4 pathway. PEP carboxylase fixes carbon at night in the mesophyll cells, and the Calvin cycle occurs during the day in the same cells (mesophyll).

9.1.5 Photorespiration

It is process which involves oxidations of organic compounds in plants by oxygen in the presence of light.

➥ This pathway is called photorespiration because it consume O_2 and releases CO_2

Rubisco is a carboxylase (adding CO_2 to RuBP) as well an oxygenase (adding O_2 to RuBP).

Rubisco acts as an oxygenase because hight levels of O_2 compete with CO_2 for the active site of rubisco.

➥ So when RuBP reacts with O_2, it cannot react with CO_2.

➥ This reaction reduces the overall CO_2 that is converted to carbohydrates, and therefore limits plant growth.

➥ When O_2 is added to RuBP, one of the products is a two carbon compound, phosphoglycolate:

RuBP + O_2 → phosphoglycolate + Phosphoglycerate

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It is common in C3 plants but highly reduced in C4 plants, and absent in CAM plants.

One molecule of phosphoglycolate /GP/ that is formed in photorespiration re - enter the Calvin cycle.

Phosphoglycolate must be conversed in to phosphoglycerate to enter the Calvin cycle in a series of chemical reactions in the chloroplast, peroxisome and mitochondrion

- The phosphoglycolate forms glycerate, which diffuses into membrane - enclosed organelles called peroxisomes. There, a series of reactions converts it to the amino acid glycine:

Glycolate → → glycine

- The glycine then diffuses into a mitochondrion, where two glycine molecules are converted to glycerate (a 3 C molecule), and CO_2:

2 Glycine → Glycerate + CO_2

- Glycerate finally join the calvin cycle Glycerate + ATP → GP + ADP

Photorespiration reducing the efficiency of photosynthesis by 25% because:

- It uses ATP in the re - synthesis of RUBP.
- Phosphoglycolate is removed in a long path.
- The carbon is oxidized which is the reverse of photosynthesis

Note

The reasons for photorespiration are not understood, but it may reflects the origin of rubisco at an ancient time when CO_2 levels were high and molecular oxygen levels were low.

The following diagram shows the whole process of photo respiration

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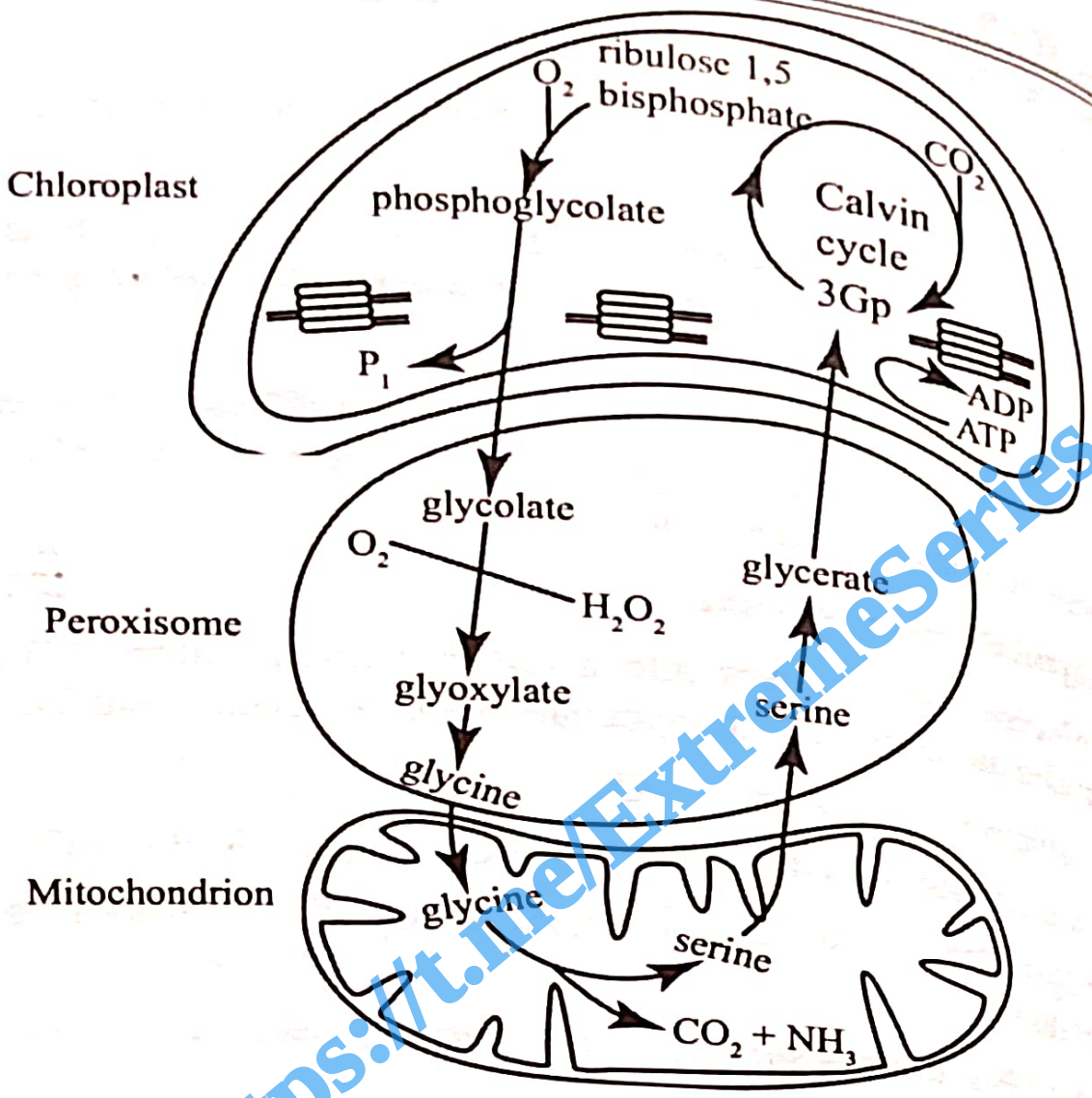

9.1.6 Contributions of Photosythesis for the Continuity of Life

The oxygen in the air comes from photosynthesis:

- ➥ Plants continue to release oxygen in the air.
- ☑ All of our food comes directly from photosynthesis.
- ☑ Human beings are also dependent on ancient products of photosynthesis (fossil fuels, natural gas, coal & petroleum)
- ☑ Carbon, oxygen, and hydrogen atoms are recycled in the environment where a constant input of solar energy to continue flowing to support life that removes carbon dioxide from the atmosphere (inhibit global warming).

Illustrative Example

22. Decarboxylation of malate in C4 plants takes place in:

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A. Mesophyll cell
B. Cortex
C. Bundle sheath cells
D. Stomata

Explanation: Decarboxylation of malate is takes place in the bundle sheath cells

malate $\rightarrow CO_2$ + pyruvate.

Answer: C

23. How does photorespiration decrease the efficiency of photosynthesis?
A. Carbon is oxidized instead of reduction to make glucose
B. It passes a lot of steps to remove phosphoglycolate
C. Extra ATP is used to resynthesize the RuBP
D. All are possible answers.

Explanation: All the above listed cases are the reason for photorespiration, decreasing the efficiency of photosynthesis.

Answer: D

24. Photo respiration involves the oxidation of:
A. PGA
B. RuBP
C. Chlorophyll
D. Chlorophyll B

Explanation: Photorespiration oxidation of **RuBP**

RuBP + oxygen $\xrightarrow{\text{Rubiso}}$ **GP** + phosphoglycolate.

Answer: B

25. Which of the following statements about C_4 and CAM photosynthesis is true?
A. In CAM photosynthesis the C_4 stage and the Calvin cycle are separated
B. In CAM photosynthesis the C_4 stage and the Calvin cycle are separated in pace.
C. In C_4 photosynthesis, the C_4 stage and Calvin cycle are separated in space.
D. In C_4 photosynthesis, the C_4 stage and Calvin cycle both occur in the same cell

Explanation: In C_4 photosythesis, the C_4 stage and Calvin cycle are separated in space. While in CAM photosynthesis the C_4 stage and

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the Calvin cycle are separated in time, but occur at the same space (mesophyll).

Answer: C

9 . 2 Cellular Respiration

Cellular respiration is the complex process in which cells, make adenosine triphosphate (ATP) by breaking down organic compounds.

- Both autotrophs are heterotrophs undergo cellular respiration to release energy.
- Some of the energy is used to make ATP, used by cells to do work.

9.2.1 What is the ATP Molecule like?

- The full name for ATP is Adenosine Tri - Phosphate
- Made of nitrogenous base (adenine), a pentose sugar and a phosphate.
- The three phosphate groups are liked to one another by two high energy bond called **phosphoanhydride**.
- The second and third phosphate bonds of ATP are unstable.

The hydrolysis of ATP produces ADP and a free phosphate molecule (Pi)

- $ATP + H_2O \rightarrow ADP + Pi + energy$
- $ADP + H_2O \rightarrow AMP + Pi + energy$
- When this phosphate bond is broken by hydrolysis, energy is released.
- This energy is needed for many cell reactions.
- This process is **exothermic reaction**, the reaction that releases energy

$ADP + Pi \rightarrow ATP$.

- The energy used to combine the two molecules is gained from food.
- This process is **endothermic reaction**, the reaction that requires energy

ATP is an energy currency of the cell, immediate source of energy for the cell, needed to do the cell work.

For example, ATP is required for the contraction of muscles

ATP is adapted to this role because it:

- Release energy in relatively small amounts that are closely matched for

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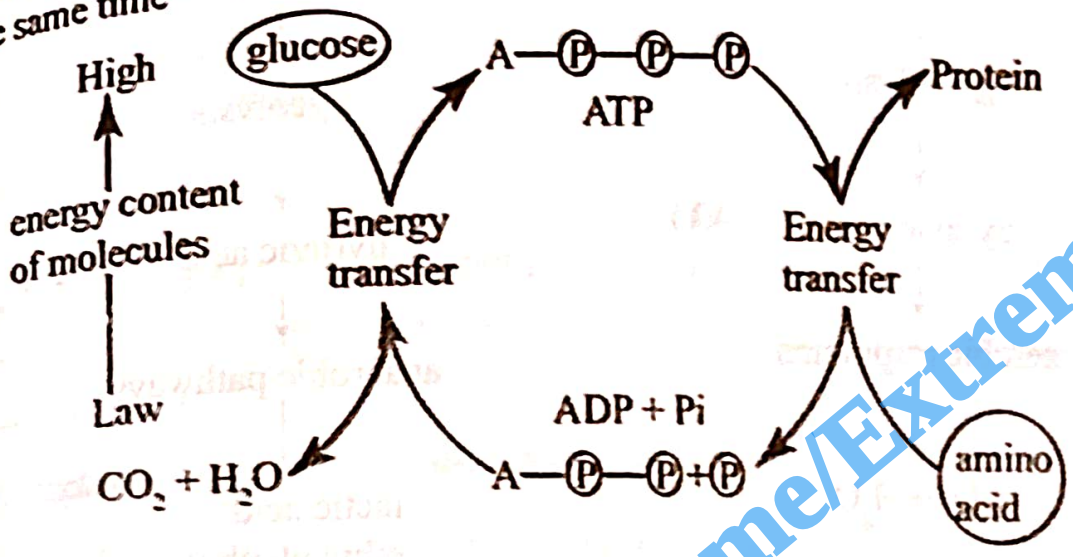
cell need

- Releases energy in a single - step of hydrolysis reaction, so the energy can be released quickly
- Able to move around the cell easily, but cannot escape from cells

ATP synthesis and ATP hydrolysis are coupled reaction, which occur at the same time



9.3 Types of Cellular Respiration

Cellular respiration can be divided into two stages: glycolysis (anaerobic) and aerobic respiration

1. **Glycolysis** - Organic compounds such as glucose are converted into three carbon molecules of **pyruvic acid.** Producing a small amount of ATP and NADH (an electron carrier molecule).
   - Glycolysis is anaerobic process because it does not require the presence of oxygen.
2. **Aerobic Respiration** - If oxygen is present in the cell's environment, pyruvic acid is broken down to produce NADH and FADH, which is used to make a large amount of ATP through the process known as aerobic respiration.
   - Pyuvic acid can enter other pathways if there is no oxygen present in the cell's environment.
   - The combination of glycolysis and these anaerobic pathways is called fermentation.

Although many kinds of organic compounds can be oxidized in cellular respiration, it is customary to focus on the simple sugar called glucose

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($C_6H_{12}O6$).

The following equation summarizes cellular respiration:

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy (ATP)}$$

(A) cellular respiration

organic compounds

↓

glycolysis

↓

pyruvic acid + ATP

↓

aerobic respiration

↓

$CO_2 + H_2O$

+

ATP

(B) Fermentation

organic compounds

↓

glycolysis

↓

pyruvic acid + ATP

↓

anaerobic pathways

↓

lactic acid, ethyl alcohol, or other, compunds

9.3.1 Glycolysis

Glycolysis is a biochemical pathway in which one glucose molecule is oxidized to produce two molecules of pyuvic acid (three - carbon molecule).

Glycolysis is a series of chemical reactions catalyzed by specific enzymes.

All of the reactions of glycolysis take place in the cytosol and occur in four main steps.

Step 1: two phosphate groups are attached to one molecules of glucose, forming a new six - carbon compound that has two phosphate groups.

- The phosphate groups are supplied by two molecules of ATP, which are converted into molecules of ADP in the process.
- Glucose + 2ATP → Fructose - 1 - 6 bisphospate

In step 2: the six - carbon compound is split into two three - carbon

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molecules of glyceraldehyde 3 - phosphate (G3P)

Fructose - 1 - 6 bisphospate → 2 glyceraldehyde 3 - phophate → 2 (G_3P)

In step 3: the two G3P molecule are oxidized, and each receives a phosphate group.

- The product of this step is two molecule of new three - carbon compound.
- The oxidation of G3P is accompanied by the reduction of two molecules of nicontinamide adenine dinucleotide (NAD^+) to NAD^+ is an organic molecule that accepts electrons during redox reactions. $2G_3P \rightarrow 2$ phosphoenol pyruvate.

In step 4: the phosphate group added in step 1 and steps 3 are removed from the three - carbon compounds formed in step 3.

$2G_3P \rightarrow 2$ pyruvate

- This reaction produces two molecule of pyuvic acid
- Each phosphate group is combined with a molecules of ADP to make a molecule of ATP.
- Because a total of four phosphate group were added in step 3, a total of 4 ATP are produced.
- In glycolysis, glucose is activated by the addition of phosphates from 2 ATP molecules and converted ultimately to 2 pyruvate + 2 NADH + 4 ATPs, yielding a net profit of 2 ATPs
- $C6H_{12}O6 + 2\,ATP + 2\,ADP + 2\,NAD^+ \rightarrow 2$ pyruvate $+ 4\,ATP + 2\,NADH + H_2O$

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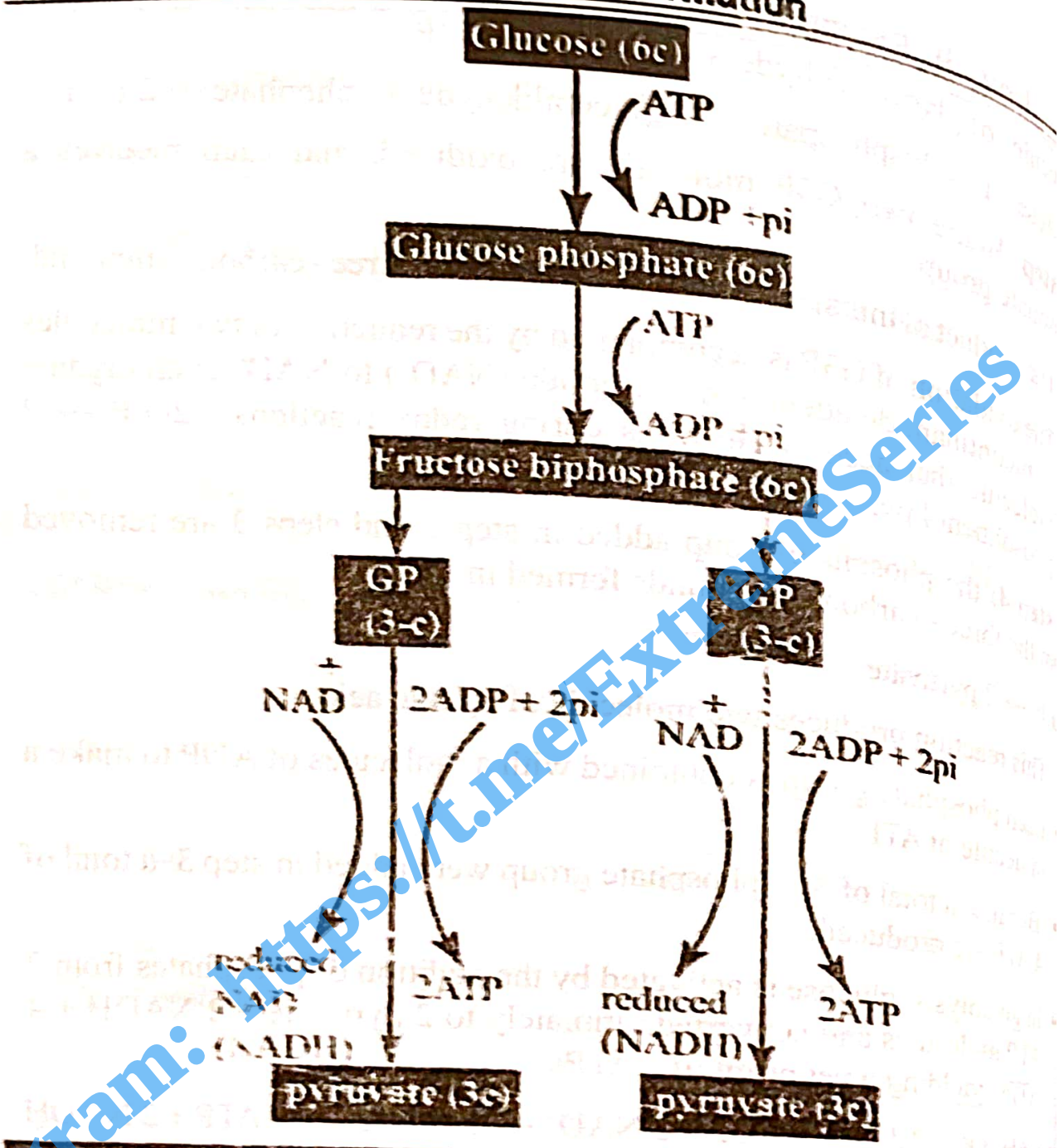

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Illustrative Example

26. Which of the following molecules is known as the energy currency of the cell?

A. Glucose B. Starch C. Fat D. ATP

Explanation: ATP (Adenosine triphosphate) is the direct source of energy which is also known as:

- Energy currency of the cell
- Cash source of energy

Answer: D

27. Which of the following defines ATP?

A. Adenine + adenosine + phosphate
B. Adenosine + Ribose sugar + double phosphate
C. Adenine + ribose sugar + three phosphate
D. Nitrogen + sugar + monophosphate

Explanation: ATP the immediate source of energy, which is composed of
- Adenine
- Ribose sugar
- Three phosphate

Answer: C

28. The amount of net gain of ATP that glycolysis provides to a cell is:
A. 2 ATP molecule
B. 4 molecules
C. 18 ATP molecules
D. 36 ATP molecules

Explanation: During glycolysis the total of 4 ATP molecules are produced. However, the cell uses 2 ATP to break down glucose molecule. The net of 2 ATP molecules are produced.

Answer: A

29. How many net gain of ATP is produced from 10 molecules of glucose by a cell in the process of glycolysis?
A. 40 B. 20 C. 380 D. 120

Explanation: A net gain of two molecules of ATP from the glycolysis of one glucose molecules. Thus from 10 molecules glucose 20 ATP molecules are gained.

Answer: B

30. Which one of the following steps in cellular respiration that takes places in the absences of oxygen?
A. Link reaction
B. Krebs cycle
C. Glycolysis
D. Electron transport chain

Explanation: Glycolysis is the first stage of cellular respiration that takes places in the absences of oxygen thus it is called the anaerobic stage of cellular respiration.

Answer: C

31. The reactions of _____ takes places within the cytosol of eukaryotic cells.

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A. Glycolysis
B. The oxidation of pyruvate
C. Electron transport chain
D. Chemiosmosis

Answer: A

What is the fate of pyruvic acid?

Fermentation is anaerobic process that does not use an electron transport chain.

Two common fermentation pathways result in the production of lactic acid and ethyl alcohol.

1. Lactic Acid Fermentation

In lactic acid fermentation, an enzyme converts pyruvic acid made during glycolysis into another three - carbon compound, called **lactic acid**.

Lactic acid fermentation involves the transfer of one hydrogen atom from NADH and the addition of one free proton (H^+) to pyruvic acid.

- In the process, NADH is oxidized to from NAD^+
- The resulting NAD^+ is used in glycolysis, where it is again reduced to NADH.
- Thus, the regeneration of NAD^+ in lactic acid fermentation helps to keep glycolysis operating.

Lactic acid fermentation by **micro organisms** plays an essential role in the manufacture of many dairy products.

Lactic acid fermentation also occurs in **animal** muscle cells during very strenuous exercise, such as sprinting.

- Muscle cells use up oxygen more rapidly than it can delivered to them.
- As oxygen becomes depleted, the muscle cells begin to switch from cellular respiration to lactic acid fermentation.
- Lactic acid accumulates in the muscle cells, making the cells cytosol more acidic.
- The increased acidity may reduces the capacity of the cells to contract, resulting in muscle fatigue, pain, and even cramps.
- Eventually, the lactic acid diffuses into the blood and a transported to the liver, where it can be converted back into pyruvic acid.

2. Alcoholic Fermentation

Some plant cells and unicellular organisms, such as yeast, use a process called alcoholic fermentation to convert pyruvic acid into **ethyl alcohol**.

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After glycolysis, this pathway requires two steps:

A. The CO_2 molecule is removed from pyuvic acid, leaving a two - carbon compound.

B. The two hydrogen atoms are added to the two - carbon compound to form ethyl alcohol.

Note: As in lactic fermentation, these hydrogen atoms come from NADH and H^+, regeneration NAD^+ for use in glycolysis.

Yeast cells will perform the reactions of fermentation, releasing ethyl alcohol and carbon dioxide in the process.

- The ethyl alcohol is the 'alcohol' in alcoholic beverages.
- The CO_2 that is generated in the first step of fermentation is allowed to escape
- Bread making also depends on alcoholic fermentation performed by yeast cells.
- The CO_2 makes the bread rise by forming bubbles inside the dough,
- The ethyl alcohol evaporates during baking

Illustrative Example

32. Which on of the following gives the products of fermentation in yeast cells?

A. Lactate, CO2, and 2ATP
C. Alcohol, CO_2 and 2 ATP
B. CO_2, H_2O and 36ATP
D. Alcohol, CO_2 and 36 ATP

Explanation: During anaerobic respiration by the yeast:

Pyruvate (3C) → Ethyl alcohol + Carbondioxide + 2 ATP

(C_2H_5OH) + (CO_2)

Answer: C

33. Which of the following is not rue about fermentation?

A. Net gain of only two ATP
B. Occurs in cytosol
C. NADH donates electrons to electron transport chain
D. Begins with glucose

Explanation: During fermentation NADH do not donate electrons to electron transport chain, because fermentation occurs by glycolysis in

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the cytosol of the cell.

34. When compared with aerobic respiration, fermentation of glucose by yeast:

Answer: C

A. Yield less ATP per molecule of glucose

B. Produces lactate

C. Produces more CO_2

D. None of the above

Explanation: Fermentation of glucose by yeast produces only 2ATPs. While during aerobic respiration 38 ATPs are produced from one glucose molecules.

35. During strenuous exercise, skeletal muscles use _____ and produce large amount of _____, which cause pain and fatigue in the muscle.

Answer: A

A. Pyuvic - carbon dioxide

B. Carbondioxide - pyruvic acid

C. Pyuvic acid - lactic acid

D. Oxygen - lactic acid

Explanation: lactic acid fermentation occurs during strenuous exercise when the body cannot keep up with the increased demand for oxygen by skeletal muscle. Pyruvic acid converts to lactic acid that builds up in the muscle causing fatigue and burning.

Answer: C

9.3.2 Aerobic Respiration

When oxygen is available, pyuvic acid undergoes aerobic respiration, the pathway of cellular respiration that requires oxygen.

- Aerobic respiration produces nearly **20 times as much** ATP as is produced by glycolysis alone.
- Aerobic respiration has three major stages **link reaction**, the **Krebs cycle** and the **electron transport chain**

1. Pyruvate oxidation (link reaction)

The pyruvic acid that is produced in glycolysis diffuses across the double membrane of a mitochondrion and enters the mitochondrial matrix.

- When pyuvic acid enters the mitochondrial matrix, it reacts with a molecule called **coenzyme A** to form acetyl coenzyme A, abbreviated

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acetyl CoA.

- The acetyl part of acetyl CoA contains two carbon atoms.
- The carbon atom that is lost in the conversion of pyruvic acid to acetyl CoA is released in a molecule of CO_2.
- This reaction reduces a molecule of NAD^+ to NADH.
- Summery of pyruvate oxidation

Pyruvate + 2 NAD^+ + 2 CoA → 2 acetyl CoA + 2NADH + 2 CO_2

2. The Kerbs cycle (citric acid cycle)

The Kerbs cycle is a biochemical pathway that breaks down acetyl CoA, producing CO_2, hydrogen atoms, NADH, $FADH_2$ and ATP.

The reactions that make up the cycle were identified by **Hans Krebs** (1900 - 1981), a German biochemist.

The mitochondrial matrix contains the enzymes needed to catalyze the reactions of Krebs cycle

The Krebs cycle has five main steps occur in the mitochondrial matrix.

In step 1: a two - carbon molecule of **acetyl CoA** combines with a four - carbon compound, **oxaloacetic acid**, to produce a six - carbon compound, **citric acid**. Notice that this reaction regenerates coenzyme A.

In step 2: citric acid releases a CO_2 molecules and a hydrogen atom to from a five - carbon compound. The electron in the hydrogen atom is transferred to NAD^+, reducing it to **NADH.**

In step3: the five - carbon compound releases a CO_2 molecule and a hydrogen atoms, forming a four carbon compound. Again, NAD^+ is reduced to **NADH.** A molecule of **ATP** is also synthesized from ADP.

In step 4: the four - carbon compound releases a hydrogen atom to form another four - carbon compound. This time hydrogen atom is used to reduce FAD to **FADH2.**

- FAD, or flavin adenine dinucleotide, is a molecule very similar to NAD^+
- Like NAD^+, FAD accepts electrons during redox reactions.

In step 5: the four - carbon compound releases a hydrogen atom to

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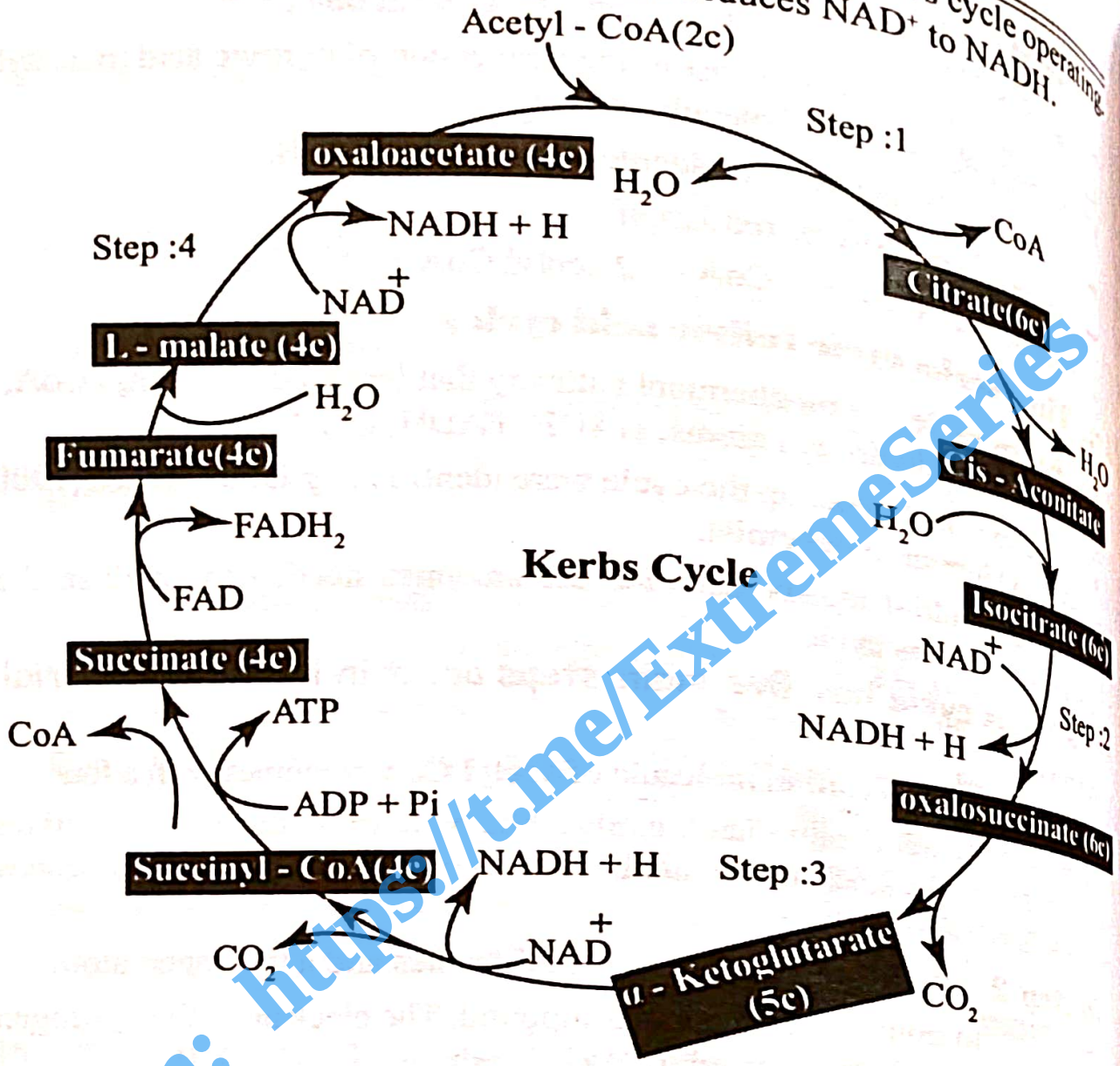
regenerate oxaloacetic acid, which keeps the Krebs cycle operating. The electron in the hydrogen atom reduces NAD^+ to NADH.



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Generally, one glucose molecule is completely broken down in two turns of the Krebs cycle.

- These two turns produce four CO_2 molecules, two ATP molecules, six NADH and two FADH2 molecules.
- The CO_2 diffuses out of the cells and is given off as waste.
- The ATP can be used for energy. The bulk of the energy released by the oxidation of glucose still has not been transferred to ATP.
- In Kreb's cycle, the 2 acetyl CoA molecules are metabolize to 4 CO_2 + 6 NADH + 2 FADH2 + 2ATPs.
- Summary of kreb's cycle:

2 acetyl CoA + 6 NAD^+ + 2FAD + 2ADP + 2Pi + $2H_2O$ →

$4CO_2 + 6NADH + 2FADH2 + 2ATP + 2\ CoA$

Lipids (fat) and proteins can also be used as respiratory substrates.

- The metabolism of proteins, lipids and carbohydrate 'converges' on the Krebs cycle or glycolysis.
- **Protein** is catabolized into amino acids, the amino group is removed by deamination process: and then amino acid joined glycolysis, the link reaction, or kreb's cycle.
- **Lipids** are catabolized into **fatty acid** and **glycerol;** and then joined glycolysis, the link reaction, or Kerb's cycle.
- Fatty acids are broken down into two pieces by the process of **beta oxidation** to inter as **acetyl CoA** to join Kreb's cycle

Note: Adding the six NADH molecules from the Krebs cycle given a total of 10 NADH molecules for every glucose molecule that is oxidized. These 10 NADH molecules and the two FADH2 molecules drive the next stage of aerobic respiration (ETC).

Illustrative Example

36. The net product of the Krebs cycles from one molecule of acetyl coenzyme are

A. ATP, 4NADH, FADH and 2 CO_2

B. 2ATP, 3NADH, 2FADH and CO_2

C. ATP, 3NADH, FADH and $2CO_2$

D. 2ATP, NADH, 3FADH and $2CO_2$

Explanation: During Krebs cycle from one molecule of acetyl COA the products are:

1ATP, 3NADH, 1FADH, and $2CO_2$.

Answer: C

37. Which one of the following molecules is not the product of Krebs cycle in cellular respiration?

A. CO_2 B. NADH C. ATP D. Pyruvate

Explanation: Pyruvate is three carbon molecule that is produced during glycolysis from the breaks down of glucose molecule. In Krebs's cycle:

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➥ 6 NADH, 2FADH2, 2 ATP and $4CO_2$ are produced during Krebs cycle.

Answer: D

38. Before pyruvate enters the citric acid cycle it is decarboxylated, oxidized and combined with coenzyme A forming acetyl CoA, carbon dioxide and one molecules of

A. NADH B. ATP C. $FADH_2$ D. ADP

Explanation: In link reaction, from one pyruvate, 1 NADH, $1CO_2$ and 1 acetyl CoA are produced.

Answer: A

39. Substrate level phosphorylation takes place in:

A. Glycolysis and Krebs cycle

B. The link reaction and electron transport chain

C. Glycolysis and electron transport chain

D. The link reaction and Krebs cycle

Explanation: Substrate level phosphorylation is the production of ATP without utilization of an enzyme ATP synthase. It takes places during glycolysis and Krebs cycle.

Answer: A

40. Which of the following is the greatest contributor of electrons to the electron transport chain?

A. Oxygen B. Glycolysis C. Krebs D. Link reaction

Explanation: Large number of electrons for electron transport chain is produced from Krebs cycle, in which 18 ATPs are formed from 6NADH and 4 ATPs are produced from $2FADH_2$.

Answer: C

41. During Krebs's cycle which of the following molecules temporarily stores most of the energy release from food molecule

A. ADP B. ATP C. NADH D. FADH

Answer: C

3. The electron transport (ETC)

Electron transport chain is a series of molecules in membrane that transfer electrons from one molecule to another

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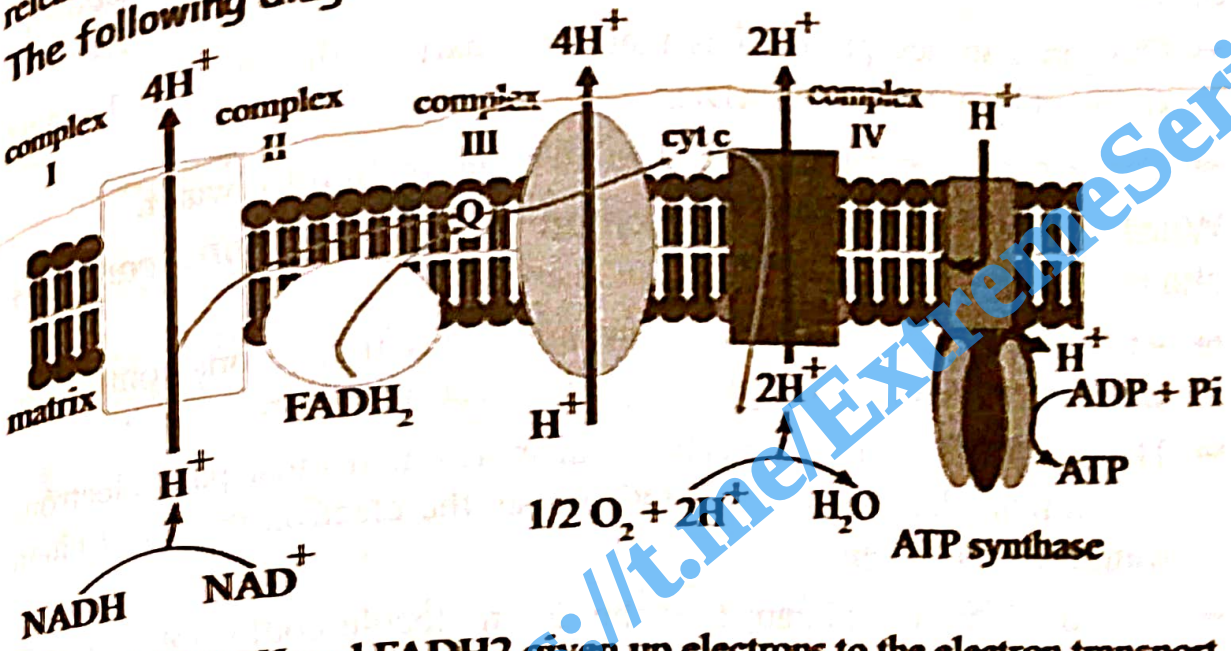
The electron transport chain, liked with chemiosmosis, constitutes the second stage of aerobic respiration.

The electron transport chain and the enzyme ATP synthase are embedded in the inner membrane of the mitochondrion in folds called cristae.

ATP is produced by the electron transport chain when NADH and FADH2 release hydrogen atoms, regenerating NAD^+ and FAD.

The following diagram shows the ETC and chemiosmosis.



In step1, NADH and FADH2 given up electrons to the electron transport chain.

- NADH donates electrons and protons (hydrogen ions, H^+) at the beginning
- FADH2 donates electrons and protons (hydrogen ions, H^+) farther down the chain

In step2, the electrons are passed down the chain

- As they move from molecule to another molecule, they lose energy.

In step3, the energy lost from the electrons is used to pump protons from the matrix, building a high concentration of protons to the inner membrane space (between the inner and outer membranes).

- Thus, a concentration gradient of protons is created across the inner membrane.

In step 4, the concentration gradients of protons drive the synthesis of ATP by chemiosmosis.

- In oxidative phosphorylation, the redox reactions in the electron

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transport chain are coupled to synthesis of ATP through the mechanism of chemiosmosis

- The same process that generates ATP in photosynthesis
- As protons move through ATP synthase and down their concentration gradients, ATP is made from ADP and phosphate.

In step 5, oxygen is the final acceptor of electrons that have passed down the chain.

- Oxygen also accepts protons that were part of the hydrogen atoms supplied by NADH and FADH2.
- The protons, electrons, and oxygen all combine to form water.

What is the importance of oxygen in the electron transport chain?

- ATP can be synthesized by chemiosmosis only if electrons continue to move from molecule to molecule in the electron transport chain.
- The last molecule in the electron transport chain must pass electrons in to a final electron acceptor. Otherwise, the electron transport chain would come to a halt.
- Consider what would happen if the last molecule could not "unload" the electrons it accepts, then no more electrons could enter the electron transport chain and ATP synthesis would stop.
- By accepting electrons from the last molecule in the electron transport chain, oxygen allows additional electrons to pass along the chain.
- As a result, ATP can continue to be made through chemiosmosis

How many ATP molecule are made in cellular respiration?

- Recall that glycolysis and the Krebs cycle produces a total of **four** ATP molecules directly for every glucose molecule that is oxidized.
- This process is known as substrate level phosphorylation. Which accounts 10% ATP production.
- Furthermore, **each NADH** molecules that supplies the electron transport chain can generate **three ATP** molecules, and **each FADH2** molecule can generate two **ATP molecules.**
- Thus, the **10 NADH** and **2 FADH2** molecules made through glycolysis, pyruvate oxidation and the Krebs cycle can produce up to 34 ATP molecules by the electron transport chain and chemiosmosis. It is called **oxidative phosphorylation, it accounts 90% of ATP producing/**

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glucose

- Alding the 4 ATP molecules from glycolysis and the Krebs cycle, and ETC gives a maximum yield of 38 ATP per molecule of glucose.
- Only 4 ATPs, which accounts 10% of ATP production/glucose are formed by substrate - level phosphorylation in glycolysis and the citric acid cycle

In most eukaryotic cells, the NADH that is made in the cytosol during glycolysis cannot diffuse through the inner membrane of the mitochondrion. Therefore

- The active transport of NADH consume ATP.
- As a result, most eukaryotic cell produce only abut 36 ATP molecule per glucose molecule

The following equation summarizes the complete oxidation of glucose in cellular respiration:

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy (36 ATP)}$$

Summary of Aerobic Respiration

i. **Glycolysis (in cytosol):-** Series of reactions in which glucose is degraded to pyuvate: net profit of 2 APTs hydrogen atoms are transferred to carrier: can proceed anaerobically.

Some Starting Materials are:

- Glucose ATP, NAD, ADP, P;

Some End products are :

- Pyruvate, ATP, NADH

ii. **Pyuvate Oxidation (in mitochondria):-** Pyuvate is degraded and combined with coenzyme A to form acetyl CoA: hydrogen atoms are transferred to carriers: CO_2 is released.

Some Starting Materials are:

- Pyuvate, coenzyme A, NAD

Some End products are :

- Acetyl CoA, CO_2, NADH

iii. **Kerb's cycle (in mitochondria):-** Series of reactions in which the acetyl portion of acetyl CoA is degraded to CO2, hydrogen atoms are transferred to carried to carriers: ATP is synthesized.

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Some Starting Materials are:

➥ Acetyl CoA, H_2O, NAD, FAD, ADP, P;

Some End products are :

➥ CO_2, NADH, $FADH_2$, ATP

iv. Electron transport and chemiosmosis (in mitochondria).

Chain of several electron transport molecules, electron are passed along chain, released energy is used to form a proton gradient, ATP is synthesized as protons diffuse down the gradient, oxygen is final electron acceptor.

Some Starting Materials are:

➥ NADH, FADH2, O_2 , ADP, P

Some End products are :

➥ ATP, H_2O, NAD^+, FAD

Illustrative Example

42. Choose the final electron acceptor in the electron transport chain during aerobic respiration of eukaryotic cells?

A. H_2O B. O_2 C. CO_2 D. NADP

Explanation: The final electron acceptor in the electron in the electron transport chain is oxygen (O_2).

Answer: B

43. If a cell contains 15 NADH and 10 FADH molecules: how many ATP molecules are produced with in the cell?

A. 65 ATP B. 80 ATP C. 30 ATP D. 70 ATP

Explanation: during electron transport chain from 1 NADH 3 ATP molecule and from 1 FADH2 2ATP molecules are produced, thus:

➥ 15 NADH × 3 ATP = 45 ATP

➥ 10FADH × 2 ATP = 20 ATP

Total = 65 ATP

Answer: A

44. Of the 36 ATP molecules the produced during the complete breakdown of glucose most are due to the action of

A. Chemiosmotic phosphorylation

B. Oxidative phosphorylation

C. Substance level phosphorylation

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D. Both A and B are correct

Explanation: About 90% of ATP molecule which are produced by oxidative phosphorylation and chemiosmosis in electron transfer chain

Answer: D

45. In the electron transport chain electron are passed:

A. From the lumen of the mitochondria inter membrane space.

B. From the interim membrane space to the lumen of the mitochondria.

C. Through ATP syntheses.

D. Along a series of electron carries.

Explanation: during electron transport chain electron pass through electron carries such as ubiquinone, cytochrome, etc.

Answer: D

46. Choose the pair that is correctly matched.

A. Kerbs cycle - cytoplasm

B. Electron transport chain - outer compartment of mitochondria

C. Glycolysis - mitochondria

D. Electron transport chain - cristae membrane

Explanation

Stages of cellular respiration	Takes place in
Glycolysis	Cytoplasm
Kerbs cycle	Matrix of mitochondria
Electron transport chain	Inner mitochondrial membrane celled cristae

Answer: D

47. Describe the process by which the protons gradient drives ATP synthesis in chemiosmosis?

Explanation: the diffusion of proton through proton channels from inner membrane space of mitochondria to the matrix to spin the rotor, then enzyme ATP synthase produced ATP.

Review Exercise

1. The Calvin cycle begins when CO_2 reacts with

A. Phosphoenolpvruvate

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B. Glyceraldehyde - 3 phosphate
C. Ribulose bisphosphate
D. Oxaloacetate

2. The enzyme directly responsible for almost all carbon fixation on Earth is
A. Rubisco
B. PEP carboxylase
C. ATP synthase
D. Phosphofrutokinase

3. In C4 plants, C4 and C3 pathways occur at different location: whereas in CAM plants, CAM and C3 pathways occur at different.
A. Times of day: locations within the leaf
B. Locations, times of day
C. Seasons: locations
D. Locations: season

4. An organism characterized as a photoautotrophs obtains energy from ___ and carbon from _____ ?
A. Light: organic molecules
B. Light: CO_2
C. Organic molecules organic molecules
D. Organic molecules: CO_2

5. Type of phosphorylation in which a phosphate group is transferred to ADP from another substance to make ATP.
A. Substance level phosphorylation
B. Chemiosmosis
C. Oxidative phosphorylation
D. Electron transport chain

6. Type of phosphorylation that occurs at the end of the electron transport chain.
A. Substance level phosphorylation
B. Chemiosmosis
C. Oxidative phosphorylation
D. Electron transport chain

7. In which part of mitochondria does the Kerb's take places?
A. On the outer membrane surface
B. In the inter - membrane space

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C. On the inner membrane surface
D. In the matrix

8. Which of the stages in the aerobic respiration of glucose does not produce ATP?
A. Electron transport & chemiosmosis
B. Glcolysis
C. Kerb's cycle
D. The link reaction

9. Which one is first inhibited if a cell contains excess of ATP?
A. Glycolysis
B. Kreb's cycle
C. Electron transport
D. Oxidative phosphorylation

10. Which of the following molecules can provide the cell the least amount of energy per molecule?
A. ATP
B. Glucose
C. Reduced NAD
D. Reduced FAD

11. During alcohol fermentation, pyruvic acid in the presence of pyruvate decarboxylatase is first converted to _____.
A. Lactic acid
B. Lactose
C. Acetyldehyde
D. Glucose

12. The conversion of one molecule of glucose to two molecules of pyruvate results in the net formation of:
A. 38 ATP molecules
B. 4 ATP molecules
C. 6 ATP molecules
D. 2 ATP molecules

13. During which stage of aerobic respiration does oxidative phosphorylation occur?
A. Chemiosmosis
B. Kreb's cycle
C. Link reaction
D. Glycolysis

14. Which of the following class of organic molecules is the least importance source of energy for cellular respiration?
A. Proteins
B. Nucleic acid
C. Lipids
D. Carbohydrates

15. Which of the following takes place under normal conditions, as electrons flow down the electron transport chain of the mitochondria?

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A. NADH and $FADH_2$
B. An electron chemical gradient
C. The PH of the matrix increases
D. The electrons lose free energy

16. Which of the following happens in cyclic and non - cyclic photophosphorylation?
A. Oxygen is generated
B. ATP is formed
C. Water molecules splits
D. NADP is reduced

17. Which is the purpose of the in foldings of the inner - cyclic member of the mitochondrion?
A. Increasing the surface area for ATP production
B. Speed up the loss of CO_2 during fermentation
C. Increasing the photosynthetic capacity of the cell
D. CO_2 H_2O and 36 ATP

18. What does the fermentation of glucose by yeast normally yield?
A. Lactic acid, CO_2 and 2 ATP
B. Alcohols, CO_2 and 36 ATP
C. Alcohols, CO_2 and 2ATP
D. CO_2, H_2O and 36 ATP

19. When athletes take part in short distance running, how do the cells generate most of the energy that is quickly needed?
A. Yeast fermentation in the stomach
B. Aerobic respiration in muscle cells
C. Mitochondrial respiration in any cell
D. Anaerobic respiration in muscle cells

20. Photosynthesis is:
A. Anabolic and endergonic
B. Catabolic and exergonic
C. Anabolic and exergonic
D. Catabolic and endergonic

21. Which of the following is not true about mitochondria and chloroplasts?
A. Both contain nuclei
B. Both transduce energy
C. Both contain chlorophyll
D. Both have double membrane

22. Cluster of photosynthetic pigments on a thylakoid membrane:

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A. Photorespiration
B. Photophosphorylation
C. Photosystem
D. Stroma

23. What is the ultimate source of electrons that replaces those lost from the photosystem II during photosynthesis?
A. Chlorophyll - a
B. Chlorophyll - b
C. Water
D. Photosystem I

24. Which of the following results in the production of oxygen during photosynthesis?
A. Electron transfer system of photosystem I
B. Reducing NADP
C. Electron transfer system of photosystem II
D. Splitting of the water molecule

25. During photosynthesis which of the following is the first step in the transduction of light energy to chemical energy?
A. Glucose formation
B. Photolysis of water
C. ATP formation
D. Carbon fixation

26. During photosynthesis which of the following is the first step in the transduction of light energy to chemical energy?
A. Glucose formation
B. Photolysis of water
C. ATP formation
D. Carbon fixation

27. In the process of photosynthesis light is necessary to:
A. Product ATP and a reducing substance
B. Combine CO_2 and H_2O
C. Split CO_2
D. Release energy

28. Enzyme that cataylses the reaction between RuBP and carbon dioxide.
A. ATP synthesis
B. Pepco (PEP carboxylase)
C. Rubisco
D. Photosynthesis

29. The process in which RuBP combines with oxygen rather than carbon dioxide
A. Carboxylation
B. Photolysis
C. Phosphorylation
D. Photorespiration

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30. The types of photosynthesis found in may tropical plants is.
A. C3 B. C4 C. CAM D. Fluid stroma

31. Among the following crops, which one uses an non - C4 photosynthetic pathway?
A. Sugarcane C. Maize
B. Sorghum D. Pineapple

32. Which of the following stages of photosynthesis can take place in the dark (light independent phase)?
A. Cyclic photo phosphorylation
B. Calvin cycle
C. Non cyclic photo phosphorylation
D. Photolysis of water

33. Which of the following plants uses CAM photosynthesis?
A. Cactus B. Maize C. Sugarcane D. Sorghum

34. ATP and NADPH are used to drive the synthesis of organic molecule in
A. Photo system - I C. Calvin cycle
B. Photo system II D. Light dependent reactions

35. Which one of the following is not true about both mitochondria and chloroplasts?
A. They have nucleic acid C. They have double membrane
B. They have transducer of energy D. They have cristae

36. The lost electron from PSII in non - cyclic photophorylation is replaced by _____ ?
A. Reduced NADP C. Photolysis of water
B. ATP D. Chlorophyll molecule

37. If a cell contains 8 NADH and 6 FADH molecules, how many ATP molecules would be produced in ETC.
A. 14 ATP B. 36 ATP C. 28 ATP D. 14 ATP

38. Which process of respiration generate most of the ATP molecules stored in Glucose
A. Electron transport chain C. Kerbs cycle
B. Anaerobic respiration D. Glycolysis

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39. The path ways of carbon fixation used by some plants to survive in hot and arid climate is
   A. C - 4 photosynthesis and C - 3 photosynthesis
   B. CAM photosynthesis and C - 3 photosynthesis
   C. C - 4 photosynthesis and CAM photosynthesis
   D. C - 3 photosynthesis only

40. Which of the following is not happens during the light dependent stages of photosynthesis
   A. Oxygen is split from water
   B. NADPH is produced
   C. Carbon dioxide fixation
   D. ATP is produced

41. The role of oxygen gas in oxydative phophorylation is
   A. Catalyzes reactions in glycolysis
   B. Form ATP
   C. Produces CO2
   D. Accepts electrons from the electron transport chain.

42. Glycolysis
   A. Takes place in the mitochondrion
   B. Is the same thing as fermentation
   C. Reduces two NAD^+ for every glucose molecule processed.
   D. Produces no ATP

43. Fermentation
   A. Takes place in the mitochondrion
   B. Does not require O_2
   C. Takes place in all animal cells
   D. Requires lactic acid

44. Which statement about pyruvate is not true?
   A. It is the end product of glycolysis
   B. It is a precursor of acetyl CoA
   C. It contains three carbon atoms
   D. It is a protein

45. Compared to anaerobic metabolism, aerobic breakdown of glucose produces

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A. More ATP
B. Pyruvate
C. Less CO_2
D. More oxidized coenzyme

46. Which statement about oxidative phosphorylation is not true?
A. It is the formation of ATP by the respiratory chain
B. It is brought about by the chemiosmotic mechanism
C. In eukaryotes, it takes place in mitochondria.
D. Its functions can be served equally well by fermentation

47. In noncyclic photosynthetic electron transport, water is used to
A. Excites chlorophyll
B. Hydrolyzes ATP
C. Reduces chlorophyll
D. Oxidizes NADPH

48. In cyclic electron transport
A. Oxygen gas is released
B. ATP is formed
C. Water donates electrons and protons
D. NADPH is formed

49. Which of the following does not happen in noncyclic electron transport?
A. Oxygen gas is released
B. Water donates electrons and protons
C. ATP forms
D. NADPH forms
E. Only PSI is involved

50. In the chloroplasts
A. Light leads to the pumping of protons out of the thylakoids
B. ATP forms when protons are pumped into the thylakoids
C. Light causes the stroma to become more basic than thylakoids
D. Protons return passively to the stroma through protein channels.

51. In C4 photosynthesis,
A. 3PG is the first product of CO_2 fixation
B. Rubisco catalyzed the first step into the thylakoids.
C. Four - carbon acids are formed by PEP carboxylase in bundle sheath cells.

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D. Photosynthesis continues at lower CO_2 levels than in C3 plants

52. Photosynthesis in green plants occurs during the day. Respiration on plants occurs
A. Only at night
B. Only when there is enough ATP
C. Only during the day
D. All the time

53. Which one of the following gives the products of fermentation in yeast cells?
A. Lactate, CO_2 and 2ATP
B. CO_2, H_2O and 36 ATP
C. Alcohol, CO_2 and 2ATP
D. Alcohol CO_2 and 36 ATP

54. The process of splitting larger molecules in to smaller ones is an aspect of metabolism called
A. Anabolism
B. Fermentation
C. Chemiosmosis
D. Catabolism

55. The pathway through which glucose is degraded to pyuvate is called
A. Aerobic respiration
B. The citric acid cycle
C. Alcohol fermentation
D. Glycolysis

56. The reactions of _____ takes places within the cytosol of eukaryotic cells.
A. Glycolysis
B. The oxidation of pyruvate
C. Electron transport chain
D. Chemiosmosis

57. Before pyruvate enters the citric acid cycle it decarboxylated, oxidized and combined with coenzyme A forming acetyl CoA, carbon dioxide and one molecules of.
A. NADH B. ATP C. FADH, D. ADP

58. In the first step of the citric acid cycle, acetyl CoA reacts with oxaloacetate to dorm:
A. Pyruvate B. NADH C. Citrate D. ATP

59. Which one of the following is a major source of electron for electron transport chain?
A. H_2O B. NADH C. Coenzyme D. ATP

60. When deprived of oxygen, yeast cell obtain energy by fermentation

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producing carbon dioxide, ATP and _____

A. Acetyl CoA
B. Lactate
C. Ethyl alcohol
D. Pyruvate

61. During strenuous muscles activity, the pyruvate in muscle accept hydrogen from NADH to become _____

A. Acetyl CoA
B. Lactate
C. Ethyl alcohol
D. Pyruvate

62. Which of the following is not true about fermentation?

A. Net gain of only two ATP
B. Occurs in cytosol
C. NADH donates electrons to electron transport chain
D. Begins with glucose

63. ATP is ideal energy molecule in cells because it:

A. Releases energy in small amount
B. Move freely in but not escape from the cell
C. Releases energy quickly
D. All of the about

64. When light strikes photosystem I (PSI), the electron carrier that pass electron to NADP is:

A. Primary electron acceptor
B. Plastocyanine
C. Ferridoxin
D. Ubiquinone

65. Which of the following statements about C4 and CAM photosynthesis is true?

A. In CAM photosynthesis the C_4 stage and the Calvin cycle are separated
B. In CAM photosynthesis the C_4 stage and the Calvin cycle are separated in space.
C. In C4 photosynthesis, the C_4 stage and Calvin cycle are separated in space.
D. In C4 photosynthesis, the C_4 stage Calvin cycle both occur in the same cell

66. Photorespiration

A. Takes place only in C_4 plants

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B. Increases the yield of photosynthesis
C. Is catalyzed by RuBP oxygenas
D. Is catalyzed by PEP carboxylase

67. In photolysis, some of the energy captured by chlorophyll is used to split
A. CO_2 B. ATP C. NADPH D. H_2O

68. Light is composed of particles of energy called:
A. Carotenoids C. Photo systems
B. Antenna complexes D. Photons

69. In plants the final electron acceptor in the light dependent reaction is:
A. $NADP^+$ B. CO_2 C. H_2O D. O_2

70. In addition to chlorophyll, most plants contain accessory photosynthetic pigments such as
A. PEP C. $NADP^+$
B. Carotenoids D. G_3P

71. The input for _____ are CO_2, NADPH2 and ATP
A. Cyclic electron transport
B. Non - cyclic electron transport
C. Carbon fixation reaction
D. Photosynthesis

72. The Calvin cycle begins when CO_2 reacts with:
A. Phosphoenol pyuvate
B. Riblose bisphospate
C. Glyceraldehyde - 3 - phosphate
D. Oxaloacetate

73. The enzyme that is directly responsible for almost all carbon fixation on earth is:
A. Rubisco C. ATP synthase
B. PEP carboxylase D. Phosphofructo kinase

74. Which compound produces ATP when oxidized?
A. Acetyl - COA C. Glyceraldehyde - 3 - phosphate
B. Glucose D. NADH

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75. At the end of glycolysis most of the energy remaining from the original glucose molecule is locked up in:
A. $NADH^+$ B. CO_2 C. Pyuvic acid D. ATP

76. Which enzyme has high affinity to CO_2?
A. PEP carboxylase C. Glyceraldehyde
B. Rubisco D. ATP synthase

77. During Krebs's cycle which of the following molecules temporarily stores most of the energy release from food molecules.
A. ADP B. ATP C. NADH D. FADH

78. What is the ultimate source of energy for all organisms living on the earth?
A. Water B. Oxygen C. Nitrogen D. Sun

79. From which of the process of cellular respiration is the majority of the ATP is produced
A. Anaerobic and fermentation
B. Glycolysis and link reaction
C. Oxidative phosphorylation and chemiosmosis
D. Kerbs cycle and glycolysis

80. Choose the final electron acceptor in the electron transport chain during aerobic respiration of eukaryotic cells?
A. H_2O B. O_2 C. CO_2 D. NADP

81. The net product of the Krebs cycle from one molecule of acetyl coenzyme are:
A. ATP, 4NADH, FADH and $2CO_2$
B. 2ATP, 3NADH, 2FADH and CO_2
C. ATP, 3NADH, FADH and $2CO_2$
D. 2ATP, NADP, 3FADH and $2CO_2$

82. If a cell contains 15 NADH and 10 FADH molecules how many ATP molecules are produced with in the cell?
A. 65 ATP B. 80 ATP C. 30 ATP D. $C_6H_{12}O_6$

83. The reaction of _____ take place within the cytosol of eukaryotic cells.
A. Glycolysis C. The citric acid cycle
B. Oxidation of pyruvate D. Chemiosmosis

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84. Before pyruvate enters the citric acid cycle, it is decarboxylated, oxidized, and combined with coenzyme A, forming acetyl CoA, carbon dioxide, and one molecule of

A. NADH B. FADH2 C. ATP D. C6H12O6

85. In the first step of the citric acid cycle, CoA reacts with oxaloacetate to form

A. Pyruvate B. Citrate C. NADH D. ATP

86. Which of the following is the major source of electrons that flow through the mitochondrial electron transport chain?

A. H_2O B. ATP C. NADH D. ATP synthase

87. The "aerobic" part of aerobic cellular respiration begin at

A. Glycolysis
B. The conversion of pyruvate to acetyl CoA
C. The citric acid cycle
D. Electron transport

88. Substrate - level phosphorylation

A. Occurs through a chemiosmosis mechanism
B. Accounts for most of the ATP formed during aerobic cellular respiration
C. Occurs during the conversion of pyruvate to acetyl CoA
D. Occurs during glycolysis and the citric acid cycle

89. Which of the following is a correct ranking of molecules, according to their energy value in oxidative phosphorylation?

A. ATP = NADP C. FAD > FADH2
B. NAD > NADH D. NADH > ATP

90. A net profit of only 2 ATPs can be produced anaerobically from the ____ of one molecule of glucose, compared with a maximum of 38 ATPs produced in _____.

A. Fermentation: anaerobic respiration
B. Aerobic respiration anaerobic respiration
C. Aerobic respiration: fermentation
D. Fermentation: aerobic respiration

91. When deprived oxygen, yeast cells obtain energy by fermentation.

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producing carbon dioxide, ATP and

A. Acetyl CoA
B. Ethyl alcohol
C. Lactate
D. Pyuvate

92. Where is chlorophyll located in the chloroplast?

A. Thylakoid membranes
B. Matrix
C) Stroma
D) Between the inner and outer membranes

93. In photolysis, some of the energy captured by chlorophyll is used to split

A. CO_2
B. ATP
C. NADPH
D. H_2O

94. Light is composed of particles of energy called

A. Carotenoids
B. Reaction centers
C. Photons
D. Photo systems

95. In plants, the final electron acceptor in noncyclic electron flow is

A. $NADP^+$
B. CO_2
C. H_2O
D. O_2

96. In electrons that has been energized by light contribute their energy to add phosphate to ADP, production ATP,

A. Crassulacean acid metabolism
B. The Calvin cycle
C. Photorespiration
D. Photophosphorylation

97. The mechanism by which electron transport is coupled to ATP production by means of a proton gradient is called

A. Chemiosmosis
B. Crassulacean acid metabolism
C. Fluorescence
D. The C3 pathway

98. During aerobic respiration what is the route through which protons return from the mitochondria inter membrane space back to its matrix?

A. Proton pump
B. ATP synthase
C. Ion channel
D. Membrane lipid

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Fill in the blank space

99. The main energy transfer molecule in a cell is ___
100. The final (terminal) electron acceptor in aerobic respiration is _____
101. Process in which RuBP combines with oxygen rather than carbon dioxide is _____
102. Cluster of photosynthetic pigments on a thylakoid membrane ____
103. Light - independent reaction of photosynthesis occurs in the ____
104. Oxygen released in photolysis in photosynthesis derived from _____
105. The products of photolysis in photosynthesis are ____, and ____
106. The Calvin cycle in C4 photosynthesis occurs in ____
107. C4 photosynthesis is most efficient in conditions ____ and ____
108. The products of the light - dependent reactions of photosynthesis are _____ and _____
109. The region on chloroplast in which the Calvin cycle (light independent reaction) in C_3 of photosynthesis take places is
110. The conversion of energy from one form to another from is called ___
111. The first molecule which accepts the excited electrons displaced from antenna complex is _____
112. The splitting of water molecules during light dependent reaction of photosynthesis is called _____
113. The diffusion of hydrogen ions (proton) through the ATP synthase enzyme is called _____
114. The raw materials of photosynthesis are ___, ___ and ___

Essay type

115. Define chemiosmosis and explain how a gradient of protons is established across the inner mitochondrial membrane?
116. Describe the process by which the proton gradient drives ATP synthesis in chemiosmosis?
117. Distinguish between the light - dependent reactions and carbon fixation reactions of photosynthesis?
118. Describe what happens to an electron in a biological molecule such as chlorophyll when a photon of light energy is absorbed?
119. Describe the physical properties of light and explain the relationship

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between a wavelength of light and its energy.

120. Explain how a proton (H^+) gradient is established across the thylakoid membrane and how this gradient functions in ATP synthesis?

121. Summarize the three phases of the Calvin cycle and indicate the roles of ATP and NADPH in the process.

122. Discuss how photorespiration reduces photosynthetic efficiency.

123. Compare the C4 and CAM pathways?

Answer To Review Exercise

I. Multiple choices

1. C	11.C	21.C	31.D	41.D	51.D	61.B	71.C	81.C	91.B
2. A	12.D	22.C	32.B	42.C	52.D	62.D	72.B	82.A	92.A
3. A	13.A	23.C	33.A	43.B	53.C	63.D	73.A	83.A	93.D
4. B	14.B	24.D	34.C	44.D	54.D	64.C	74.D	84.A	94.C
5. A	15.D	25.B	35.D	45.A	55.D	65.C	75.C	85.B	95.A
6. B	16.B	26.B	36.C	46.D	56.A	66.C	76.A	86.C	96.D
7. D	17.A	27.A	37.B	47.D	57.A	67.D	77.C	87.B	97.A
8. D	18.C	28.C	38.A	48.B	58.C	68.D	78.D	88.D	98.A
9. A	19.D	29.D	39.C	49.E	59.B	69.A	79.C	89.D	
10.A	20.A	30.B	40.C	50.B	60.C	70.B	80.B	90.D	

II. Short answer

99. Glucose
100. Oxygen
101. Photorespiration
102. Photosystem
103. Stroma
104. Water
105. Electron, hydrogen and oxygen
106. Bundle sheath cell
107. Hight temperature, low carbondioxide
108. ATP, NADPH
109. Mesophyll
110. Energy transformation (transduction)
111. Primary electron acceptor
112. Photolysis
113. Chemiosmosis
114. H_2O, CO_2, chlorophyll and sunlight

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III. Essay type

115. In chemiosmosis, some of the energy of the electrons in the electron transport chain is used to pump protons across the inner mitochondrial membrane into the into intermembrane space. This pumping establishes a proton gradient across the inner mitochondrial. Protons (H^+) accumulate within the intermembrane space, lowering the pH.

116. The diffusion of protons through channels formed by the enzyme ATP synthase, which extends through the inner mitochondrial membrane from the intermembrane space to the mitochondrial matrix, provides the energy to synthesized ATP.

117. In the light - dependent reactions, electrons energized by light are used to generate ATP and NADH: these compounds provide energy for the formation of carbohydrates during the carbon fixation reactions.

118. Photons excite biological molecules such as chlorophyll and other photosynthetic pigments, causing one or more electrons to become energized. These energized electrons may be accepted by electron acceptor compounds.

119.

- Light consists of particles called photons that move as waves.
- Photons with shorter wavelengths have more energy than those with longer wavelengths.

120.

Photophorylation is the synthesis of ATP coupled to the transport of electrons energized by photons of light. Some of the energy of the electrons is used to pump protons across the thylakoid membrane, providing the energy to generate ATP by chemiosmosis .

As protons diffuse through ATP synthase, an enzyme complex in the thylakoid membrane, ADP is phosphorylated to form ATP.

121.

A. The carbon fixation reactions

CO_2 is combined with ribulose bisphospate a five - carbon sugar, by the enzyme ribulose bisphospate carboxylase (rubisco), forming the three - carbon molecule phosphoglycerate (PGA).

B. In the carbon reduction phase

- The energy of ATP and NADP are used to convert PGA molecules

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to glyceraldehyde - 3 - phosphate (G3P).

- For every 6 CO_2 molecules fixed, 12 molecules of G3P are produced, and 2 molecules of G3P leave the cycle to produce the equivalent of 1 molecule of glucose.

C. In **the RuBP regeneration** phase

The remaining G3P molecules are modified to regenerate RuBP

122.

In photorespiration, C3 plants consume oxygen and generate CO_2 by degrading Calvin cycle intermediates but do not produce ATP.

Photorespiration is significant on bright, hot, dry days when plants close their stomata, conserving water but preventing the passage of CO_2 into the leaf.

123.

- In the C4 pathway, the enzyme PEP carboxylase binds CO_2 effectively, even when CO_2 is at a low concentration. C4 reactions takes place within mesophyll cells. The CO_2 is fixed in to oxaloacetate, which is then converted to malate. The malate moves into a bundle sheath cell, and CO_2 is removed from it. The released CO_2 then enters the Calvin cycle.
- The crassulacean acid metabolism (CAM) pathway is similar to the C4 pathway. PEP carboxylase fixes carbon at night in the mesophyll cells, and the Calvin cycle occurs during the day in the same cells.

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UNIT 10

10. Evolution

Evolution is a change in genetic composition of a population over successive generation and may lead to the origin of new species. This may be caused by: Meiosis, hybridization, natural selection and mutation.

10.1 Theories of the Origin of Life

The origin of life means the emergence of heritable and evolvable self - reproduction. There are several theories about the origin of life. These are:

i. Special creationism
ii. Spontaneous generation (abiogenesis)
iii. Eternity of life theory
iv. Cosmozoan theory
v. Biochemical origin

i. Special creationism

Special creationism stated that the formation of life on earth may have been taken place due to supernatural or divine forces. They are linked with different religions. There are many different versions of special creationism. These are:

A. Young Earth creationism

It suggests that the Earth is only a few thousand years old, and was created in six 24 - hour day.

B. Old Earth creationism

It accepts life is created by God as it was put in genesis, and explains that the earth is very old.

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C. Day - age and creationism

The six day in genesis represents a billion of geologic times, making the days just symbolic. There is a large gap between the formation of Earth and the creationism

D. Progressive creationism

They accept the fossil record showing is the series of organisms, but deny genetic relationship between ancient and modern species. Each species are considered as unique creations

E. Theistic evolution

They proposed that God 'invented' evolution and takes active part in the ongoing process of evolution.

F. Intelligent design

It is the latest version of creationism, and claims that life is developed due to a combination of natural force and the intervention of the supernatural force.

ii. Spontaneous generation (abiogenesis)

This theory suggests that life can evolve, 'spontaneously' from non - living objects. This theory was articulated by:

A. Aristotle (384 - 322), a Greek philosopher

He believed that life could come from non living things. For example:

- Mice from earth.
- Flies from bovine manure.
- Maggots from rotting meat and fish from the mud of previously dry lakes.

B. Francesco Redi (1668), an Italian scientist.

He designed a scientific experiment to test the spontaneous creation of **maggots (macro organism)** by placing fresh meat in jars.

Procedure: he prepared a few jars containing a piece of meat, one jar is open and the other were covered with a cloth.

Result: a few days later, the open jar contained maggots whereas the covered jars contained no maggots.

Contusion: the maggots from open jar were come from fly eggs, which disprove spontaneous generation.

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C. John Needham (1748), an English priest.

He demonstrated that the existence of living things are highly depends on non living organic matter

Procedure: he boiled mutton broth and then tightly stoppered the flasks.

Result: many of the flasks become cloudy and contained micro organisms.

Explanation: Organic matter contained a vital force that could confer the properties of life in non living matter.

D. **Lazzaro Spallanzani (1729 – 1799),** an Italian priest and naturalist

He demonstrated that air can bring microbes into the flask, so that boiling and sealing the flask can make free of germs

Procedure: First he sealed the glass flasks that contained water and seeds and placed it is boiling water for 3/4 of an hour.

Result and explanation: No growth took place as place as long as the flask remained sealed. If the flask were open, air would carry germs to the culture medium.

However, the supporters of spontaneous generation claimed heating the air sealed flasks destroyed its ability to support life.

E. **Louis pasteur,** a French scientist.

He showed that broth or wine only went sour if micro - organisms were allowed to enter from outside.

Procedure:

i. He prepared sterilized swan - neck flasks (an 'S' shaped), which are designed to let air in and out and stopped microbes in the air from contaminating the content of the flask.

ii. Broth was sterilized and allowed to cool slowly in the flask.

Result: sterilized broth in the swan - neck flasks has shown no growth of microbes as long as the swan necks remained intact/unbroken/. However, when the flask neck was tilted or broken, the broth was interacted with micro organisms and allowed growth of microbes

Conclusion: Pasteur articulated "Omne vivum ex vivo" (life only comes from life), and disproved the theory of spontaneous generation.

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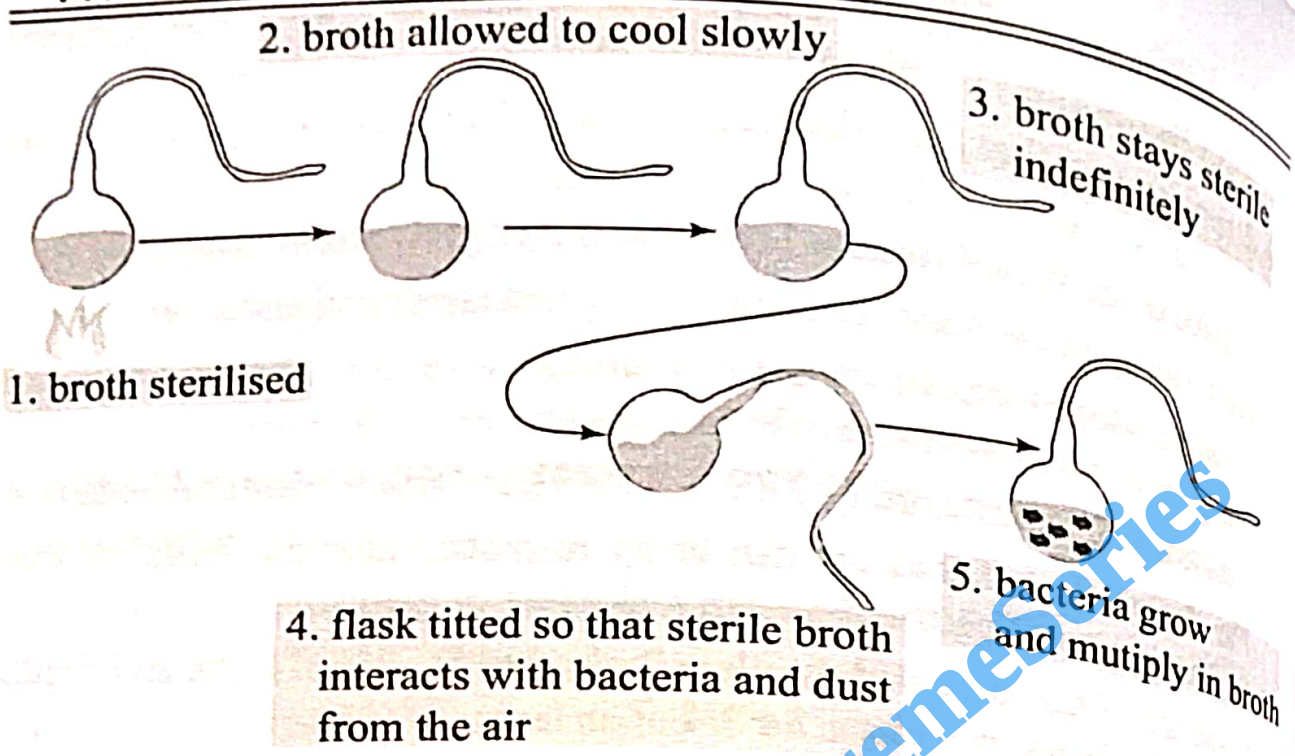

Note: F. Redi (by using macro organism such as fly), and L. Pasture (by using micro organisms) disproved spontaneous generation by showing life only comes from pre-existing life

iii. Eternity of life theory

This theory states that life on earth is eternal, meaning it has no beginning and end. Life is an inherent property of the universe and it has always existed, and no need of creation. It is supported by eminent scientists including Albert Einstein, because it coincides the idea "the universe is unchanging".

iv. Cosmozoan they

This theory states that life on earth originally came from elsewhere in the universe/possibly from other planet/ in the form of highly **resistant spores** by cosmological structure such as **meteorites** and **cosmic dust**. Then it subsequently evolves into different forms of life. This ides was first proposed by Hermann Richter in 1865 and supported by others.

v. Biochemical origin

This theory stated that life on earth was originated as the result of a number of biological reactions. Currently scientists are accepted this theory as a better way to understand the origin of life on earth. This theory was formulated by **Aleksandr Oparin,** and **John Haldane**. Both suggested that:

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a) The primitive atmosphere was reduced (had no oxygen), but had other gases such as ammonia, methane, hydrogen and water vapour.

b) These gases were combined and formed simple organic chemicals, and then more complex molecules called **coacervates/prebiotic soup**

- Coacervates/prebiotic soup are able to absorb and assimilate organic nutrients from the environment in a way similar to the metabolism of cells, then reproduces and divide similar to cells do.
- Coacervates were considered as the precursors of cells (the first true cells).

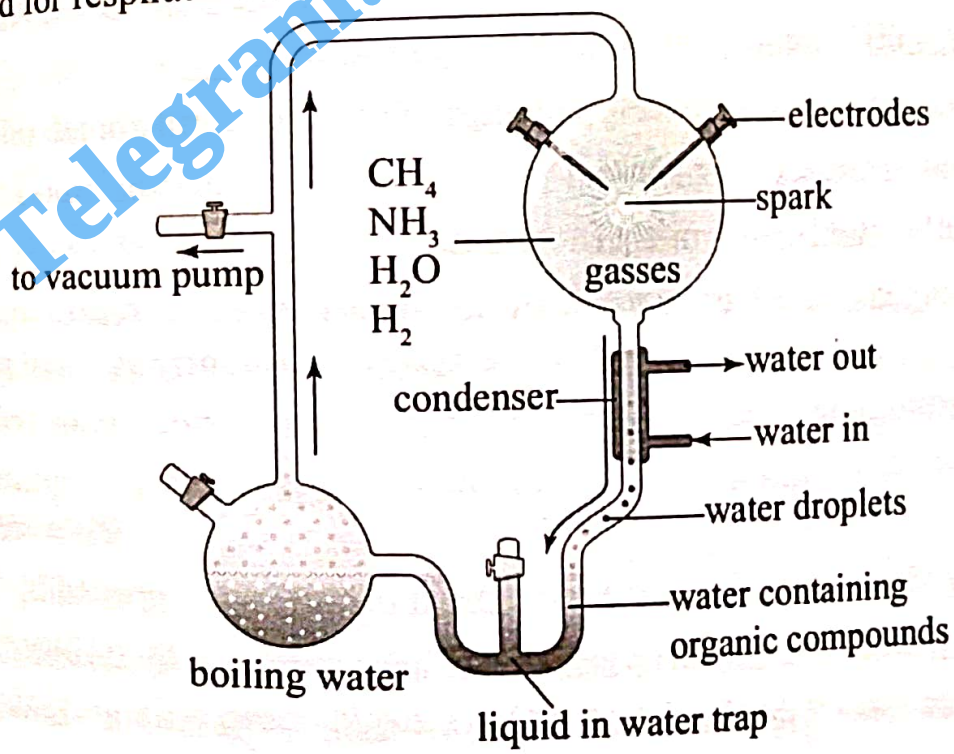
c) The source of energy for these reactions was either **lightening** or **UV** radiation.

10.1.1 What was the Evidence of Biochemical Theory ?

In 1953, Stanley Miller conducted his famous experiment known as spark - discharge experiment to demonstrate the biochemical origin in the laboratory.

He passed electric sparks repeatedly through a mixture of gases that contain: methane (CH_4), ammonia (NH_3), water (H_2O) and hydrogen (H_2).

The result was: Hydrogen cyanide (HCN) that is the precursor to synthesize nitrogenous bases; Amino acids that is essential to form proteins; Pentose sugars that are needed to form nucleic acids, and hexose sugars that are needed for respiration and to form starch and cellulose.



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However, miller could not produce either polymers from his monomers or life from the experiment.

Illustrative Example

1. A genetic change in population that passes on through successive generations n known as:

   A. Mutation
   B. Evolution
   C. Selection
   D. Adaptation

   Explanation: Any change in a gene in an individual is called mutation, but the change in the genetic composition of a population over successive generation is evolution.

   Answer: B

2. Which of the following was not included in theory of special creation?

   A. All the living organisms that see today were created as such
   B. The earth was always the same since creation and will be the same in future
   C. The earth was created within 6 days
   D. Life created by supernatural power
   E. All

   Explanation: Theory of special creation said that life has not changed a bit since its was created by God.

   Answer: E

3. The primitive atmosphere was reducing since _____

   A. Hydrogen atoms are very fer
   B. Helium atoms were more
   C. Sulfur atoms were many in number
   D. Oxygen atoms were vary few

   Explanation: The active hydrogen atoms reacted with free oxygen atoms forming water. So, the primitive atmosphere was reducing without free oxygen.

   Answer: D

4. "The organic compounds were the basis of life" were proved by?

   A. Lamarck B. Darwin C. Mendel D. Miller & Urey

   Explanation: These two scientists, in their experiment showed the

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formation of simple amino acids and organic acids. They concluded life came from organic compounds.

Answer: D

5. Which among the biologists, gave the most acceptable biochemical theory ____

A. Oparin B. Urey C. Darwin D. Lamarck

Explanation: Oparin's theory was called Naturalistic theory of chemical evolution. This theory states that life originated by the composition of chemicals, or life come from pre - existing non living organic.

Answer: A

6. During the time of evolution, the possible energy sources were _____

A. Chlorophyll C. Bacteria

B. Green plants D. U.V rays and lightening

Explanation: Volcanic eruptions and ionizing radiations like X - rays were the only available sources of energy during primitive time.

Answer: D

7. According to spontaneous generation, life originated _____

A. From micro organisms C. From air

B. From similar organisms D. Only spontaneously

Explanation: This theory states that life originates from non living things matter like mud, straw etc, spontaneously. It was theory which had no specific explanation. It was soon rejected.

Answer: D

8. Who demonstrated that life originated from pre - existing cells?

A. Louis pasteur C. Charles Darwin

B. Hugo de Vries D. Stanley Miller

Explanation: Louis pasteur in his swan neck flask experiment found out that life originated from pre - existing life.

Answer: A

9. During origin of life, which among the following was not found in free form?

A. Methane B. Ammonia C. Oxygen D. Hydrogen

Explanation: Oxygen was not found in free form. It was produced mainly by blue green algae. U.V rays broke down water vapor present

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in molten mass to oxygen and hydrogen.

10. Miller in his experiment, synthesized simple and amino - acid from

Answer: C

A. Methane, ammonia, oxygen, nitrogen
B. Hydrogen, methane, ammonia, water
C. Ammonia, methane, carbondioxide, oxygen
D. Hydrogen, water oxygen, helium

11. Which of the following scientists developed the biochemical (abiogenesis) theory of the life?

Answer: B

A. Haldane and Bernal
B. Oparin and Haldane
C. Miller and Bernal
D. Miller and Oparin

Explanation: Alexaner Oparin (a Russian) and John Haldane (an English) independently suggested the theory of abiogensis at different times.

12. What is the essence of Oparins's theory of biochemical evolution?

Answer: B

Explanation: Alexander Oparin, a Russian biologist worked intensively on a biochemical theory of evolution and explaine how life is originated on the earth. He suggested that; simple organic compound could have undergone a series of reaction to form complex molecule.

10.1.2 Evolution of Autotrophs

The earth was created 4.5 billion years ago, the primitive atmosphere was reduced, had no oxygen. One of the greatest changes that effect the evolution of early life was the evolution of oxygen, which was 2.4 billion years ago.

The first organism appeared on earth was heterotrophic prokaryote 3.5 billion years ago. They had no nucleus (DNA), but have RNA. They gave rise to three distinct line of evolution:

i. **Archeabacteria** - prokaryotes that live in extreme conditions, such as high temperature (termophilic), methane (methanobacteria), salt (halophilic), and ocean vent (sulphobacteria).

ii. **Eubacteria** - including ordinary bacteria and cyanobacteria (blue green bacteria or blue green algae)

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iii. **Eukaryotes** - protoctistans (protozoa), fungi, plants and animals. Almost all are aerobic (depends on oxygen for respiration)

Autotrophs are organisms that can produce their own food, and produces oxygen gas as a by product. The level of oxygen has increased after the evolution of photosynthetic autotrophs. The two major types of autotrophs are chemoautotrophs and photoautotrophs.

i. **Chemoautotrophs** - are organisms that obtain energy from chemical reaction (SO2, H2S, NH3, H2, S and Fe) and CO_2 for carbon source to make their own food. For example, archaea bacteria, which live in extreme conditions such as in deep - sea vents. The first autotrophic cells probably used chemosynthesis to make food & produce O_2.

ii. **Photoautotrophs** - are organisms that obtain energy from the sun by green pigment called chlorophyll, carbon from CO2 to make their own food. Most modern autotrophic cells use photosynthesis to make food. An important by product of photosynthesis is oxygen.

10.1.3 Theories of Evolution

There are three basic theories of evolution that try to explain the mechanism of evolution, how species are gradually changed over time. These are: Lamarckism, Darwin's natural selection, and Neo - Darwinism

Lamarckism

In 1809, Jean - Baptist De Lamarck, a French naturalist, proposed the theory of:

i. Inheritance of acquired characters

It stated that the traits acquired by organisms during their life time can pass to subsequence generation. For example, giraffes that had acquired long necks would have offspring with long neck

ii. New Needs

Organisms felt new need when there is a change in environment (light, temperature, food, air, and etc.) to fulfill their new needs. For example they may exert special efforts like changes in **habits** or **behavior.** These changes are heritable.

iii. Use and disuse

The new habit/behavior involves the greater use of certain organs to meet new need. If a body structure or process is continually used, it will become enlarged or more developed. For examples:

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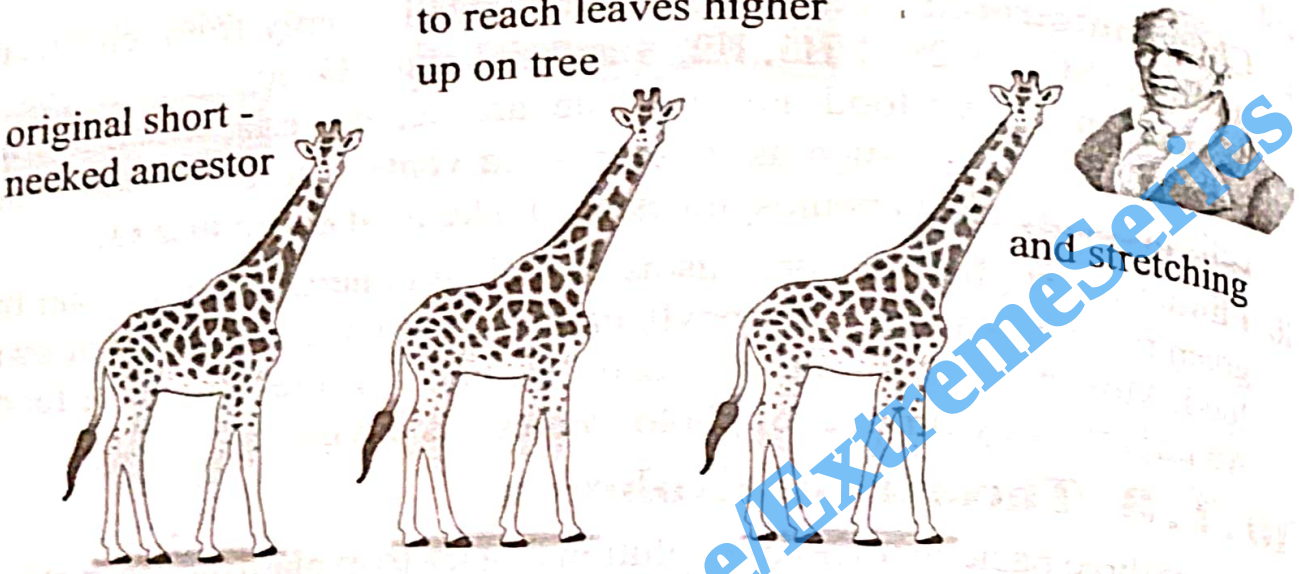
- The elongated neck of the giraffe. Over a lifetime of straining to reach high branches, develop an elongated neck. However, Lamarck could not explain how this might happen.
- The toes of water bird. Straining their toes to swim though water the birds gained elongated and webbed toes to improve their swimming.



- Lamarck Suggests that any structure or process that is not used will become reduced in size or less developed. For example: Wings of penguins, which **has not been used** for flying is get reduced and become smaller.

iv. Speciation

Speciation is the process of forming new species from existing organism. Lamarck believed that the characters accumulated in every generation is inherited and will result in the formation of a new species and transmitted to next generation.

Significance of Lamarckism

i. It was the first comprehensive theory of biological evolution.
ii. It explains the existence of vestigial organs in animals due to their continuously disused.

Vestigial organs are organs that are a part of the organism that are no longer in use. This can be used to determine the relatedness of different species. For examples.

- Tailbone and appendix are vestigial organs in humans.
- The bone structures of the front flippers of a whale contain bones of limbs that exist in mammals

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iii. It explains the development of strong jaw muscles and claws in the carnivores due to their continued use.

iv. It stimulated other biologists to look for the mechanism of organic evolution/biological evolution/.

Note: Lamak's theory was disproved: because he didn't believe evolution as a gradual process: evidences were not supported by the science or genetics.

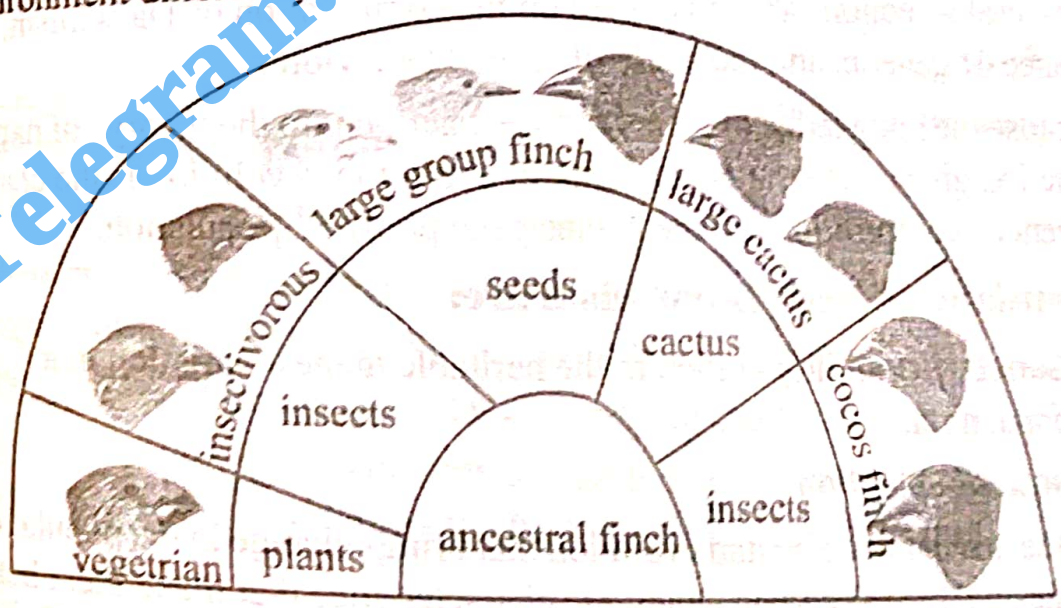
Darwinism/Theory of natural selection

This theory states that "those organisms that are best adapted to their environment would have an advantage and be able to reproduce in greater numbers and pass on their advantageous adaptation to their offspring than other types".

Darwin's evidence came from a visit to the Galapagos Islands that are found 600 miles away from the coast of Ecuador in South America.

He observed 14 species of finches on these islands that are similar to each other in many ways and similar to the species of finches that are found on mainland. The main difference in the 14 species of finches on the Galapagos Islands was the shape of their beaks.

According to Darwin an ancestral finches colonized the islands originated from mainland of South America and they are evolved to different species of finches such as seed eaters, insect eaters, fruit eaters and others. Because they faced different environment on the islands, so they are adapted to their environment differently.



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Basic postulates of Darwinism

i. **Geometric increase/over reproduction** - all species tend to reproduce more offspring than that can possibly survive.

ii. **Struggle for existence** - there is greater competition between organisms at all levels of life because the numbers of individuals produced are more than the number that can be supported. Therefore, only the fittest will survive

iii. **Variation under nature**

The is variation (difference) among organisms. No individuals of a species are exactly similar and they have some differences. These variations will given rise new characters and heredity, which is passed to the next generation.

iv. **Natural selection or survive of the fittest :** Those members of a species that better adapted to their environment will survive and reproduced in greater numbers than others (less adopted). They develop various adaptive modifications to suit the changed condition of life.

v. **Origin of species:** Over long periods of time best fitted and suitable individuals survived and adapted to the environment. As an environment is ever changing, further changes occur and thus new adaptations appear in organisms. On this way new species appear.

Neo - Darwinism theory

This theory is a revised version of Darwin's theory of evolution by means of natural selection, which is based on the combination of Darwinism, the science of genetic, and the study of animal behaviour.

Because the knowledge of genes and gene action into the theory of natural selection give a better understanding of evolution. Evolution is the change in genetic composition (all the alleles/gene pool) of population.

Postulates of Neo - Darwinism are:

i. **Genetic Variability** - refers to the heritable /genes/ variation in a population.

ii. **Natural Selection**

It is the driving force behind evolution that brings change in the population. Organisms possess certain genotypic characteristics that make them better adapted that make them better adapted to the environment, which leads into speciation.

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iii. Reproductive isolation

It refers to the inability of living organisms to interbreed due to: mechanical isolation, geographical isolation, temporal isolation, and behavioural isolation etc.

Illustrative Example

13. Absence of oxygen in earth, made the atmosphere reducing, But in later became oxidizing in nature due to the emergence of

A. Angiosperms
B. Gymnosperms
C. Eukaryotic algae
D. Cyanobacteria

Explanation: The primitive atmosphere had no oxygen and reducing (i.e no free oxygen was available). It was converted into oxidizing nature by the emergence of cyanobacteria. They released oxygen into atmosphere by converting water into hydrogen and oxygen.

Answer: D

14. In the process of evolution of life on earth, which of the following four processes evolved last? (EUEE, 2005 E.C)

A. Photosynthesis
B. Aerobic respiration
C. Chemosynthesis
D. Photo - outotrophes

Explanation: Chemosynthesis evolved first and gave rise to life. Photosynthesis (photoautotrophs) evolved later and released oxygen, which is led to the evolution of aerobic respiration.

Answer: B

15. During the course of evolution, which of the following events came before all the others? (EUEE, 2005 E.C)

A. Origin of dinosaur and other reptiles
B. Availability of free oxygen in the atmosphere
C. Origin of the oldest eukaryotic organism
D. Origin of the first multicellular animals and plants

Explanation: Oxygen availability started to rise 2.4 - 2.1 billion years ago, and animals evolved 0.7 - 0.6 billion year's age.

Answer: B

16. From the times when oxygen was first produced on the earth, it took approximately _____ years for the level to begin to rise.

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A. 1 billion B. 0.6 billion C. 0.1 billion D. 2 billion

Explanation: Oxygen was produced by cyanobacteria since 3.5 billion years ago, but its concentration started to rise to 2.4 billion years ago. So, the gap between the times oxygen being produced and the times its concentration stared to rise is about 1 billion years.

Answer: A

17. Which of the following is the first autotrophic organism in the evolution of living thing?

A. Cyanobacteria
B. Sulfur Bacteria
C. Blue - green algae
D. Green algae
E. A & C

Explanation: Cyanobacteria (blue - green algae) are fist phto - autotrophs. Archaebacteria such as sulphur bacteria are first autotrops, but not photosynthetic, rather chemosynthetic.

Answer: E

18. According to the theory of evolution as proposed by Charles Darwin, which one of the following is the most important deriving force of evolution? (UEE, 2003 E.C)

A. Mutation
B. Over - reproduction
C. Generic recombination
D. Natural selection

Explanation: Natural selection operates on the individual favouring those that best fit environment. The selection is in different forms. Example: food or space shortages, diseases, natural calamities, etc.

Answer: D

19. Which of the following theories is more accepted by scientists?

A. Darwinism
B. Lamarckism
C. Neo - Darwinism
D. A and B

Explanation: Neo - Darwinism which incorporates genetics and ethology (animal behavior) in the first Darwinian theory of natural selection is more acceptable among the scientists.

Answer: C

20. The paper entitled "**philosophie zoololique**" was written in 1809 by _

A. Darwin B. Miller C. Aristotle D. Lamarck

Explanation: "**philosophie zoololique**" describes the theory of use,

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disuse and inheritance of acquired characteristics and was written by Lamarck.

Answer: D

21. Which of the following contributes to the theory of natural selection?
A. Fecundity and variation
B. Struggle for survival
C. Survival of the fittest
D. All

Explanation: Since there is scarce resource, there is always struggle for survival. There is also natural tendency of organisms to over - reproduce (fecundity). Natural variation also exists among organisms. All these lead to selection which favors survival of the fittest.

Answer: D

22. Which of the following idea is not part of Darwin's theory of evolution? (UEE, 2004 E.C)
A. Over reproduction
B. Use and disuse of body parts
C. Existence of heritable variation
D. Competition for scarce resource

Explanation: Use and disuse of body parts is part of Lamarck's theory of transformation.

Answer: B

23. Which of the following is true about the evolutionary time of groups of organisms? (UEE, 2004 E.C)
A. The dinosaurs appeared before origin of land plants
B. The earliest homosapiens appeared before flowering plants.
C. The first photosynthetic organisms appeared before the oldest eukaryotes.
D. The first animal appeared before the formation of free oxygen in the atmosphere

Explanation: The evolutionary clock goes as follows: formation of earth - oldest fossils of prokaryotes → the first photosynthetic organisms → evolution of free oxygen → oldest eukaryotic fossils → first animals → oldest multicellular fossils → age of dinosaurs → first appearance of Homo sapiens → see on page 374, the table.

Answer: C

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24. If a new mutant allele arises in a certain population, which of the factors determines if the allele is going to be adaptive or non - adaptive? (UEE, 2005 E.C)

A. The environment in which the population livings.
B. The rates at which the gene mutates.
C. The population in which the gene is found.
D. The use and disuse of the gene by the population

Explanation: The environment in which the population lives determines the fate of a mutation. For example resistance against penicillin, becomes adaptive (advantageous) when penicillin is constantly utilized. In penicillin - free environment, that mutation is non - adaptive.

Answer: A

25. Which of the following factor is not important for evolutionary change of a population? (EUEE, 2005 E.C)

A. Over population
B. Survival of all that are born
C. Existence of heritable variation
D. Insufficiency of natural resources

Explanation: Survival of all offspring that are born in population is not important for evolutionary change because it means that natural selection did not operate on the population

Answer: B

26. Which of the following aspects of evolution do Lamarckism and Darwinism agree?

A. Source of variation among organisms
B. Existence of heritable variation
C. Mechanisms of survival of all that are born
D. Insufficiency of natural resources

Explanation: Survival of all offspring that born in population is not important for evolutionary change because it means that natural selection did not operate on the population.

Answer: C

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1.1.4 The Evidences for Evolution

There are more evidence of evolution in living things, which proves the creation of new species from common ancestors. The most common are: comparative anatomy, embryology, paleontology, comparative biochemistry, and breeding

Comparative anatomy

The evolutionary relationship between organisms can be studied based on the structural similarities (anatomical features). Organisms which have similar body structure are closely related evolutionarily & they probably share a common ancestor. It is one of the strongest forms of evidence for evolution. It can be analogous & homologous.

A. Analogous structures

Analogous structures are structures with similar functions but are different in morphology and anatomy. For example: The wings of a bird and a mosquito. These organisms are evolutionarily unrelated, and have no common ancestor.

B. Homologous structures

Homologous structures are structures which have the same basic anatomy and a common evolutionary origin but they have a different function. Organisms with homologous structures are evolutionary related and probably share a common ancestors.

For example: Pterodactyl/having five digits/ on fore limbs of mammals (arm - human, leg - cat, wing - birds, and flipper - whale is an example of homologous structure. In different animals, these structures have similar anatomy, but different function.

i. Arm for manipulation in humans
ii. Leg for running in cats
iii. Flipper for swimming in whales
iv. Wing for flying in bats

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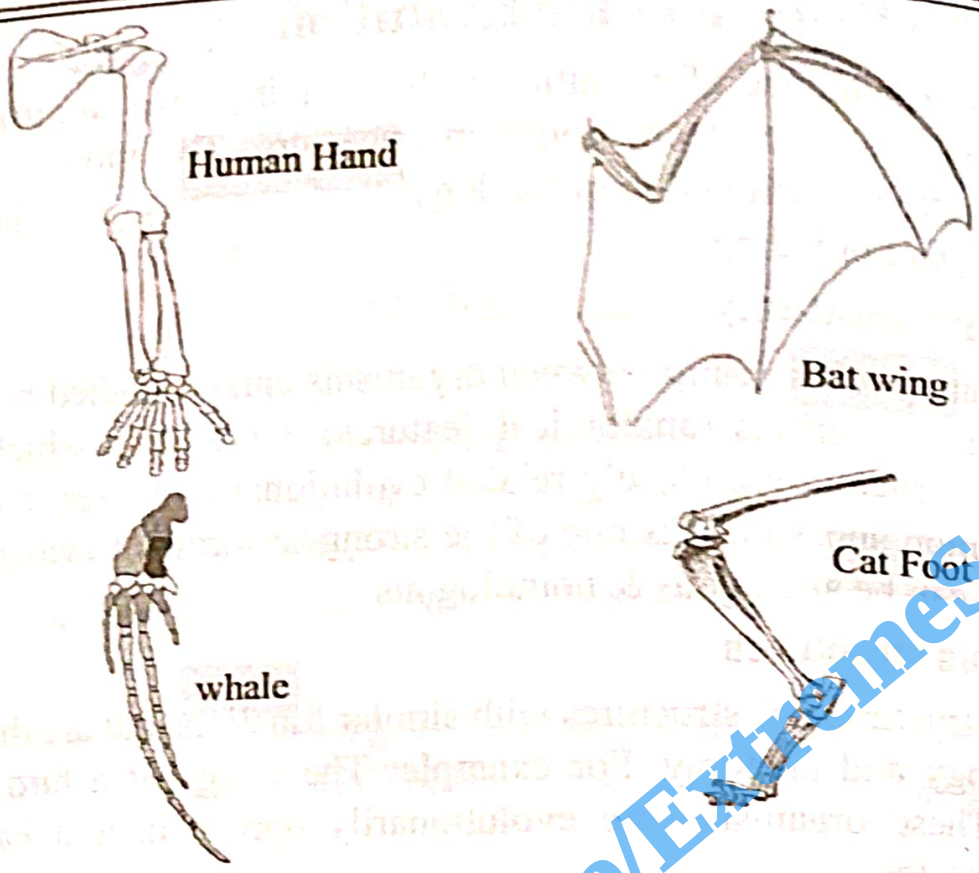

Comparative Embryology

Comparative embryology studies the way in which the embryos of vertebrates develop before they hatch or born. The embryonic developmental similarities of a common ancestry.

For example:

- ☑ All vertebrate embryos have gill slits and tails at the early stage, but disappeared at later stage of development.
- ☑ The embryonic tail does not develop in to all species, in human it is reduced to the **coccyx** or **tailbone**

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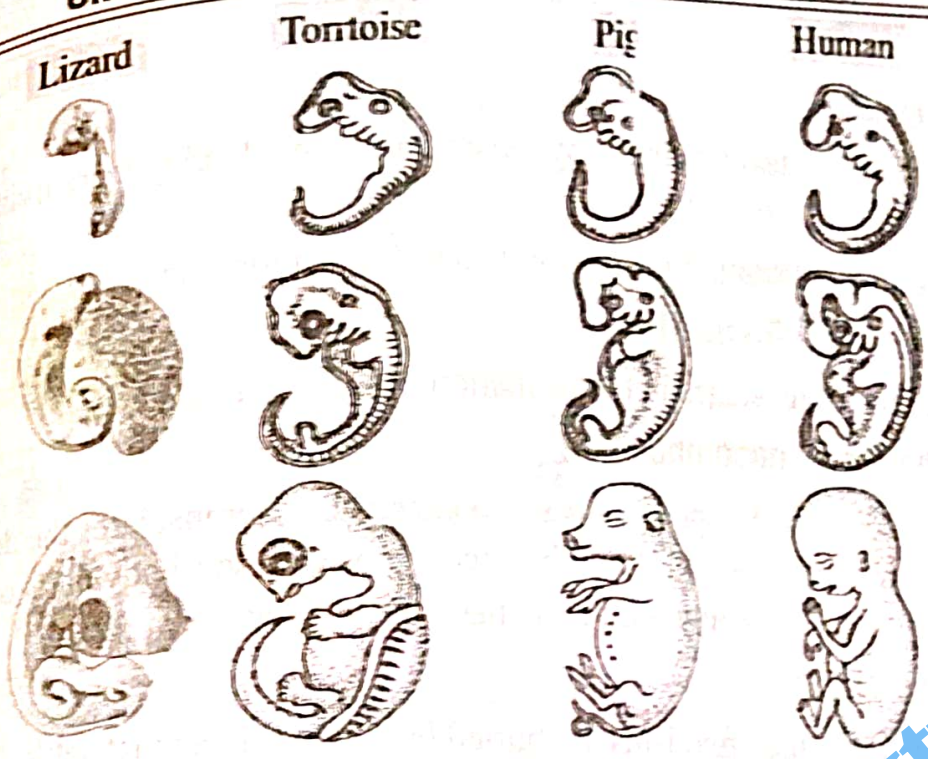

Evidence from paleontology

The word paleontology comes from the Greek word 'palaios which means ancient and logos' which is study. It is the study of ancient life mainly based on fossils.

What are fossils and how it can be used as an evidence of evolution?

- Fossils are remains of animals or plants, which was embedded in the soil or water and preserved for many hundreds of years.
- The study of fossils is used establish similarities and differences among the organisms with their common ancestor in the past.

What are the different categories of fossils?

- Fossils are categorized into:

Category I

They are the remains of dead animals, body parts or the imprint left from the remains, they include: bones, teeth, skin impressions, and hairs. Practical example includes:

i. The hardened shell of ancient invertebrate such as a trilobite or an ammonites

ii. The impression of an animal or plant, even if the actual parts are missing.

iii. A spider preserved in amber. Amber is fossilized resin from coniferous

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trees.

Category 2

These are fossil categories made by the organisms while they were living. They are called trace fossils.

They include: foot print, burrows/, and coprolite/ animal faeces/

Stages in fossil formation

There are four main stages in the formation of fossils. These are:

i. Death without decomposition

The dead bodies are dumped into water bodies, the water insulates the dead remain from decomposition. Microbes decompose the soft body parts over a long period but leave any hard body parts unchanged.

ii. Sedimentation

The hard parts of the organisms are buried by tiny particles that settle out of water over a long period. The sedimentation process insulates the remains from complete decay.

iii. Permineralisation

As the sediments accumulate, the lower layers become compacted by the weight of the layers on top as the result over time pressure turns the sediments into rock.

iv. Uplift

Earth movement such as folding and earth quake expose rocks that contains fossils

Dating fossils

Fossil dating means determining the age of fossils. There are two type of dating fossils: These are relative dating and absolute dating

A. Relative dating

The relative dating of fossil is also called **stratigraphy**. Is is based on the strata or layer of the sedimentary rock.

The fossil in the oldest strata (the deepest layer) are the oldest fossils: whereas more recent rocks (fossil), nearest to the surface of the earth are the youngest fossil.

It is an old method of dating and it uses chronological sequences of fossils in sedimentary rocks to estimate the age of the fossil.

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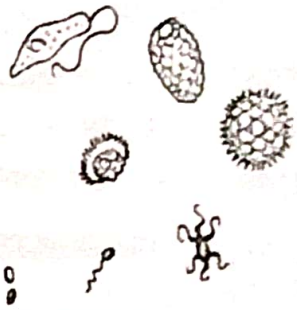
Key events in the fossil record of animals evolution is listed in the following chart. Geological time periods are divided into eras and periods.

Era	Time (millions of years ago)	Important events	
Cenozoic	0.0	Present time	
	less than 0.1	advent of modern humans	
	2.4	ice age	
	66.4	mass extinction	
Mesozoic	141	first flowering plants	
	195	birds evolve from reptiles	
	245	first dinosaurs	
	230		
Paleozoic	280	mass extinction	
	340	reptiles appear	
	360	first insects	
	370	amphibians appear	
	420	plants colonize land	
	540		
Precambrian	700	simple multicellular organisms evolve	
	2,100	oldest eukaryotic fossils	
	2,500	oxygen begins to accumulate in atmosphere	
	3,500	oldest prokaryotic fossils	

B. Absolute sating or radiometric dating

Radiometric dating of fossil uses radioactive elements to determine the actual age of fossils. Radioactive elements are unstable, and they constantly decay into another form, and have half - life. Half is the time needed to decompose half of the parent radioactive isotope into daughter isotope.

The age the fossil is calculated based on:

- The half life time of isotopes.
- The original amount of isotope presented in the fossil should be known.

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➥ The amount of isotope that is left in the fossil should be determined.

Some isotopes Used in Radiometric Dating	
Isotope	Half - life (years)
Carbon - 14	5,730
Uranium - 235	704,000,000
Potassium - 40	1,250,000,000
Uranium 238	4,500,000,000

There are different techniques of radioactive dating. The most known are: **radiocarbon dating and potassium - argon dating**

I. Radiocarbon dating

By this method fossil is dated by comparing the amount of carbon - 14, a radioactive isotope, with the amount of carbon - 12, a stable isotope. Living things take carbon into their bodies constantly.

Most of the carbon is in form of carbon - 12. However, very small proportion is in the form of carbon - 14, which undergoes decay.

The ratio of carbon - 14 to carbon - 12 is a known quantity for living organisms, which is **1: 1 trillion**, respectively. Thus, over time the amount of carbon atom (C^{14}) decays into non - radioactive nitrogen atom (N^{14}), the half - life time of C14 is 5730 years.

Thus, over time, the amount of carbon - 14 declines with respect to the original amount of the stable carbon 12. After 5,730 years, half of the carbon - 14 in a sample will have decayed.

Radiocarbon dating can only be used to date fossil remains less than about 60,000 years old

For example

What is the age of the fossil that contains 3.125% of the original amount of C - 14?

Answer: Just look at the following chart, the original amount of C - 14 was 100%, then continuously decays by half until 3.125% of C - 14 was left $\left(\begin{array}{l} \text{i.e } 100\% \xrightarrow{1 \text{ half - life}} 50\% \xrightarrow{2 \text{ half - life}} 25\% \xrightarrow{3 \text{ half - life}} \\ 12.5 \xrightarrow{4 \text{ half - life}} 6.25\% \xrightarrow{5 \text{ half - life}} 3.125\% \end{array}\right)$

which took a total of five half - life. Each half - life of C - 14 is 5730

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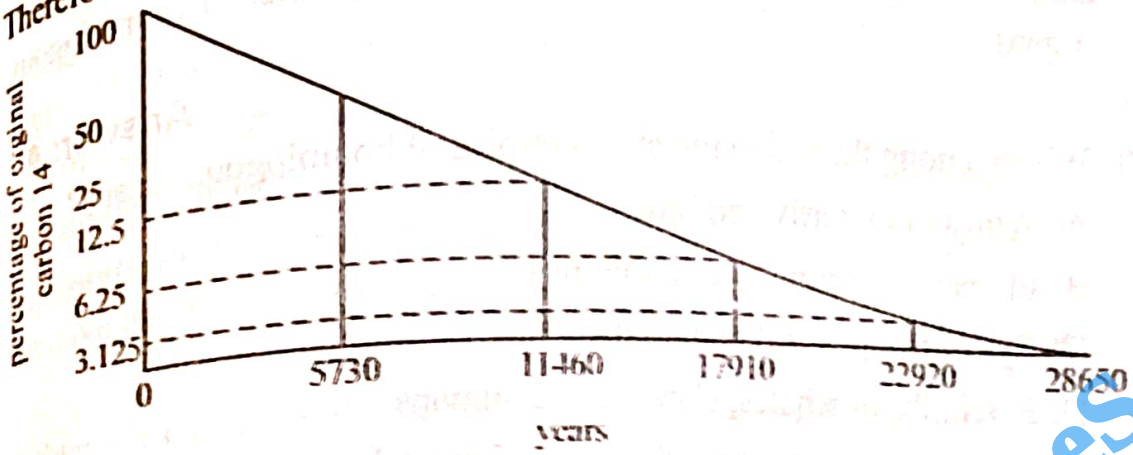
years.

Therefore, the age of the fossil is $5_{\text{half-life}} \times 5730$ year = 38650 years old.



II. Potassium - Argon dating

In this method, radioactive potassium atom (K40) is used which decays into argon atom (Ar40), with the half - life of 1.25 million years. Since it has longer half - life, it can be used to date older fossil of age more than thousands of years

Comparative Biochemistry

Organisms that share very similar molecules such as (DNA, protein, haemoglobin, amino acids) and biochemical pathways are closely related evolutionarily, and have common ancestor. DNA hybridization is a technique that used to compare the DNA or the base sequences of different animals for comparison.

For example

- ☑ 98% of the DNA of chimpanzee and human are similar, indicating how they are closely related and have common ancestor.
- ☑ Human and chimpanzee has no differences in the sequences of amino acids for haemoglobin but have differences with gorilla and Gibbon: this indicates chimpanzee is the closest relative to human being than gorilla, and gibbon.

Illustrative Example

27. In some animals, the same structures develop along different directions due to adaptations to different needs, this called as _____

A. Divergent evolution
B. Convergent evolution
C. Parallel evolution
D. Saltation

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Explanation: It is called divergent evolution and structures are called homologous organs. Homology indicates that the animals have common ancestors. Vertebrate heart or brains is an examples of homologous organs.

Answer: A

28. Which among the following are examples of homologous organs?

A. Wings of butterfly and bird

B. Flippers of penguins and dolphins

C. Eye of octopus and mammals

D. Forelimbs of whales, cheetah and humans

Explanation: Forelimbs of whales, bats, cheetah and humans are homologous organs. They all have similar structure but perform different functions. The remaining options are examples of analogous organs.

Answer: D

29. What is the age of a fossil containing 25% of the original carbon -14 atoms?

Explanation: Twenty five percent of the original carbon - 14 atoms remain after the second half - life has passed. The half - life of C - 14 is 5730 years. So $2 \times 5730 = 11{,}460$ years. The fossil is about 11,460 years old.

30. What is the age of a fossil containing 3.125% of the original carbon - 14 atoms?

Explanation: The about investigated amount of carbon - 14 remains in the fossil after 5 successive half - lives have passed. So, $5 \times 5730 = 28{,}650$ years. The age of the fossil is about 28,650 years old.

31. Why do you think that radiocarbon dating becomes less accurate for older fossils?

Explanation: The half - life of C -14 is relatively shorter (5730 years). Radiocarbon dating will be accurate with fossil that are up to 60,000 years old only.

32. What percentage of the amount of carbon - 14 originally present in a fossil would be left its second half - life? (UEE, 2003 E.C)

A. 75% B. 50% C. 25% D. 12.5%

Explanation: After the first half - life 50% will be left. After the second

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half - life of this will decay, and 25% of the original will be left.

Answer: C

33. Choose the breeding method used by human to produce new and improved varieties of plants and animals? (UEE, 2003 E.C)

A. Crossing - over
B. Natural selection
C. Artificial selection
D. Independent assortment

Explanation: Humans improves varieties of organisms usually by selective breeding (Artificial selection).

Answer: C

34. Carbon - 14 has a half - life 5730 years, suppose a fossil contains only 12.5 % of the amount of carbon - 14 normally present in livings organisms, how old is the fossil? (UEE, 2004 E.C)

A. 5730 years
B. 11,460 years
C. 17,190 years
D. 22,920 years

Explanation: After 3 half - live, 12.5% of the original C - 14 remains. So $3 \times 5730 = 17,190$ years. What remained in the fossil is 12.5% therefore, the age of the fossil is 17,190 years old.

Answer: C

35. Which of the following pairs of molecules can give information about how much two species are evolutionary related to one another? (UEE, 2004 E.C)

A. DNA and proteins
B. Starch and cellulose
C. Lipids and carbohydrates
D. Carbohydrates and proteins

Explanation: Molecules founds in all, or at - least in the most organisms are used in comparative biochemistry. In this regard, DNA and proteins is the best molecules to be used.

Answer: A

36. When two species are compared, which of the following sources of evidence is **least** informative about the degree of relationships between the species? (EUEE, 2005 E.C)

A. Nucleotide sequence of DNAs
B. Amino acid sequence of proteins
C. Glucose sequence of polysaccharides
D. DNA hybridization

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Explanation: Glucose unit does not show similarities amount different organisms. But, other molecules such as DNA, proteins, and hemoglobin can bear more information about similarities between species.

Answer: C

37. Which of the following characteristics can show the evolutionary relationships among organisms?

A. Structures having similar function

B. Structures having common origin

C. Structure having same sizes.

D. Structure having different origins.

Explanation: Structures having common origin (Homologous Structures) shows common ancestor.

Answer: B

10.1.5 Natural Selection

Natural selection is the process that results in the adaptation of organism to its environment by means of advantageous genes gained through the process. It is the 'driving force' behind evolution that brings changes over time in populations and leads to speciation (formation of different species).

For example

i. The creation of antibiotic resistant bacteria is due mutation, which enables them changing their genetic composition and become new species to resist antibiotics.

ii. Why all human in different parts of the world are the same species? Gene flow between people all over the world plays a greater role to decrease differences. Humans all over the world are under the same species, but with different races. Currently Species is defined as a group of similar organisms that can breed together and produce fertile offspring, and with similar biochemistry, physiology, and evolutionary history. Therefore, all human in the world fulfill the definition of species.

Types of natural selection

All types of natural selection operate in the same manner, but they have different influences on a population. There are three types of natural selection. These are direction selection stabilized selection, and disruptive selection.

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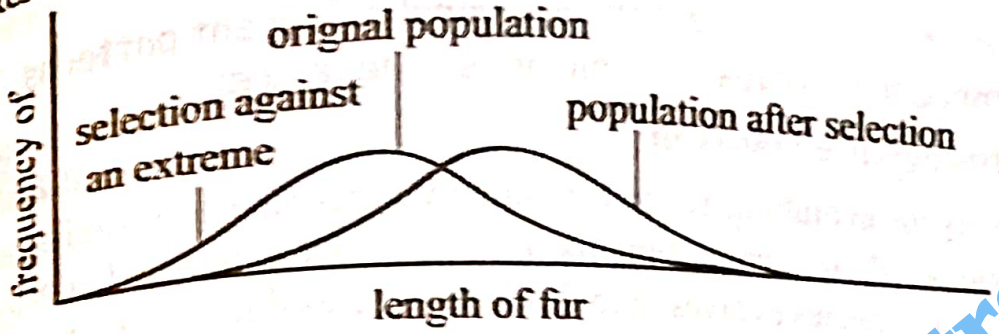
i. Directional Selection

It is the types of selection that works in favors of one extreme feature (phenotype) of a population. It shifts the curve in one direction, changing the average value of the character.

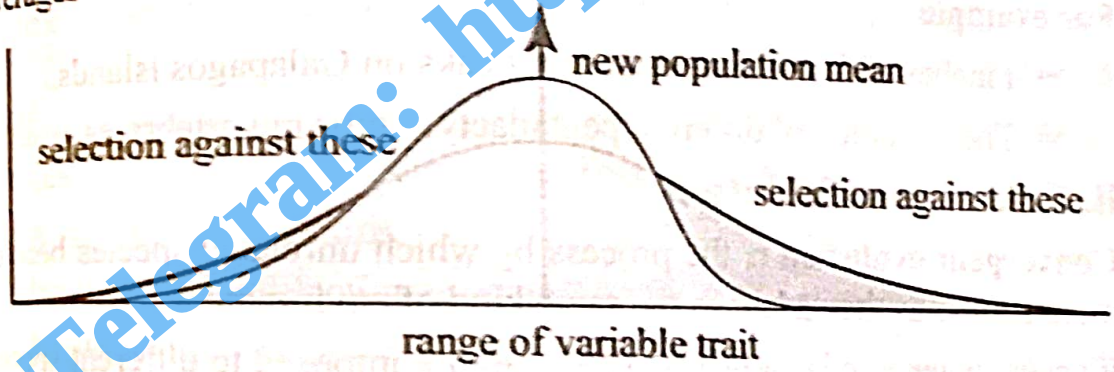
For example: in cold climate thicker fur foxes are favored over thinner fur foxes.



ii. Stabilizing selection

In stabilizing selection, individuals with the average form of a trait have the highest advantage (fitness) than the extreme forms, which confer lower fitness. The curve looks like a bell shape

For example: Birth mass correlation in human, babies who are very heavy or light show a high neonatal mortality rate than those of medium mass / averages mass/.



iii. Disruptive selection

In disruptive selection, individuals with extreme variation of a trait have greater advantage (fitness) than individuals with the average form of the trait. It works in favors of the two extreme features than mean value.

For example: A finch with a medium length of beak may not be able to obtain insects out of cracks in the bark of trees than the one with longer and shorter beaks

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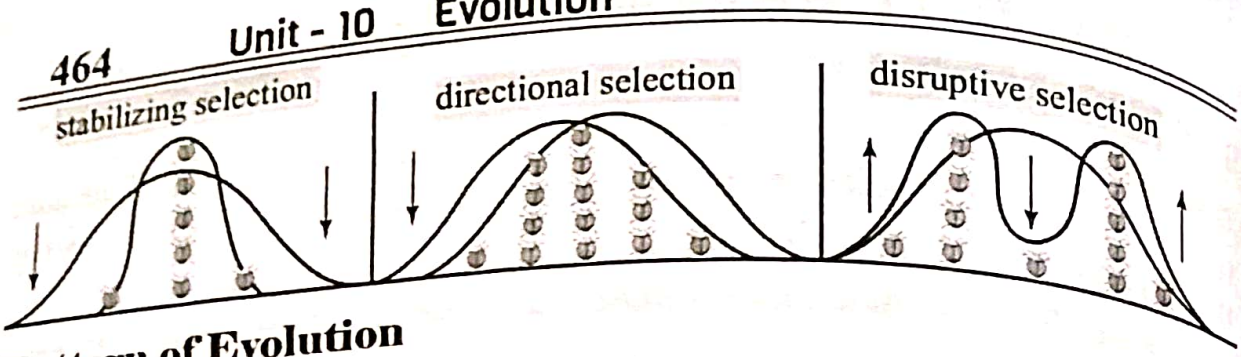

Pattern of Evolution

It refers to the different types of evolutions.

Evolution over time can follow several different patterns. There are three main types of evolutions. These are:

i. Divergent Evolution

In divergent evolution, related populations become less similar as they respond to different environments. It is the process in which a population of the same species evolves in to different species due to different selection pressure on each of them in their environments. It occurs when closely related species diversify to new habitats as the result it leads to the development of new species.

- It is responsible for the creation of the current diversity of life on earth from the first living cells.
- It is responsible for the evolution of humans and apes from a common primate ancestor.

For example

- Finches with different shapes of beaks on Galapagos islands.
- The evolution of different pentadactyl limbs in vertebrates.

ii. Convergent Evolution

Convergent evolution is the process by which unrelated species become more similar as they adapt to the same kind of environment

It occurs when similar selection pressures are imposed to different species that occupy similar niche.

It take place when species of different ancestry begin to share analogous traits because of a shared environment or other selection pressure. For examples:

- Whales and fish have some similar characteristics both evolve methods of moving through the same medium that is water.
- Wings of birds and insects.

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- Different ant eaters of the world (giant armadillo, giant pangolin, giant anteater and spiny anteater) are evolutionary different but they all have pointed mouth to dig narrow cracks in the ground and eat insect.

III. Parallel evolution

Parallel evolution is the similar development of trait in distinct species that are not closely related, but share asimilar original trait in response to similar evolutionary pressure.

For example

1. North America cactus and African euphoria /shrubs/ develop similar adaptation that is thick stem to survive in hot arid climate.
2. Elephants and extinct mammoths. These animals are two distinct species, occupying different niches but share a similar evolutionary challenge or pressure

Illustrative Example

38. The wings of a birds and a bat are result of

A. Divergent evolution
B. Convergent evolution
C. Stabilizing selection
D. Disruptive selection

Explanation: Birds and bat unrelated, they are different species, but they have wings to fly. So they have evolved similar structures because they have similar selection pressure in the environment.

Answer: B

39. Sparrows with averages sized wings survive serve storms better than those with longer or short wings. This illustrates:

A. Disruptive selection
B. Directional selection
C. Stabilizing selection
D. A and B are correct

Explanation: medium sized wings are selected against the extremes (very long short wings), this is stabilizing selection.

Answer: C

40. Species which have diverged after origin from common ancestor giving rise to new species adapted to new habitats and ways of life is called as ____

A. Adaptive radiation
B. Divergent evolution
C. Convergent evolution
D. Mutation

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Explanation: Adaptive radiation (divergent evolution) is a process that given rise to a variety of species that are originated from its original species.

Answer: A

41. If an infant of 4Kg and 2Kg are likely to survive than infant born with 3 kg. What kind of natural selection is operated?

A. Directional
C. Disruptive
B. Stabilizing
D. None

Explanation: Disruptive selection individual with extreme phenotypes have selective advantage over the mean

Answer: C

42. What is the difference between parallel evolution and convergent evolution?

Explanation: Parallel evolution is the development of a similar trait in related, but distinct species descending from the same ancestor.

- Convergent evolution - the development of a similar trait in unrelated evolutionary origin species because of smiler selective pressure.

10.1.6 Speciation

Speciation is the process by which new species are formed from the population a species. There are two main types of speciation: **allopatric speciation** and **sympatric speciation**

Allopatric speciation

It occurs when a population from an existing species becomes geographically isolated and the isolated population develops into a new species.

Geographical isolation is the physical barrier as river, mountain, landmass, and lakes etc. that prevent interbreeding between the same species (interrupts gene flow between the populations)

If mutations arise in one population, it cannot be passed to the other due to geographic separation, thus the two population will become different species and unable to interbreed.

For examples:

- 14 species of Finches on the Galapagos Islands.
- Shrimp population of the Caribbean Sea and North Pacific Ocean, which are separated by /isthmus of panama/

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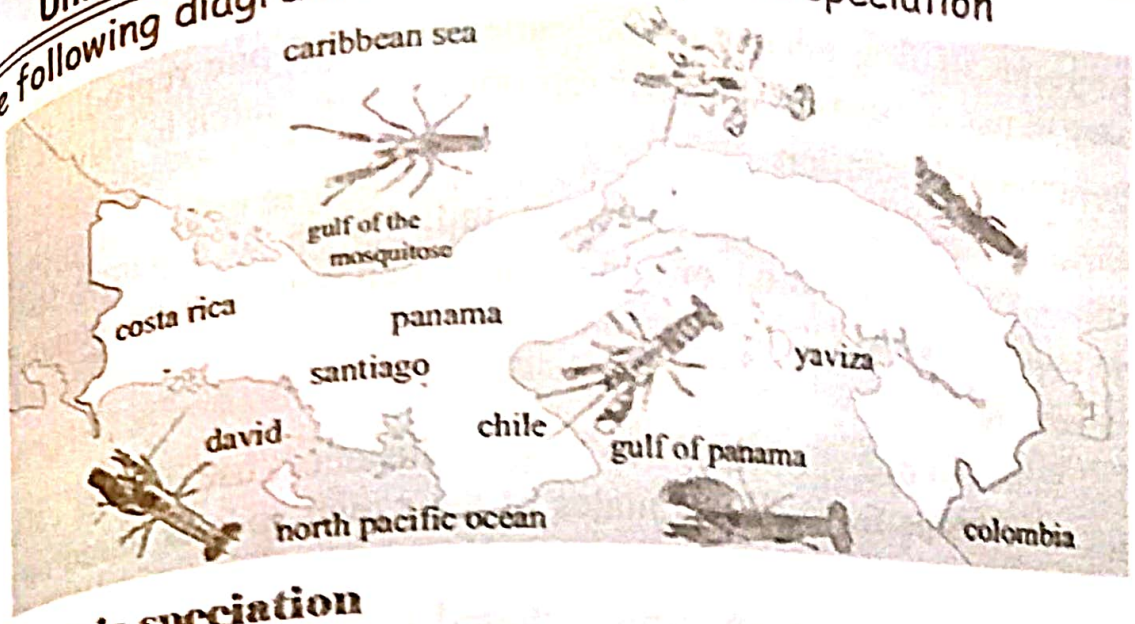
The following diagram shows that allopatric speciation



Sympatric speciation

It is the type of speciation that occurs in a population that live in the same geographic area, the mechanism of speciation is due to reproductive isolation.

Type of reproductive isolating mechanism :

i. Prezygotic reproductive isolating mechanism
ii. Postzygotic reproductive isolating mechanism

I. Prezygotic reproductive isolating mechanism

It is a barrier that block sexual mating, blocks fertilization from happening, and affects the ability of fertilization between gametes and form a zygote.

Per zygotic reproductive isolating barriers include:

I. **Ecological (or habitat) isolation** - a difference in habitat preference/ selection/ can separate two populations in the same geographic al area.

The difference in habitat preferences between the two population creates different gene pools between the two species.

For example:

One species of ladybird beetle eats one type of plant, while the other closely related species eats a different plant. The two species never occur on the same host plant (habitat).

II. **Temporal isolation**

Two closely related species living in the same area have different breeding season, which preventing gene flow between them.

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For example

Two species of frog inhabiting the same area, but one reproduce from January to march whereas the other reproduce from march to may.

III. Behavioural isolation

Absence of cross - attraction between individuals of separate species inhibits courtship initiation or individual behavior during copulation doesn't allow normal fertilization to occur.

For example

- The males of two species of tree frog for instance, use distinct calls to attract. Female frogs choose males of their own species based on the unique calls.
- Sexual selection in many birds is based on unique intricate mating dances .

IV. Mechanical isolation

If the reproductive structures are incompatible between species, it inhibits or stops copulation (sexual intercourse) or pollination.

In animal species, the male and female reproductive parts need to fit together almost like a key in a lock, if not it may prevent groups from interbreeding.

V. Gametic isolation

Gametic that are transferred during sexual intercourse are ineffectual for fertilization due to problem with storage or transfer of gametes or sperm partiality.

For example

Many marine organism such as sea urchins simply release sperm and egg cells into the water. These gametes display unique surface molecules that enable an egg recognize sperm of the same species.

Post zygotic reproductive isolating mechanism

It is also called hybrid incompatibility, which occurs after fertilization or zygote is formed, but it can't be developed into viable offspring. It occurs when there is sexual mating between two different species or populations.

Postzygotic reproductive barrier includes:

i. **Hybrid inviability:** embryo may die before reaching reproductive maturity because the genes of its parents are incompatible.

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Example :

The hybrid embryo of sheep and goats die in the early development stage before birth.

4. **Hybrid infertility or sterility** - it is the condition in which some hybrids are infertile due to the absence of normal meiosis.

Example:

Mule is a hybrid offspring of a female horse and a male donkey. Mule is infertile because horse's egg has one more chromosome than a donkey's sperm cell.

The mule can grow and develop, but meiosis does not occur because the chromosomes are not homologous in the mule germ cell (2n of donkey = 62 Chromosomes) to (2n of horse = 64 chromosomes), then 2n of mule becomes = 63 chrs.

5. Hybrid breakdown

It is the type of reproductive failure that appears after the second filial generation of crosses between different species or sub - species. Some species produce hybrid offspring that are fertile, but having some abnormalities.

It is caused by incompatibility between interacting genes or second generation will weak or sterile.

For example

The second - generation of the hybrid offspring of the mosquito species *Aedes aegypti* and *Aedes mascarensis* develop abnormal genitalia that make mating difficult.

What is Polyploidy and why is it Important in Plant Evolution?

Polyploidy is a condition formed when a cell have many sets of chromosomes per cell. A single cell contains three or more sets of chromosomes. It is a major factor in sympatric speciation in plants.

What causes polyploidy?

It is due to **chromosomal non - disjunction** (failure of separation of homologous or sister chromatids) during anaphases of meiosis I and II.

Why polyploidy is important in **plant evolution**? Because:

1. It changes infertile hybrids to fertile one, done by doubling the

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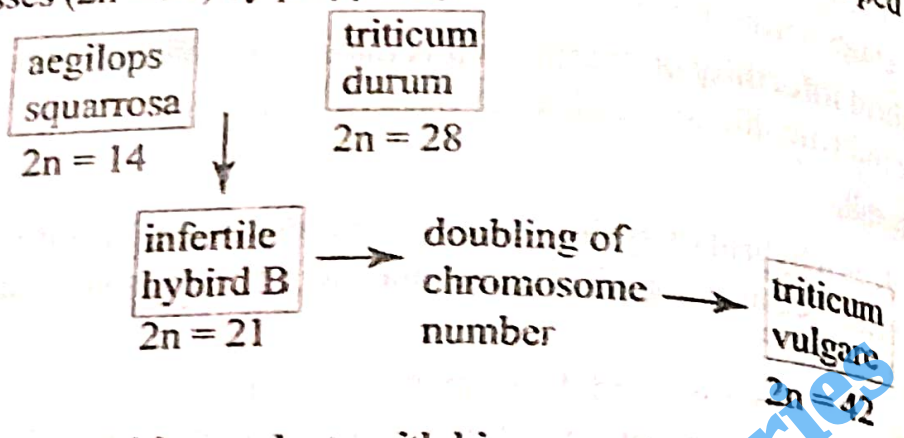
chromosomes number.

For example: the modern wheat *Triticum vulgare* were developed from infertile grasses (2n = 21) by polyploidy.



2. It is used to forms bigger plants with bigger seeds.

Illustrative Example

43. What is the different between allopatric and sympatric speciation?

Explanation: In allopatric speciation, geographical isolation is the cause for speciation. Sympatric speciation is the type of speciation that occurs in a population that live in the same geographic area, the mechanism of speciation is due to reproductive isolation (prezygotic or post zygotic).

44. Saint Bernard and Chihuahuas (two breeds of domestic dogs) cannot normally mate because they differ so much in size. Thus, they are reproductively isolated to some extent. What type of isolating mechanism is operating in this case?

A. Artificial
B. Prezygotic
C. Postzygotic
D. Geographic

Answer: B

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Matching: the term listed under column "A" with discription listed under column "B"

A	B
45. Hybrid inviability	A. It is the condition in which some hybrids are sterile due to the absence of normal meiosis.
46. Hybrid infertility	B. Embryo may die before reaching reproductive maturity
47. Hybrid breakdown	C. Two closely related species living in the same area have different breeding season, which preventing gene flow between them
48. Gametic isolation	D. Reproductive failure that appears after the second filial generation of crosses between different species or sub - species.
49. Mechanical isolation	E. Gametes that are transferred during sexual intercourse are ineffectual for fertilization due to problem with storage or transfer of gametes or sperm partiality.
50. Behavioral isolation	F. Reproductive structure incompatibility between species, it inhibits or stops copulation (sexual intercourse) or pollination
51. Temporal isolation	G. Absence of cross - attraction between individuals of separate species inhibits courtship initiation or individual behavior during copulation doesn't allow normal fertilization to occur
52. Ecological isolation	H. A difference in habitat preference /selection/ can separate two populations in the same geographic al area.

Answer: 45. B 46. A 47. D 48. E 49. F 50.G 51. C 52. H

10.1.7 Human evolution

Anthropology deals with the study of the evolutionary history of man. Human didn't evolved from monkeys, apes and chimpanzees, but have common ancestors in the line of evolution.

There has been a line of evolution for millions of years that has given rise to old world monkeys, new world monkey's, the great apes and the

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different species of human that have lived.

Homo sapiens - are modern and the latest human species live on the plant. They differ from primates or great apes into two major features. These are:

- They have large brain size
- Bipedalism, the ability to truly walk on just two legs.

When did the common ancestors of human, chimpanzee, gorilla, and bonobo is diverged into independent species?

A line of primate evolution in the diagram shows that:

- The common ancestor of chimpanzee and bonobo were extinct 3 million years ago; then, chimpanzee and bonobo were evolved as an independent species.
- The common ancestor of chimpanzee, bonobo and human were extinct as about 6 million years ago: then, human was diverged from other primate and evolved as an independent species

The diagram shows the evolutionary time line tree of humans and the great apes.

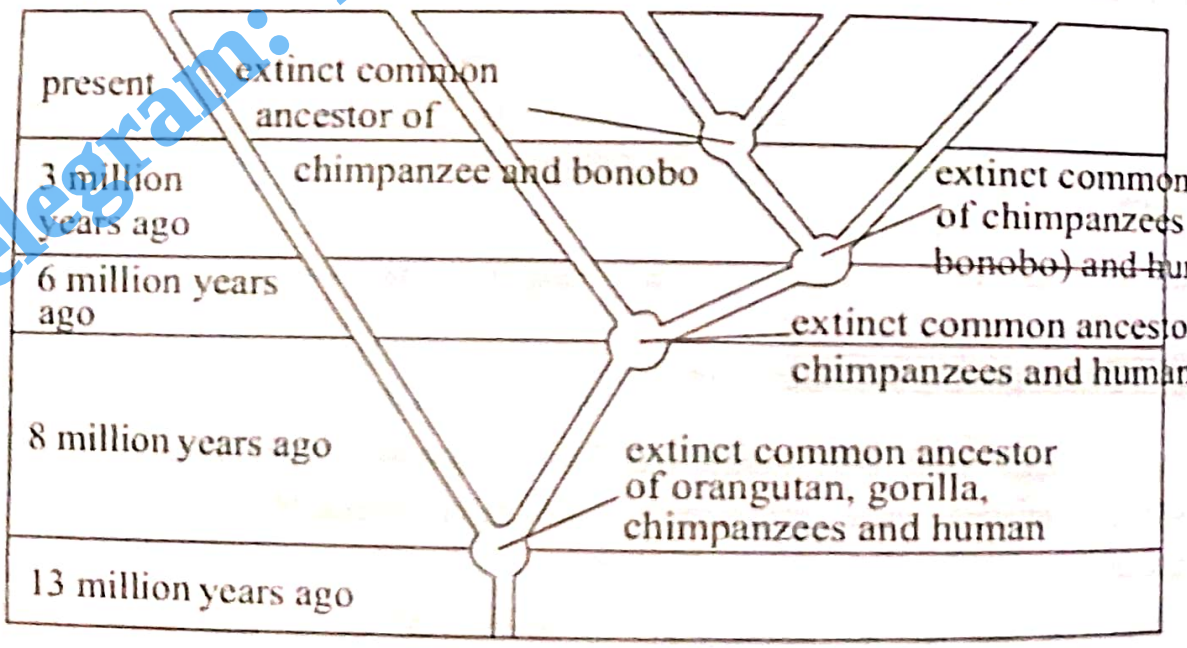
orangutan 48 chromosomes (24 pairs)

gorilla 48 chromosomes (24 pairs)

chimpanzee 48 chromosomes (24 pairs)

bonobo 48 chromosomes (24 pairs)

human 46 chromosomes (23 pairs)



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Hominids and Hominins

Hominids - are all current and extinct great apes (human, gorilla, orangutan, and chimpanzee); it includes all species of human in the genera of "homo" and extinct common ancestors.

Examples of hominids includes:

1. Homo sapience, Homo erectus, and Homo habilis
2. Australopithecus, Ardipithecus kaddabba, A. Ramidus, Australopithecus anamnesis, A, afarensis, A. gorhi, A. africanus, A aeriopicus, A. rudolfernsis, A robustus, A boisei.

Hominin - are any species of early humans (**family hominidae, order primate**) that is more closely related to modern humans, it encompasses all the genus homo. Homo sapience is the only species exists today.

Examples of hominin includes: Homo sapience, Homo Neanderthals, H. erectus, H. hasilis, and various species of Australopithecus.

Note: All hominin are included under hominids, but all hominids are not included under hominin

Ardi and Lucy

i. Lucy (**Australopithecus afarensis**) is a famous fossil dated at about 3.2 million years, discovered by Donald Johanson and Tom Gray in 1974 at Hadar in Afar, Ethiopia. Some of the characteristics of Lucy are:

- She was an adult female of about 25 years old.
- Her skeleton was about 40% complete.
- She was bipedal
- She was partly arboreal (tree - dwelling)
- She was about 107 cm tall
- She was 28 kg in weight

What is the significance of the discovery of Lucy for the study of human evolution?

Lucy was bipedal; this shows that in human evolution bipedalism came before the development of big brains.

ii. **Ardi (Ardipithecus ramidus)** is a female fossil dated 4.4 million years old, discovered in 1992, in Afar, Ethiopia. Some of the characteristics

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of Ardi are:

- She was bipedal and had arboreal life
- She was 4.4 million years old
- She is the nearest fossil to the common ancestor of humans and chimpanzee.

What is the significance of the discovery of Ardi in human evolution?

It proved that the common ancestor of human and chimpanzee is more similar to human than chimpanzee

How Brain Size Changed During Human Evolution?

During the course of human evolution, the brain has got bigger. Australopithecus have a brain that is between 0.7% and 1.0% of their body mass whereas modern humans have a brain size between 1.8% and 2.3% of their body mass.

Homo sapience (modern human) have a cranial capacity of 1500 cm^3 compared to 500 cm^3 of Australopithecus (such as that of Lucy).

A larger brain allows humans to:

- Run faster in a more upright posture
- Plant in advance to avoid attack
- Develop and use tools and weapons.

The diagram shows the increment of cranial capacity in the course of human evolution.

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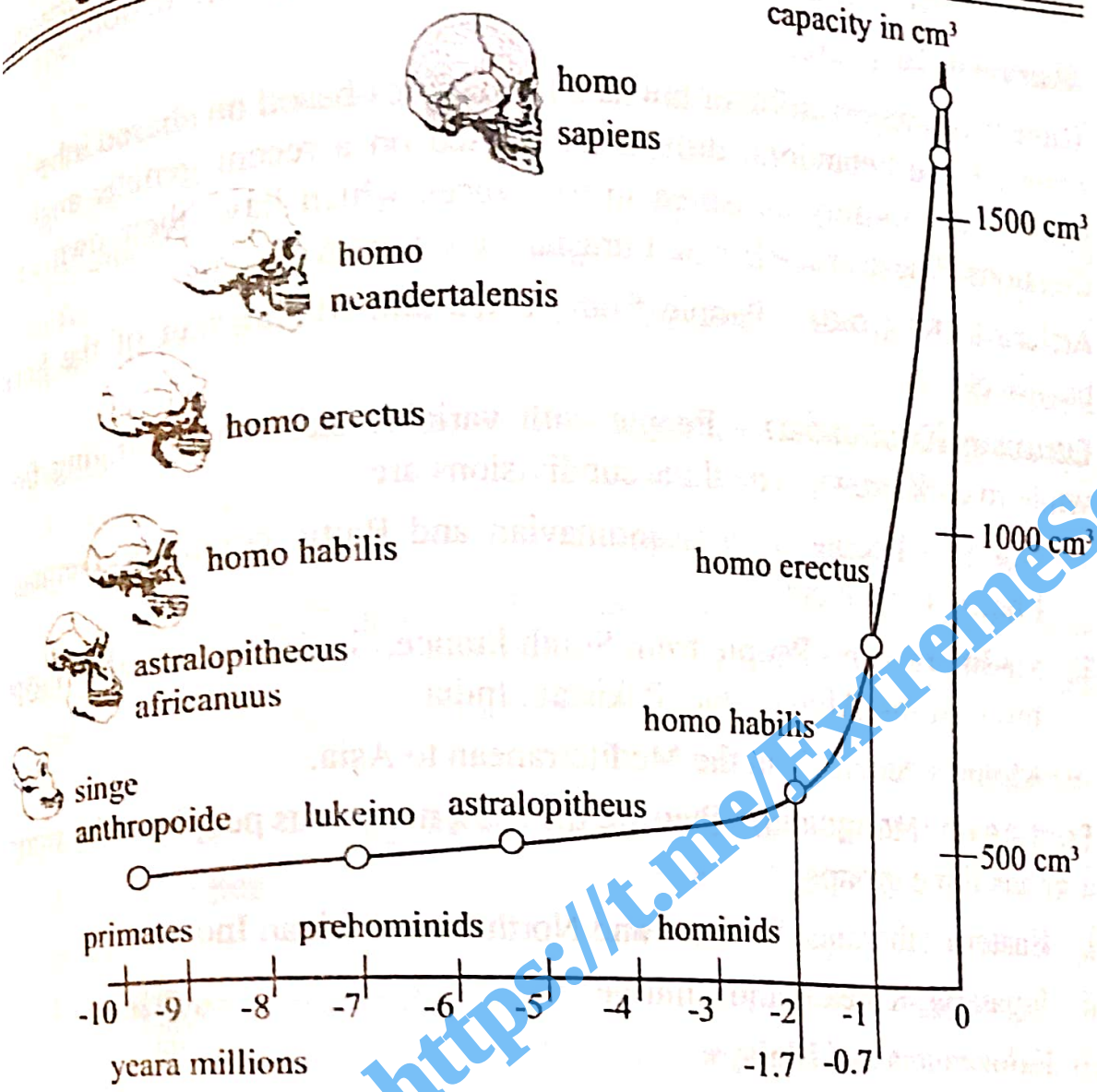

Are human still evolving?

The simplest answer is yes, because evolution is a gradual and continuous process that never stops.

Homo sapiens (modern humans) first appeared in Africa and migrated to other parts of the world about 10,000 year ago.

As human moved from Africa to different areas of the world, they encountered different environmental challenges. As a result, human population evolves into different lines.

For example: humans encountered colder climates, developed body features that gave a survival advantage to conserve heat, such as:

- A shorter, squatter body shape, this reduces the surface - area - to - volume ratio and so reduces the rate of heat loss by radiation
- An increased layer of adipose tissue under the skin to act as insulator.

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➥ Increased hairiness (fur), this reduces heat loss by convection.

Races of humane

Race is a categorization of humans into groups based on shared inherited, physical and behavioral differences. Based on a recent genetic analysis, humans are loosely classified in to 3 races which have their own sub - divisions. These are: African, Eurasian, and East Asia.

Africans (Negroid) - People from Africa and Melanesian of the South pacific etc.

Eurasian (Caucasoid) - People with variable skin color ranging from white to dark brown. The three subdivisions are:

i. Nordic - People from Scandinavian and Baltic countries, Germany, France, Britain etc.

ii. Mediterranean - People from South France, Spain, Italy, Wales, Egypt, Jews, Arabs, Afghanistan, Pakistan, India

iii. Alpine - People from the Mediterranean to Asia.

East Asian (Mongoloid) - they are the most numerous people in the world, and has three groups:

i. Eastern Siberians, Eskimos and Northern American Indians.

ii. Japanese, Koreans and Chinese.

iii. Indonesians and Malays.

Illustrative Example

53. Man belongs to the super family

A. Hominin
B. Hominoidae
C. Primates
D. Mammalia

Explanation: Hominoidae is the super family of man. Man belong to the family Hominidae. The order of man is primates and class is Mammalia.

Answer: B

54. The cranial capacity of Modern man was _____ c.c.

A. 1300 - 1600
B. 1000 - 1100
C. 850 - 1300
D. 1300

Explanation:- The cranial capacity of Modern man is almost same as that of Neanderthal mans i.e 1300 - 1600 c.c. They had a fully developed

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brain.

Answer: A

55. It was believed that modern man evolved in Africa.
A. True B. False

Explanation: It is true modern man evolved in Africa. Later its members migrated to eastwards, westwards and southwards. Modern man originated and evolved ten thousand years ago.

Answer: A

56. What is known as the study of the evolutionary history of man?
A. Anthropology C. Cytology
B. Ethology D. Humanology

Explanation: Anthropology deals with the study of the evolutionary history of man. Ethology deals with the study of animal habits and behavior. Cytology is the study of cells.

Answer: A

57. Which of the following includes all the others?
A. Greats apes C. Primates
B. Hominins D. Hominids

Explanation: Hominids include all species and extinct common ancestors; hominin includes only the human species in the genus Homo. All hominines are included under hominids, but all homininds are not included under hominine.

Answer: D

58. Which of the following hominid has largest brains?
A. Homo erectus C. Homo sapiens
B. Asutralopitheus D. Ardithecus ramidus

Explanation: Homo sapiens has the largest brain (1500 cm^3).

Answer: C

59. To which genus of human - like organisms does Lucy belong? (UEE, 2004 E.C)
A. The genus homo C. The gens sahelanthropus
B. The genus ardithecus D. The genus Australopithecus

Explanation: the scientific name of Lucy is Australopithecus afarensis.

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60. Which of the fossil is the nearest to the common ancestor of the hominids and the apes? (UEE, 2004 E.C)

Answer: D

A. Homo habilis
B. Homo erectus
C. Adripithecus ramidus
D. Australopithecus afarensis

Explanation: Ardi (4.4 million years old fossil) is the closest common ancestor of hominids. Others are relatively recent.

10.1.8 Mutations

Answer: C

Mutation is the structural change in DNA, chromosome, a gene or a nucleotide base and the end products (type of protein made and its function) as well. It is a random change in genetic materials of the organisms, caused by several agents. The factors (agents) that cause mutation could be: spontaneous and induced mutations.

i. Mutation that have no known cause are known as a **spontaneous mutation.**

ii. Mutations that result from the action of mutagens called **induced mutations.**

➥ Agents that increase the mutation rate are called **mutagens (mutagenic agent)**

➥ Mutagenic agent includes: radiation, x - ray, UV - radiation, nuclear radiation, and many chemical substances

Types of Mutation

There are two major types of mutations. These are: **point mutation**, and **chromosomal mutation.**

i. Point Mutation

Point mutation is a change in single nucleotide base in a gene or DNA base sequences through substitution, addition, or deletion.

➥ It affects the process of protein synthesis.

➥ It is less serious than chromosomal mutation

A. Substitution

In a substitution mutation, one nucleotide replaces another, forming a new codon that may signal the insertion of the wrong amino acid.

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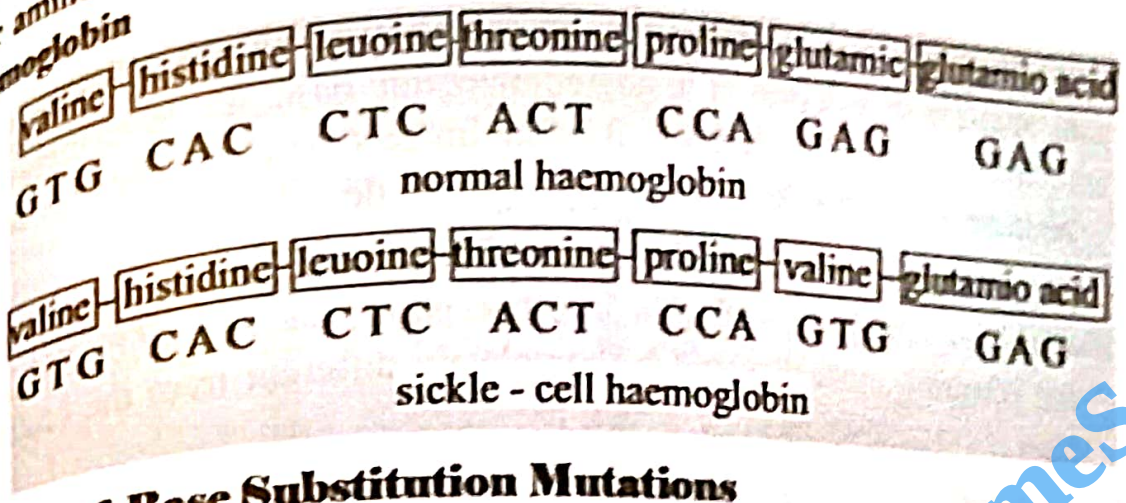
For example

A substitution of one nucleotide at the sixth codon (A by T) changes the amino acid **glutamic acid** into **valine**, which results in sickle cell haemoglobin



Types of Base Substitution Mutations

VI. **Silent mutation** is a type of mutation occurring in a gene, but doesn't affect the final protein.

For example: If the codon AAA were changed to AAG, the amino acid would not change; both of these codons represent the amino acid lysine.

5'TCTCAAAAATTTACG 3' 5'TCTCAAAAGTTTACG 3'

3'AGAGTTTTTAAATGC 5' 3'AGAGTTTTCAAATGC 5'

—ser Gin Lys Phe Thr— —ser Gin Lys Phe Thr—

VII. **Missense mutations** - is a type of mutation occurring in a gene due to substitution of a single base pair in the DNA, which result in change in the amino acids of a protein.

For example: If a change of codon from AAA to GAA, the amino acid would change from **lysine to Glutamin**

5'TCTCAAAAATTTACG 3' 5'TCTCAAGAATTTACG 3'

3'AGAGTTTTTAAATGC 5' 3'AGAGTTCTTAAATGC 5'

---ser Gin Lys Phe Thr--- ---ser Gin Glu Phe Thr—

VIII. **Nonsense mutations** - Is a type of mutation occurring in a gene when a stop codon from DNA is introduced into mRNA.

From example: the codon AAA represents the amino acid tryptophan. If a mutation occurred to change AAA to TAA, then the protein would stop early. Nonsense mutations typically have severe consequences for protein function.

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5' T C T C A A A A A T T T A C G 3' 5' T C T C A A T A A T T T A C G 3'

3' A G A G T T T T T A A A T G C 5' 3' A G A G T T A T T A A A T G C 5'

— ser Gin Lys Phe Thr — — ser Gin Stop

IX. Frameshift mutations is a type of mutation occurring in a gene due to addition or deletion of a single nucleotide base, which changes the reading frame of the mRNA. It is the biggest type of point mutation, because it can change the type of protein made.

For example: If there is an extra addition of one nucleotide into the mRNA, all the codons after the insertion point would be off by one, and the resulting protein would be completely different from the original plan.

B. Addition

It is the type of point mutation in which a nucleotide base is added to DNA sequences. The whole sequence after the added based is changed, which leads to frame shift mutation that is the change of the reading frame of the genetic code.

5' T C T C A A A A A T T T A C G 3' 5' T C T C A A G A A T T T A C G 3'

3' A G A G T T T T T A A A T G C 5' 3' A G A G T T C T T A A A T G C 5'

— ser Gin Lys Phe Thr — — ser Gin Glu Ite Thr —

C. Deletion

It is the type of point mutation in which a nucleotide base is lost from DNA base sequences. The out come is similar to that of base addition, which cause frame shift mutation.

ii. Chromosomal Mutations

Chromosome mutations involve changes in the structure and arrangement of a chromosome or the deletion or addition of a piece of chromosome. They occurs most often due to error in meiosis during crossing over at prophase 1.

Types of chromosomal mutations

Chromosomal mutations could be either chromosomal mutation I or chromosomal mutation II.

A. Chromosomal mutations I

These involve mutations that change the structures of chromosomes. This happened when a piece of chromosome is break up and then forms sticky

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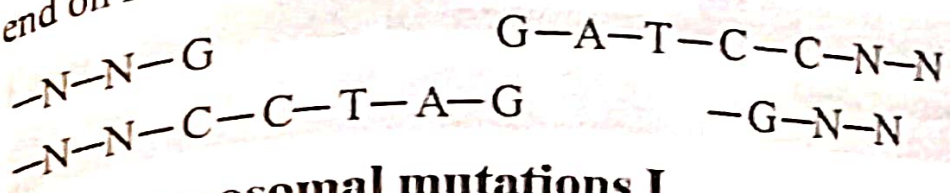
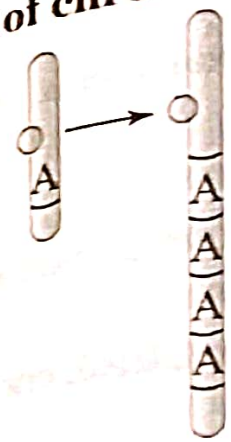
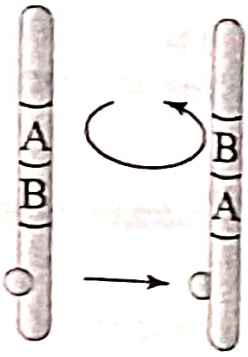
ends /have a number of unpaired bases/. The following structure shows sticky end on DNA.



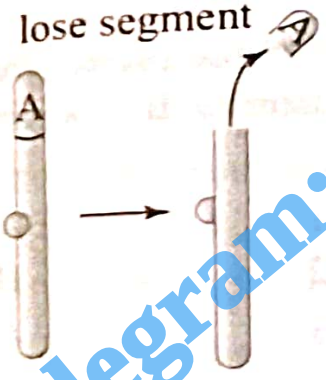
Types of chromosomal mutations I



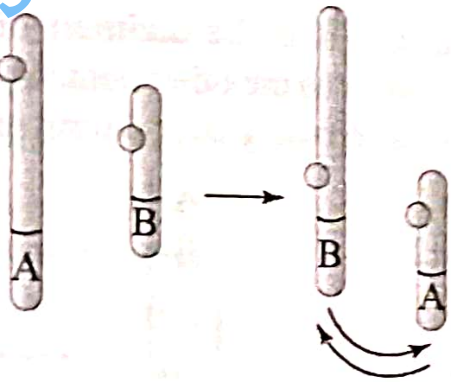
(a) a **duplication** is a repeated segment of a chromosome. In this example segment A is repeated



(b) an **inversion** is a chromosome segment with a reversed orientation. an inversion does not change the amount of genetic material in the chromosome , only its arrangment.



(c) a **deletion** is the loss of a chromosome segment. a deletion can occur at the tip (shown) or within the chromosome



(d) a **reciprocal translocation** occurs when two nonhomologous chromosomes exchange segments

1. Inversion

In an inversion, chromosomal segment is breaks off and rotates by 180° on its own axis before getting reattached to the chromosome but in the opposite direction than it was originally.

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- When the inversion does not include the centromere of chromosome, it is called **paracentric inversion**
- When the inversion does the centromere of chromosome, it is called **pericentric inversion**

2. Deletion

It occurs when apiece of chromosome is accidentally removed, cause a decrease in gene numbers. It is also known as partial **monosomies**, because there is a missing part of chromosome.

Types of deletion

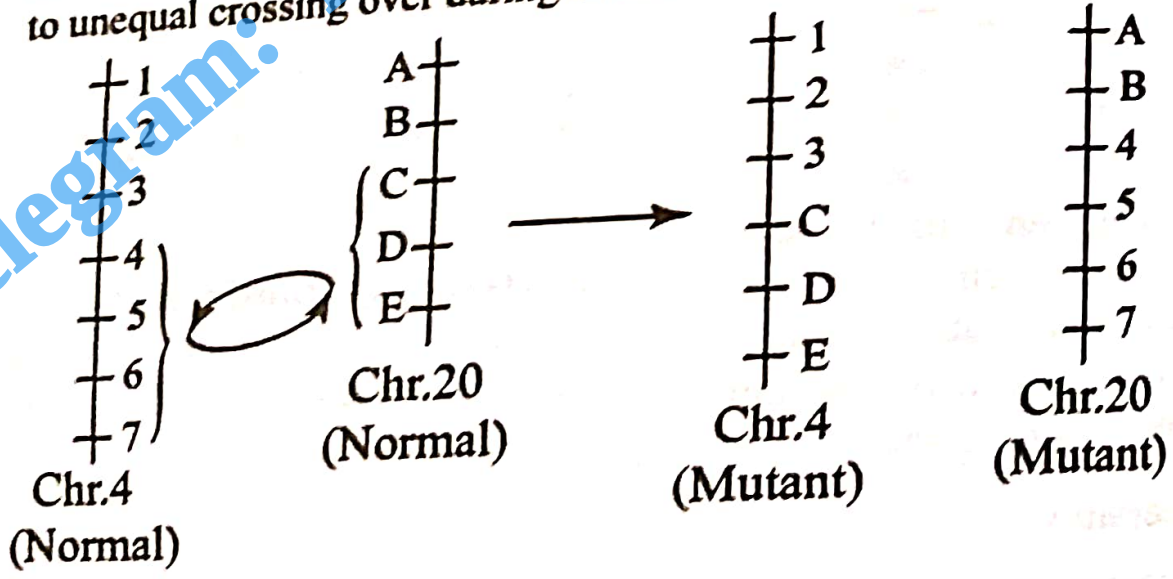
i. Terminal deletion - it occurs when the terminal (end) part of the chromosome is deleted.

ii. Interstitial deletion - two pieces of chromosomes are deleted, one at the end and one within the chromosomes

iii. Micro deletion - it occurs when a very small portion of chromosome is deleted from the chromosome.

3. Duplication/Amplifications/ - is a mutation due to the occurrence of multiple copies of a piece of chromosome or the whole chromosome is present in the nucleus. It is also called partial trisomy.

4. Insertion - It is the addition of a piece of chromosome from one chromosome to the other homologous chromosomes. It is occurred due to unequal crossing over during meiosis1.



5. Translocation

In a translational, a piece of one chromosome breaks off and reattaches to a nonhomologous chromosome

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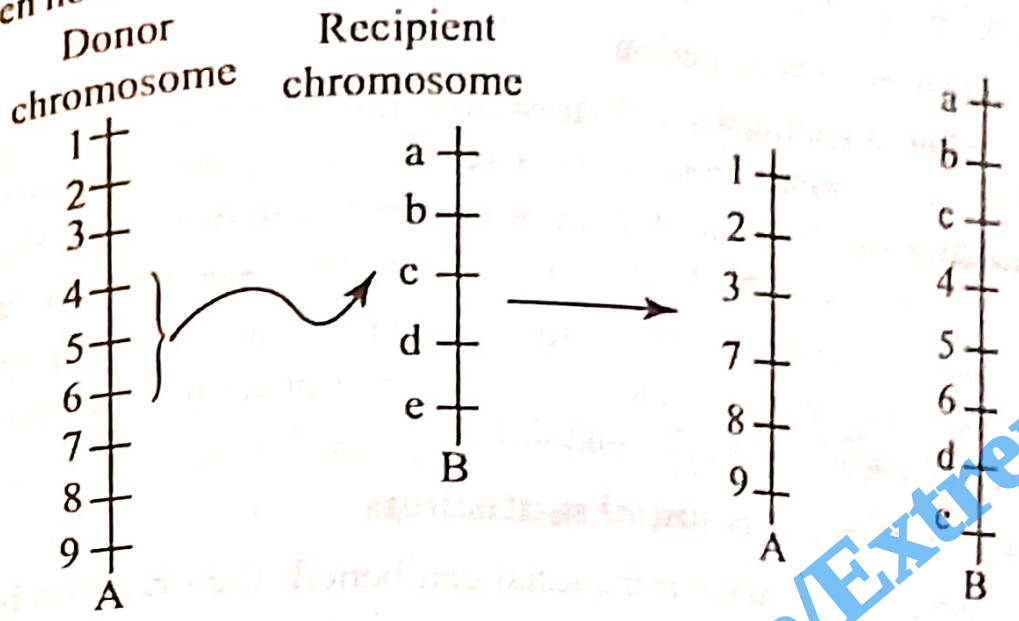
Types of translocation

I. Reciprocal translocation

It occurs when pieces of chromosomes "swap" or exchanges places in between non - homologous chromosomes.



II. Robertsonian translocation

It occurs when a segment of a chromosome is attached to another chromosome and causing an elongation of it. It occurs between acrocentric chromosomes (chromosomes with centromere located very close to one end).

Robertsonian translocation includes:

1. Balanced Robertsonian translocation - it occurs when there is no net loss or gain of genetic materials and it has not apparent effect on individual.
2. Unbalanced Robertsonian translocation - it occurs when there is net loss or genetic materials and it has apparent effect individual.

   Example: the translocation of chromosomes 21 into the 14th chromosome and causes the common genetic disorder down's syndrome.

B) Chromosomal Mutations II

It is a mutation that changes the number of chromosome the cell (heteropliody). It is further divided into aneuploidy and polyploidy.

A. **Aneuploidy** - the type of chromosomal mutation alters parts of a chromosome set, causing either the addition or loss of one or more chromosomes. It is any deviation from the human diploid number of

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46 chromosomes.

Example: Downs syndrome - it occurs when there is three copies (trisomy) of chromosome 21.

Downs syndrome is characterized by: mental retardation, heart defects, and stunted growth.

Chromosomal non - disjunction

It occurs when homologous chromosomes fail to separate successfully to opposite poles during meiosis as a result one of the gamete lacking a chromosome and the other having an extra chromosome. It results in daughter cells with abnormal chromosome numbers, known as aneuploidy

B. Polyploidy - is a type of chromosomal mutation, when there is multiple sets of chromosomes, such as tetraploid (3n) or tetraploid (4n) within the cells. It causes gigantism and inferility.

Advantages of chromosomal mutations

In some cases, chromosomal mutational can benefit the organism because some mutations help the organisms to survive better than others.

Example

1. People who have a lactose tolerance gene are more advantageous than others that don't have it, because this gene enables us to feed milks from cows and goats as sources of food.
2. Resistant gene for malaria (sickle cell)

Disadvantage of chromosomal mutations

Chromosomal mutations can be dangerous and even detrimental /harmful/ to the life of living organisms. Some of theses causes problem, such as: genetic disorder, other disease (cancer, diabetes, asthma, etc.).

Genetic Disorders - they are specific disorders or disabilities caused by mutations within the organism's DNA. Example: trisomy

Trisomy : is a genetic disorder in which a person has three copies of chromosome instead of two.

Some of the examples of trisomy are listed below:

- Trisomy 21/Down Syndrome/ - results is developmental and metal retardation
- Trisomy 13 /Patau syndrome/ - results in malformation of brain, defects of eye, additional toes, sex organ

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- Trisomy 18 /Edward's syndrome/ - results in malformation of sex organ, lip, skull, hand and feet, heart, etc.
- Klinefelter syndrome - occurs in male, which possess a sex chromosome XXY: have a feminine feature.
- XYY Syndrome - happen in male, characterized by very tall height, severe acne, learning disabilities
- Turner Syndrome - occurs in a female that possess a sex chromosome XO: have a masculine feature.
- Triple X syndrome /XXX/ - abnormality due to extra copy of the X chromosome in some female. It is characterized by tall stature, low muscle tone, learning disabilities, behavioral problem, flat feet, kidney, etc.

Illustrative Example

61. Which of the following is incorrect with respect to mutation?
    A. Sudden
    B. Continuous
    C. Change in chromosomes and genes
    D. Leads to variation in DNA

Explanation: Mutations are not continuous. Instead, mutations are discontinuous variations in the genotype and phenotype of an organism. This is due to the changes in chromosomes and DNA. Mutations are sudden and they are a cause of the variation in DNA.

Answer: B

62. Which of the following is not a type of mutation?
    A. Gene mutation
    B. Chromosomal aberrations
    C. A & B
    D. Colourful mutations

Explanation: Mutations can be of two different types depending upon their cause and environment. These are: Gene mutation, Chromosomal aberrations.

Answer: D

63. Which of the following is a classic example of point mutation?
    A. Phenylketonuria
    B. Sickle cell anaemia
    C. Genomatic mutations
    D. Colourful mutations

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Explanation: Sickle cell anaemia is a classic example of point mutation,. Point mutation generally means that when a single base pair of DNA changes, a gene mutation occurs that causes the alternation of DNA.

Answer: B

64. Which of the following is also known as the removal of one or more bases from the nucleotide chain?

A. Deletion
B. Duplications
C. Translocation
D. Inversion

Explanation: The removal of one more bases from the nucleotide chain is also known as deletion. Deletion is a type of frame - shift mutation. Addition of one or more bases in a nucleotide chain, then it is called insertion.

Answer: A

65. What does the given type of mutation (aberrations) in the figure represent?

PQRSTU → PQRSTUTU

A. Deletions
B. Duplications
C. Translocation
D. Inversion

Explanation: The given figure represents the duplication aberrations where a part of the chromosome gets added so that a gene or a set of genes is duplicated or is represented twice.

Answer: B

66. What is translocation?

A. Shifting of part of chromosomes to the outside wall of nucleus
B. Shifting of a chromosome to another chromosome
C. Shifting of a part of a chromosome to another homologous chromosome
D. Shifting of a part of a chromosome to another non - homologous chromosome

Explanation: Translocation is the shifting of a part of a chromosome to another non - homologous chromosome. It results in the formation of new recombinant chromosomes as this induces faulty pairing of chromosomes during meiosis.

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Answer: B

67. Paracentric is a type of inversion involving centromere.
A. True B. False
Explanation: Paracentric is not a type of inversion involving centromere. Instead, it is an inversion without involving centromere. Pericentric is a type of inversion which involves centromere.

Answer: B

68. On which of the following chromosomal mutation are based on?
A. Mutant allele and their defective products
B. Imbalance in chromosome number and chromosome arrangement
C. Mutant allele and chromosome arrangement
D. Mutant allele and imbalance in chromosome number
Explanation: The chromosomal mutations are the disorders that on the imbalance in chromosome number and chromosomal arrangement.

Answer: B

69. Which of the following is the characteristic feature of Down's syndrome?
A. Congenital heart disease C. Underdeveloped gonads
B. Short stature D. All
Explanation: Congenital heart disease, and underdeveloped gonads are some features of Down's syndrome in people.

Answer: D

70. What is the genotype of the person suffering from Klinefelter's syndrome?
A. 44 + XXX B. 42 + XXX C. 44 + XXY D. 42 + XXY
Explanation: The genotype of the person suffering from Klinefelter's syndrome is 44 + XXY. This deviation from a normal person's genotype is due to the additional copy of the X - chromosome.

Answer: C

71. Which of the following is TRUE about the characteristic of a mutation?
A. Randomness C. Accidental mistakes
B. Spontaneous D. All
Explanation: All refers to the properties of mutations.

Answer: D

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72. Which of the following is the cause of mutation?
A. UV - rays B. Some drug C. X - ray D. All
Explanation: Accidental mistakes during DNA replication are the basic causes of mutations. Mutagen such as X - rays, pesticides drugs increase the rate of such mistakes and complicates the matter.
Answer: D

73. In some cells if there are 4 sets of chromosomes per cell. What is the polyploidy level of such cells?
A. Haploid B. Tetraploid C. Diploid D. Hexaploid
Explanation: Tetraploid means 4 sets of chromosome.
Answer: B

74. What exactly happen during a substitution mutation? (EUEE: 2005 E.C)
A. One base is replaced by a different base.
B. An extra base is added during DNA replication.
C. A base is missed out during DNA replication
D. Change occur in the arrangement or structure of a chromosome.
Explanation: Substitution mutation refers to the replacements of a base by another.
Answer: A

75. Which one of following is not mutation (UEE: 2004 E.C)
A. DNA replication to form 2 daughter molecules
B. Gain of an extra chromosome by a cell
C. Deletion of a base pair from DNA
D. Loss of a chromosome by a cell.
Explanation: DNA replication is a normal life process that should take when needed.
Answer: A

76. Which of the following refers to a change in a gene? (UEE: 2003 E.C)
A. Mutation C. Protein synthesis
B. Replication D. Transcription
Explanation: Mutation refers to gene changes: replication refers to DNA synthesis on itself translation is protein synthesis (conversion of

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the information residing in DNA into protein) and transcription is the copying of the information in DNA by synthesized RNA.

Answer: A

77. Which of the following types of mutations is sickle cell anaemia? (EUUEE: 2005 E.C)

A. Addition of a base pair
B. Deletion of a bas pair
C. Substitution of abase pair
D. A shift in the reading frame of the genetic code

Explanation: Sick - cell anemia is caused by a substitution mutation that result in the replacement of glutamate by valine in one of the 4 polypeptide chains of haemoglobin consequently. RBCs with such as a haemoglobin molecule will attain sickle - shape and fail to carry enough oxygen.

Answer: C

78. What is the type of mutation that arose from a change in the nucleotides sequence of the DNA as a particular locus f the chromosomes?

A. Gene mutation
B. Numerical mutation
C. Structure mutation
D. Chromosomal mutation

Explanation: Mutation in a locus of a chromosome are called point (gene) mutations.

Answer: A

79. Which of the following types of gene mutation results in a replacement of just one amino acid by another in a protein? (CEE 1995 E.C)

A. Addition of a base pair
B. Substitution
C. Deletion of abase pair
D. None sense mutation

Explanation: a base - pair substitution may cause the replacement of one amino acid by another. Addition and deletion of base - pair results in profound change called frame - shift .

Answer: B

80. Which of the following types of chromosome mutations involves loss of chromosome segment?

A. Aneuploidy
B. Duplication
C. Translocation
D. Deleting

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Explanation: Loss of chromosome segment is deletion.

Answer: D

81. Which of the following is not an example of point mutation in which two non homologous chromosome exchange a segment.

A. Substitution
C. Deletion
B. Addition
D. Translocation

Explanation: Translocation is a chromosome mutation in which two non homologous chromosome exchange a segment.

Answer: D

82. What is mutation? Explain how it can be caused.

Explanation

- Mutation – is any spontaneous change in the amount and arrangement or structure of the DNA organism.
- Some mutations can be caused by mutation factors such as chemical and high energy radiations.

10.1.9 Genetic Drift

Genetic drift is an evolutionary change in allelic frequencies of a population as a matter of chance or random events. It is through to be another possible mechanism for evolution of new species, because it can result in significant changes within a population. It is caused by population decline or the creation of a new population.

- Small populations can undergo greater changes in allele frequencies, exhibiting a large degree of genetic drift, whereas large populations retain fairly stable allele frequencies, maintaining a small degree of genetic drift.

In a small population, a particular allele may disappear completely over a few generations. In a larger population, a particular allele may vary widely in frequency due to change but still be present in enough individuals to be maintained in the population

For example

1. Small populations of bird have an allele for two different sized beaks (small and medium). If natural disaster or a new disease wipes out the entire population of small beak size from the population, the genetic variation and allelic frequencies of birds will be decreased from the gene pool.

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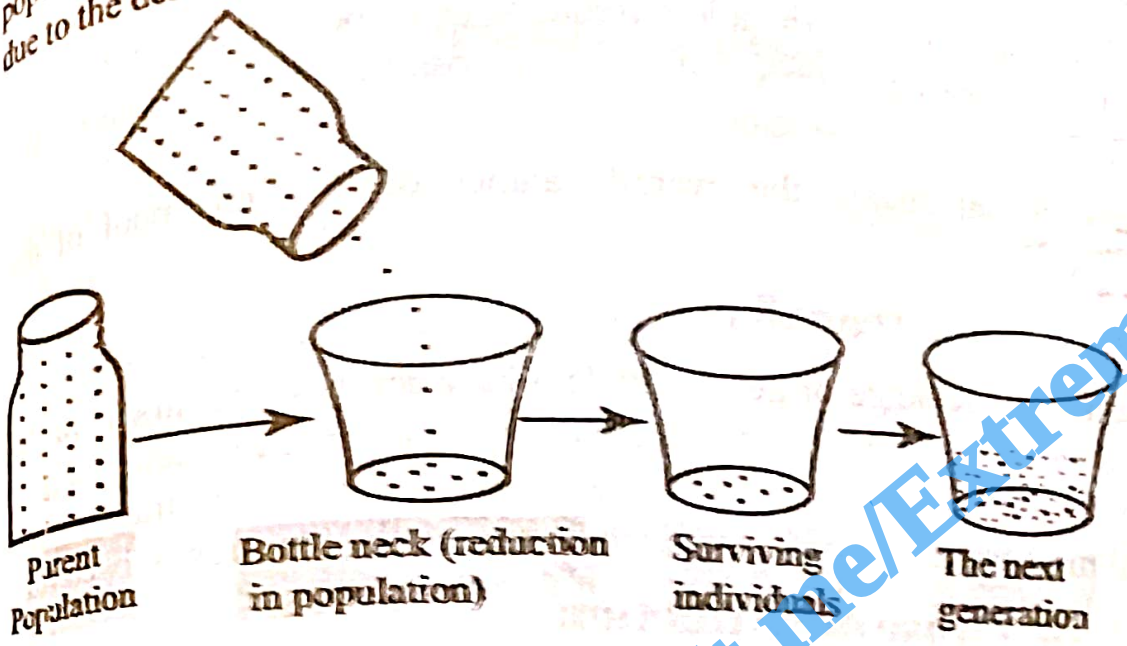
Types of genetic drift

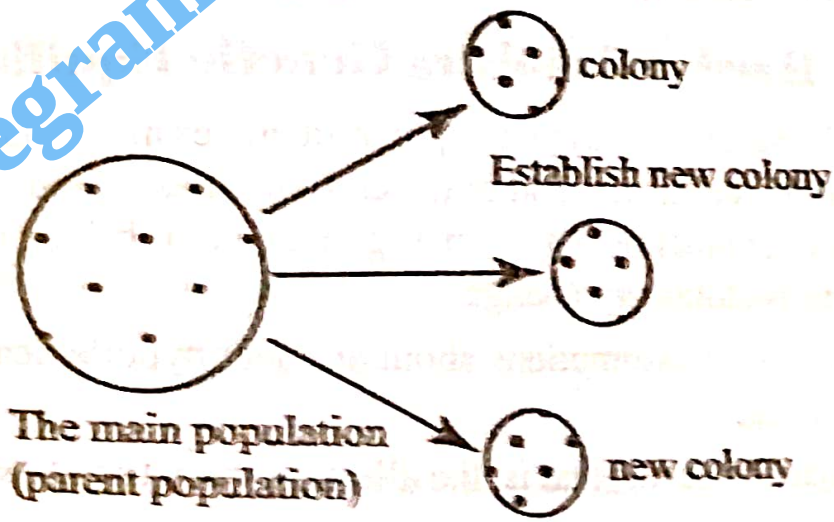
a) Bottle neck effect

In the bottle neck effect the population size severly decrease due to natural disaster (earthquakes, floods, fires). The frequency of certain alleles in a population changes because the organisms that carry them are eliminated due to the death of most of the population.



b) Founder effect

It occurs when a small group of individuals breaks off from a larger population to establish a colony. The new colony is isolated from the original population, thus it will not have the full genetic diversity of the original population. As a result, the allelic frequencies of the new population will be different from the original population.



Note: The founder effect is similar in concept to the bottleneck effect, but occurs by colonization of new area rather than

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catastrophe/disaster.

10.1 Gene Flow

Gene flow is the movement of genetic material between populations, or organisms of the same species. It occurred when organisms are migrating (immigration or emigration) into new population

Gene flow can't evaluate the allele frequencies of population, but has the function of deceasing genetic differences across the groups, delaying the development of new population.

However, it can change the overall balance of the gene pool of the population

10.2.1 Gene transfer

Gene transfer exchange of genes (alleles) between two organisms or the introduction of new DNA in to an existing organism call. Gene transfer enables the development of a new species. Horizontal gene transfer/non - sexual movement of genetic materials/is especially common in bacteria.

10.2.2 Cause of Extinction

Extinction is the complete disappearance (dying out) of a species.

These could be due:

1. Anthropogenic effects/environmental change due to human activities/
2. Natural disasters are played significant role for the direct and indirect cause species extinction.

10.2.3 Hardy - Weinberg Genetic Equilibrium

It is a principle that showed as genotypic frequencies in a population tend to remain the same from generation to generation unless acted on by outside influences. Such population is known as genetic equilibrium, meaning will not undergoing evolutionary change

It is based on a set of assumptions about an ideal hypothetical population that is not evolving.

i. No net mutations occur; that is, the alleles remain the same
ii. Individuals neither enter nor leave the population (no migration)
iii. The population is large (ideally, infinitely large).

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iv. Individuals mate randomly.
v. Selection does not occur.

Note: If one or more of these conditions is /are/ not met, change in gene frequency may occur then a deviation from the Hardy - Weinberg law shows a change in the gene pool has occurred and evolution has take place. Thus the law is a means of measuring evolution.

Hardy - Weinberg equilibrium equations

There are two simple equations that represent the relationship between allele and genotype frequencies.

If all the alleles are the same in a population then the frequency of the allele is one.

I. The first equation represents the frequencies of both alleles in the population is $\mathbf{P + q = 1}$ where:
   - p represents the frequency dominant allele
   - q represents the frequency of recessive allele.

II. The second equation is used to calculate the genotype frequencies.
   $p^2 + 2pq + q^2 = 1$, where
   p^2 represents the frequency of dominate genotype.
   2pq represent the frequency of heterozygous genotype frequencies.

Example

There are 90 recessive black fruit flies in the total population of 1000.

a) What is the frequency of recessive alleles.
b) What is the frequency homozygous dominant alleles?
c) What is the frequency of dominant alleles?

Answer:

a) The recessive allele frequency of black fruit (bb) flies is:

$$= q^2 = \frac{90}{1000} = \mathbf{0.09}$$

The value of q is expressed by $q^2 = 0.09$ and $q = \sqrt{0.09} = 0.3$.

$p + q = 1$ this implies, $p + 0.3 = 1$, $p = 1 - 0.3 = 0.7$

b) The homozygous dominant allele (BB) frequency $= p2 = (0.7)2 = \mathbf{0.49}$

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c) The dominant allele is represented by

$p^2 + 2pq = 0.49 + 2(0.7)(0.3) = \mathbf{0.91}$

Illustrative Example

83. What is the term used for the total genetic information in a population?

A. Gene pool
B. Allele frequency
C. Distribution of traits
D. Phenotype frequency

Answer: A

84. In a population of butter flies, pink wing is recessive to purple wings. In a population of 600 butter flies, 54 have pink wings.

a) How many alleles are in the gene pool for wing color?
b) What is the frequency of recessive allele?
c) What is the frequency of homozygous dominant allele?
d) What percentage of butterfly population is heterozygous?

Answer:

a) 600 individuals each have two alleles for wing color,
$600 \times 2 = 1{,}200$ alleles in this gene pool

b) Remember the Hardy - Weinberg equation

$\boxed{P + q = 1}$ and $P^2 + 2pq + q^2 = 1$

(frequency of Homozygous dominant) (Frequency of Heterozygous dominant) (frequency of recessive)

☑ The frequency recessive allele = $q^2 = 54/600 = \mathbf{0.09}$

$q = \sqrt{0.09} = 0.3$

c) The frequency of homozygous dominant allele is P^2

$P + q = 1 \Rightarrow P + 03 = 1 \Rightarrow P = 0.7 \Rightarrow P^2 = \mathbf{0.49}$

Therefore, 49% of the population is homozygous dominant

d) The frequency of heterozygous dominant is = 2Pq

$2(pq) = 2(0.7)(0.3) = \mathbf{0.42}$

85. In a population of 350 horse, 16% express the recessive phenotype. How many are homozygous dominant?

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Answer

- The frequency of recessive allele = $q^2 = 0.16$, then
  $q = \sqrt{0.16} = 0.4$
- The frequency of homozygous dominant = $P^2 = (0.6)^2 = 0.36$
- Therefore, the percentage of homozygous dominant will be
  $0.36 \times 350 = 126$

86. If 84% of the population express the dominant phenotypes
a) What is the frequency of dominant allele?
b) What is the frequency of recessive allele?

Answer: Remember the Hardy - Weinberg equations
$P + q = 1$ and $P^2 + Pq + q^2 = 1$

a) Both P^2 and 2Pq express the dominant phenotype, which accounts 84% . P^2 - is homozygous dominant, & 2 Pq is heterozygous
Therefore, the frequency of dominant allele is $P^2 + 2Pq = 0.84$

b) From the equation $P^2 + 2Pq + q^2 = 1$, the frequency of recessive allele is $q^2 = 1 - 0.84 = 1 \Rightarrow q2 = 0.16$

87. How do mutations affect genetic equilibrium?

A. Mutations cause emigration
B. Mutations cause immigration
C. Mutations introduce new alleles.
D. Mutations maintain genotype frequency

Answer: C

88. The sum of allele frequency in a population should be

A. 1
B. Less than 100
C. Equal to genotypic frequency
D. One half of genotypic frequency

Answer: A

89. In the Hardy - Weinberg equation p2 + 2pq + q2 = 1, what does the term 2pq represent?

A. Frequency of heterozygous individual
B. Frequency of homozygous recessive individuals
C. Frequency of individuals with two allele
D. Frequency of homozygous dominant individuals

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90. Genetic drift has greatest impact on

Answer: A

A. Large population
B. Small population
C. An individual organism
D. Migrating population

91. In evolution, extinction describes the end of

Answer: B

A. An allele
B. A single species
C. An individual organise
D. A population of organism

92. The major source of new alleles in natural population is

Answer: B

A. Mutation
B. Polyploidy
C. Genetic drift
D. Natural selection

93. Random change in allele frequency due to change alone is called

Answer: A

A. Gene flow
B. Genetic drift
C. Natural selection
D. Sexual selection

Answer: B

Review Exercise

1. Suppose that the population has 1000 individuals and that each one produces two gametes. What is the frequency of allele A (Assume that allele AA = 490, aa = 420, and Aa = 90)

A. 0.7 B. 0.3 C. 1 D. 0.5

2. Disruptive selection _____

A. Eliminate uncommon form of allele
B. Shifts allele frequencies in one direction only
C. Doesn't favor intermediate forms of trains
D. B and C

3. Individuals move their alleles into and out of a population by way of migration tend to counter (reverse) the change caused by different evolutionary factors is _____

A. Gene flow
B. Genetic drift
C. Mutation
D. Natural selection

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4. Which ancient organism were most likely responsible for the development of the ozone layer?
   A. Protists
   B. Sulfur bacteria
   C. Cyanobacteria
   D. Plants
5. Palaeontologist can estimate the absolute age of the fossil by measuring the concentration of
   A. Radioisotopes
   B. Stable isotopes
   C. Radioisotopes in the surrounding
   D. Stable isotopes in the area
6. How is information from the field of embryology used as evidence for evolution?
   A. It shows that organisms from the same species have similar DNA
   B. It shows that organisms from the different species have similar DNA
   C. It shows that organisms from the different species have similar development patterns.
   D. It shows that organisms from the same species have similar development patterns.
7. Which idea bout evolution was yield by Darwin but not by Lamarck?
   A. Species are fixed and do not change
   B. Species are moving toward perfection
   C. Organisms best suited to their environment survive and reproduce
   D. The use or disuse of an organ determines its appearance in a species.
8. Which of the following would be considered as a useful mutation to organism?
   A. Antibiotic resistant bacterial
   B. Formation of cancer cells
   C. Sickle cell anemia for malaria
   D. A and C
9. Which one is a conclusion drawn from the observed DNA similarities found between apes and chimpanzees?
   A. Apes and chimpanzees are the same species.

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B. Apes and chimpanzees are the products of artificial selection.

C. Apes and chimpanzees are thought to products of artificial selection

D. Apes and chimpanzees are thought to have evolved from a common ancestor.

10. What assumption is made during the relative dating of fossil?

A. Remains of extinct organisms can be compared to organisms living today.

B. Comparing age of the rock layers

C. Upper rock layers contain younger fossil and are found in deeper layer.

D. It is based on radioactive materials to estimate the age of the fossil.

11. Traces of starch from wild barely (hordem) and wheat (triticum) species were found on a 23,000 - years - old seed - grinding stones used by ancient humans in Israel. The method which would have been used to data these ancients starch grains is.

A. Radioactive carbon - 14 dating

B. Radioactive uranium - 235 dating

C. Relative dating by comparing rock strate

D. The use of a molecules clock.

12. According to the Hardly - Weinberg theory, which of the following represents homozygous dominant individual?

A. p^2 B. q C. q^2 D. 2pq

13. All of the following are the assumption of Hardly - Weinberg equilibrium that population is not evolving, except

A. No natural selection C. No mutation

B. Random mating D. The population must be small.

14. The giant anteater of South America, the giant armadillo of North Africa, and the spiny anteater of Oceania share may characteristics in common. They all eat ants, have long narrow snouts, few teeth, and large salivary glands, yet they do not have evolutionary origins. This an example of ____

A. Genetic drift C. Adaptive radiation

B. Founder effect D. Convergent evolution

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15. All of the following not true of about homologous structures?
A. They demonstrate common ancestry
B. An example is the wing of bat and human arm.
C. An example is the wing of bat and the wing of fly.
D. An example is the whale's lateral fin and wings of birds.

16. What is the best term used to refer to all of the genes present in a population?(UEE general science, 1995 E.C)
A. Biochemistry
B. Gene pool
C. Genome
D. DNA

17. Which one of the following modes of natural selection was responsible for the fast increase in the number of the black forms of moths in Europe during industrial revolution? (UEE, 2003 E.C)
A. Stabilizing selection
B. Disruptive selection
C. Normalizing selection
D. Directional selection

18. Which one of the following is the most important for speciation to occur? (UEE, 2003 E.C)
A. Hybridizing among population
B. Isolation of the gene pool
C. Migration between populations
D. Free flow genes between population

19. Animals without close evolutionary relations are sometimes seen to have similar structures adapted for the same function.
A. Convergent evolution
B. Diverge evolution
C. Sympatric evolution
D. Allopatric evolution

20. What do the structural similarity between the flippers of whales and arms of human are shows?
A. Whales evolved from than human species
B. Whales are older than human's species
C. The human species began life in the ocean
D. Whales and humans had a common ancestor

21. A population is separated into two group by geographic barrier overtime so that they cannot interbreed when reunited

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A. Extinction
C. Hybridization
B. Immigration
D. Speciation

22. Heterozygous advantage should be most closely linked to which of the following?
A. Sexual selection
C. Diploid
B. Stabilizing selection
D. Directional selection

23. Dog breeders maintain the purity of their doges by keeping them parts when they are fertile. This kind of isolation is most similar to which of the following reproductive isolating mechanisms?
A. Hybrid sterility
C. Habitat isolation
B. Hybrid breaks down
D. Gametic isolation

24. If two species of frogs mate and their offspring fail to develop and hatch, the mechanism keeping them separate is:
A. Hybrid sterility
C. Hybrid infertility
B. Hybrid inviability
D. Hybrid fertility

25. Approximately how old is the time does the fossil ardi (*Ardipithecus ramidus*) recode extend?
A. 3.2 million years
C. 4.4 million years
B. 3.2 million years
D. 3.5 million years

26. Approximately how old is the fossil Lucy (*Australopithecus afarensis*)?
A. 8 million years
C. 6.0 million years
B. 3.2 million years
D. 4.5 million years

27. In earliest fossils found in the geologic record are
A. Fungi B. Seed plants C. Bacteria D. Protozoa

28. In which hominids species did scientists find the smallest brain size (cranial capacity)? (UEE, 2005 E.C)
A. *Homo sapiens*
C. *Homo erectus*
B. *Homo habilis*
D. *Homo Neanderthals*

29. What is the reproductive isolating mechanism called if two species of frogs do not inter breed because they cannot understand the mating calls of one another? (UEE, 2005 E.C)
A. Seasonal isolation
B. Behavioral isolation

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C. Temporal isolation D. Isolation by distance

30. Among the following which one is the best criterion to show that two populations belong to the same species?

A. Morphological similarity C. Production of fertile offspring

B. Physiological similarity D. Inhabiting the same habitat

31. What kind of natural selection is at work in a situation where individuals at both extreme ends of the range get better survival advantages than those around the middle?

A. Stabilizing selection

B. Direction selection

C. Disruptive selection

D. Selection that removes the extremes

32. Which one of the following statement describe Charles Darwin's theory of evolution

A. Acquired characteristic

B. Evolution occurs by the use and disuse of body parts

C. Natural selection leads to evolution and speciation

D. Living things do not change through time

33. Which one of the following alternative shows the gases that were believed to be presents in the ancients atmosphere before the origin of life on earth?

A. Ammonia, oxygen, methane C. Methane, oxygen, hydrogen

B. Hydrogen, ammonia, methane D. Oxygen, hydrogen ammonia

34. The half life of a certain isotope is about 5000 years. What percent of the original amount of the isotope is expected to be presents in a fossil that is 20,000 years old?

A. 12.5% B. 6.25% C. 50% D. 25%

35. The two scientists who develop the biochemical theory of the origin of life were:

A. Richter and Arrhenius C. Disruptive selection

B. Haldane and Oparin D. Directional selection

36. In a certain plants population, nature favors the shortest and tallest plants while it eliminates plants with averages height.

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A. Stabilizing selection
B. Normalizing selection
C. Disruptive selection
D. Convergent evolution

37. Lift is said to begin on earths and transformed from simple forms to complex ones in millions of years. Branch of biology is concerned with the study of this process?
A. Microbiology
B. Astrobiology
C. Paleontology
D. Entomology

38. The process of by which Unrelated organism living in different parts of the world evolve similar structure adapted four the same function is
A. Parallel function
B. Adaptive radiation
C. Divergent evolution
D. Convergent evolution

39. Vertebrates that have similar basic structural pattern such as the pentadacty limb are assumed to have common ancestor. Such structures are called:
A. Analogous structures
B. Vestigial structures
C. Homologous structures
D. Adaptive structures

40. Which of the following animals are evolutionary the closest to Homo sapiens?
A. Gorillas
B. Chimpanzees
C. Orangutan
D. Gibbons

41. What is the term for the total genetic information in a population
A. Gene pool
B. Allele frequency
C. Distribution of traits
D. Phenotype frequency

42. Saint Bernard and Chihuahuas (two breeds of domestic dogs) cannot normally mate because they differ so much in size.
A. Artificial
B. Prezygotic
C. Postzygotic
D. Geographic

43. How do mutations affect genetic equilibrium?
A. Mutations cause emigration
B. Mutations cause immigration
C. Mutations introduces new alleles
D. Mutations maintain genotype frequency

44. The sum of allele frequency in a population should be

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A. 1
B. Less than 100
C. Equal to phenotype frequency
D. Frequency of homozygous dominant individuals

45. In the Hardy - Weinberg equation $p^2 + 2pq + q^2 = 1$, what does the term q^2 represent?
A. Frequency of heterozygous
B. Frequency of individual with two allele
C. Frequency of homozygous recessive individuals
D. Frequency of homozygous dominant individuals

46. Genetic drift has the greatest impact on
A. Larger population
B. Small population
C. Growing population
D. Migrating population

47. In evolution, extinction describes the end of
A. An allele
B. A single species
C. An individual organism
D. A population of organism

48. The major source of new alleles in a natural population is
A. Mutation
B. Polyploidy
C. Genetic drift
D. Natural selection

49. Random change in allele frequent due to chance alone is called
A. Gene flow
B. Genetic drift
C. Natural selection
D. Sexual selection

50. Saint Bernards and Chihuahus (two breeds of domestic dogs) cannot normally mate because they differ so much in size. Thus they are reproductively isolated to some extent. What type of isolating mechanism is operating in this case?
A. Artificial
B. Prezygotic
C. Postzygotic
D. Geographic

51. How do mutations affect genetic equilibrium?
A. Mutations cause emigration
B. Mutations cause immigration
C. Mutations introduces new alleles.

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D. Mutations maintain genotype frequency

52. The sum of allele frequency in a population should be:
   A. 1
   B. Less than 100
   C. Equal to genotypic frequency
   D. One half of the genotypic frequent

53. In the Hardy - Weinberg equation $p^2 + 2pq + q^2 = 1$, what does the term 2pq represent?
   A. Frequency of heterozygous individual
   B. Frequency of homozygous recessive individuals
   C. Frequency with two allele
   D. Frequency of homozygous dominant individuals

54. Genetic drift has the greatest impact on
   A. Large population
   B. Small population
   C. Growing population
   D. Migrating population

55. In evolution, extinction describes the end of
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   C. An individuals
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56. The major source of new alleles in a natural population is
   A. Mutation
   B. Polyploidy
   C. Genetic drift
   D. Natural selection

57. Random change in allele frequency due to change alone is called
   A. Gene flow
   B. Genetic drift
   C. Natural selection
   D. Sexual selection

58. Suppose that the population has 1000 individuals and that each one produces two gametes. What is the frequency of alleles A (Assume that allele AA = 490, aa = 420, and Aa = 90)
   A. 0.7 B. 0.3 C. 1 D. 0.5

59. Disruptive selection _____
   A. Eliminate uncommon form allele
   B. Shifts allele frequencies in one direction only
   C. Doesn't have intermediate forms of traits

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D. B ans C

60. Individuals move their alleles into and out of a population by way of migration; tend to counter (reverse) the change caused by different evolutionary factory is _____
   A. Gene flow
   B. Genetic drift
   C. Mutation
   D. Natural selection

61. Individuals move their alleles into and out of a population by way of migration; tend to counter the change caused by mutation, natural selection, and genetic drift is _____
   A. Gene flow
   B. Mutation
   C. Genetic drift

62. Which ancient organism are most likely responsible for the development of the ozone layer?
   A. Protists
   B. Sulfur bacteria
   C. Cyanobacteria
   D. Plants

63. Palaeontologist can estimate the absolute age of the fossil by measuring the concentration of
   A. Radioisotopes
   B. Stable isotopes
   C. Radioisotopes in the surrounding
   D. Stable isotopes in the area

64. Which of the following is incorrect with respect to mutation?
   A. Sudden
   B. Continuous
   C. Change in chromosomes and genes
   D. Leads to variation in DNA

65. Which of the following is not a type of mutation
   A. Gene mutation
   B. Chromosomal aberrations
   C. Genomatic aberrations
   D. Colourful mutations

66. Which of the following is a classic example of point mutation?
   A. Phenylketonuria
   B. Sickle cell anaemia
   C. Haemophilia
   D. Thalassemia

67. Which of the following is also known as the removal of one or more bases from the nucleotide chain?
   A. Deletion
   B. Insertion
   C. Transition
   D. Transversion

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68. What does the given type of mutation (aberrations) in the figure represent? PQRSTU → PQRSTUTU

A. Deletion
B. Duplications
C. Transition
D. Inversion

69. What is translocation?

A. Shifting of part of chromosomes to the outside wall
B. Shifting of part of chromosomes to another chromosome
C. Shifting of part of chromosomes to another homologous chromosome
D. Shifting of a part of chromosome to another non - homologous chromosome

70. On which of the following chromosomal mutation are based on?

A. Mutation in allele and their defective products
B. Imbalance in chromosome number and chromosome arrangement
C. Mutant allele and chromosome arrangement
D. Mutant allele and imbalance in chromosome number

71. Which of the following great apes is evolutionarily the closest relative to humans?

A. Banobo
B. Orangutan
C. Gorilla
D. Chimpanzee

72. Which of the following is the best source of fossil? (CEE, general science, 199 5E.C)

A. Volcanic rocks
B. Sedimentary rocks
C. Sand dunes
D. Forest floors

73. All forms of the cosmozoan (panspermia) theory suggest that.

A. Life arrived on earth from other outer spaces
B. Life forms arrived on earth from moon
C. Organic molecules arrived on earth by UFOs
D. Organic molecules from the earth is important in the origin of life earth

74. All of the following can cause frame - shift mutations, expect _____

A. Deletion
B. Addition
C. Substitution
D. Insertion

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75. The only source of new genetic material to act as a raw material for evolution is _____
A. Mutation
B. Recombination
C. Synapses
D. Segregation

76. Which of the following parings between genetic disorders and mutation that causes them is correct?
A. Sickle cell anemia - substitution
B. Down's syndrome - trisomy of chromosome - 21
C. Pau syndrome - trisomy of chromosome - 13
D. All are correct

77. Which of the following is chromosome mutation?
A. Aneuploidy
B. Translocation
C. Polyploidy
D. All

78. If the age of the fossil is 11,460 years, what was the ratio of C^{14} to C^{12} in the fossil
A. 1 : 4 B. 4 :1 C. 1 : 2 D. 2 : 1

79. The eyes of mole rat is lost, because the animal spent most of their time in the dark. This statement is supported by ____ theory of Lam ark
A. Theory of need
B. Theory of acquired characteristics
C. Use and disuse
D. Natural selection

80. The discovery of Lucy (a fossil) proves:
A. Bipedalism comes after large
B. Large brain comes after bipedalism
C. The common ancestor of human & chimpanzee looks like human
D. Ethiopia is a cradle of human being.

81. Among the following fossil (hominid species), which one is the oldest of all
A. Homo erectus
B. Ardipithecus ramidus
C. Australopithecus
D. Homo sapience

82. Which of the following reproductive isolation mechanisms keeps the

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horse and donkey as two independent species

A. Hybrid inviability
B. Hybrid infertility
C. Ecological isolation
D. Habitat isolation

83. Which of the following is prevented from taking place if populations are separated by a geographic barrier

A. Mutation
B. Evolution
C. Gene flow
D. Natural selection

84. Which of the following feature do analogous structures mostly share?

A. Phylogenetic similarity
B. Functional similarity
C. Ancestor-al similarity
D. Developmental similarity

85. If a substance that weighs 2,000 grams and has a half - life of 100 years is left with only 250 grams, for how long has the radioactive decrying activity been undergoing?

A. 200 years B. 250 years C. 300 years D. 500 years

86. Which of the following combination is known as Neo - Darwinism?

A. Natural selection and Mendelian law of inheritance
B. Natural selection and acquired inheritance
C. Acquired inheritance and Mendelian law of inheritance
D. Natural selection and Lamark's theory of evolution

87. Which of the following expression is more related to the phrase "survival of the fittest"?

A. Natural selection
B. Gene mutation
C. Mendelian inheritance
D. Use and dis use

88. The statement "life is an inherent property of the universe" is demonstrated by the evolution theory known as _____

A. Eternity
B. Cosmozoan
C. Spontaneous generation
D. Biochemical

89. In the process of evolution of life on earth, which of the fount processes evolved last?

A. Photosynthesis
B. Aerobic respiration
C. Chemosynthesis
D. Photo - autotrophs

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Answer key - unit Eight

1. D	10.B	19.A	28.B	37.C	46.B	55.B	64.B	73.A	82.B
2. C	11.A	20.D	29.B	38.D	47.B	56.A	65.D	74.C	83.C
3. A	12.A	21.D	30.C	39.C	48.A	57.B	66.B	75.A	84.B
4. C	13.D	22.B	31.C	40.B	49.B	58.D	67.A	76.D	85.C
5. A	14.D	23.C	32.C	41.A	50.B	59.C	68.B	77.D	86.A
6. C	15.C	24.B	33.B	42.B	51.C	60.A	69.D	78.A	87.A
7. C	16.B	25.C	34.B	43.C	52.A	61.A	70.B	79.C	88.A
8. D	17.D	26.B	35.B	44.A	53.A	62.C	71.D	80.B	89.B
9. D	18.B	27.C	36.C	45.C	54.B	63.A	72.B	81.B	

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UNIT 11

11. Human Body System (II)

What is body Coordination ?

Body coordination is the working together of the different system of an organism to produce appropriate reactions to the stimuli.

There are two main forms of coordination in humans. These are **nervous** and **hormonal coordination.**

- The nervous system uses nerve cells/neurons/ to pass electrical impulses along their length.
- The endocrine system produces chemicals are called **hormones** that are transported in the blood plasma to their target cells.
- The target cells have specific receptors on their cell - surface membranes and the change in concentration of hormones stimulates them.

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The comparison between nervous system and endocrine system:

Feature	Nervous system	Endocrine system
Speed of action	Very fast, takes only seconds	Slower, takes minutes to hours
Duration of action	Seconds to minutes	Minutes to days
Method of transmitting message	Electro - chemical	Chemical
Transport method	Neurons	Hormones
Actors	Brain and spinal cord are involved	Glands and organs are involved
Voluntary/involuntary	It is either under voluntary or involuntary control	It is under involuntary control
Response	Responses are localized	Responses are wide spread

11.1 The Nervous System

The nervous system is responsible for the reception and processing of sensory information from both external and internal environments

The nervous system has two major divisions. These are

i. The central nervous system (CNS) consists of the brain and spinal cord.

ii. The peripheral nervous system (PNS) consists:

- The sensory receptors, the nerves that link these receptors with the CNS.
- The nerves that link the CNS with effector (muscles and gland)

Function of the nervous system

- **It receives sensory input.**

Sensory receptors in skin and other organs response to external and internal stimuli by generating nerve signals that travel by way of PNS to the CNS.

- **It performs information processing and integration.**

Analysing or summing up the information it receives from all over the body.

- **It generates motor output.**

Nerve signals from the CNS go by way of the PNS to the muscles, glands

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and organs.

Neurons and their function

The **nervous system** is composed of only two principal types of cell These are: neuron and supporting cells.

i. Neurons

Neurons are the basic structural and functional units of the nervous syste

- They are specialized to respond to physical and chemical stimuli.
- They release chemical regulars.
- Conduct electrochemical impulses.

ii. Supporting cells (glial cells)

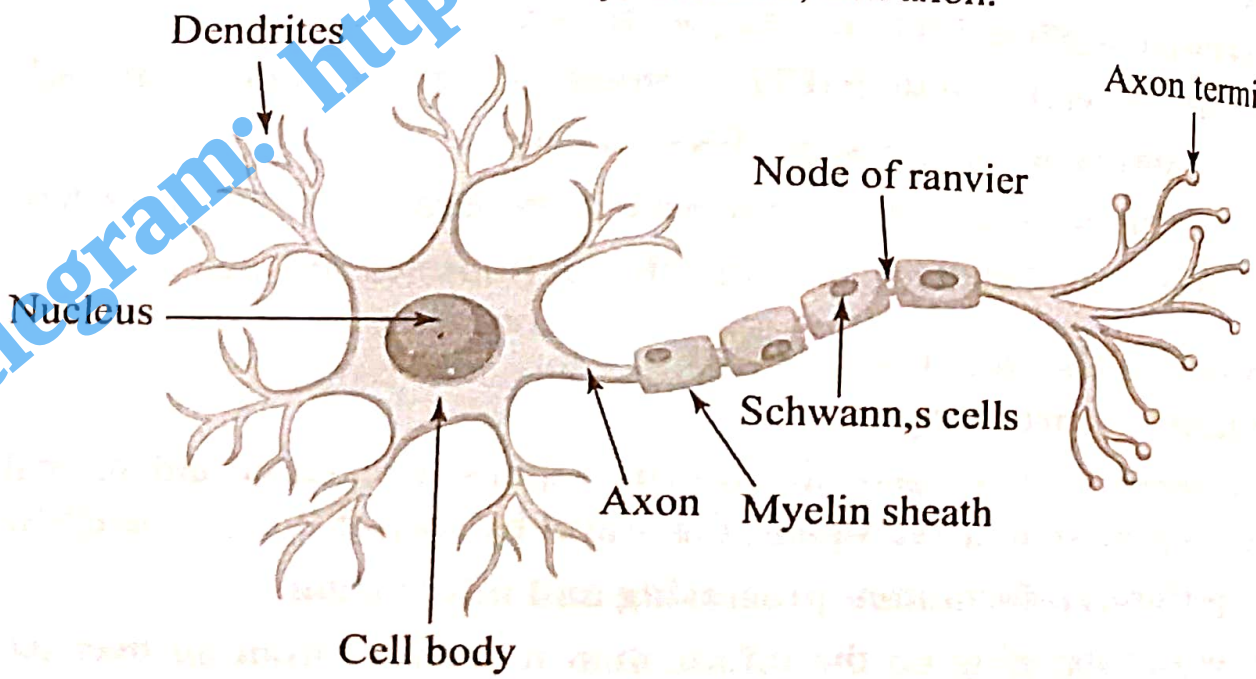
They are collectively known as neurological or glial cells (from the Middl Greek glial which means glue/attach)

- They aid the functions of neurons and are about five times more abundant than neurons.
- The providing support and nourishment to the neurons.

II . I . I Parts of Neurons

Neurons are the basic unit of communication in the nervous system.

All neurons consist of cell body, dendrite, and axon.



i. Cell body

Cell body is the compact section of a nerve cell that contain nucleus, cytoplasm and organelles

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- It is also known as a soma or the neutron's core/center
- Most cell bodies of neurons are concentrated in CNS and from Grey matter
- The cell bodies that are found in PNS are called **ganglia**
- The cell body contain genetic material, mitochondria that provide energy

ii. Axon:

An axon or nerve fibre is a long, slender projection of a nerve cell neuron, which conduct electrical impulse away from the cell body.

- Each neuron has only one axon that conducts information away from the cell body.
- It will also branch at its terminal into many axon terminals.
- It delivers the impulse to another nerve fibres.
- Individual axons are termed as nerve fibers and collectively they form a nerve.
- Axons are insulated with a fatty substances are called **myelin sheaths.**

iii. Dendrites

Dendrites are short extension of the cell body of a neuron, used to recieve electrical impulse from other neuron and transfer to the cell body.

Neutrons can have more than one set of dendrites known as dendritic trees. The number of dendritic tree neurons have generally depends on their roles.

iv. Myelin Sheath

It is fatty substance and insulating layer around the nerves, it allows electrical impulses to transmit quickly and efficiently along the nerve cells.

- The axon of peripheral never and large axons are covered by a myelin sheath.
- Within the peripheral nervous system, Schwann cells wrapped in layers around the neuron form the myelin sheath.
- The gap where there is no myelin sheath is called **node of ranvier.**
- Some nerve fibers are unmyelinated and this makes nerve impulse transmission significantly slower.

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11.1.2 The three Type of Neurons

Neurons vary in structure and function, based on the function of neurons they are grouped into three broad types.

i. Sensory (affecter or afferent) neurons

- They carry massage from sensory receptors to the central nervous system.
- They have typically a long dendrite and short axon
- They are triggered by the physical and chemical inputs from environment such as sound, touch, heat, and light /stimuli/

ii. Motor (effector or efferent) neurons :

- They transmit massage from the central nervous system to the effectors such as muscles or glands.
- They have a long axon and dendrites.
- They play role in voluntary and involuntary movements.
- They allow the brain and the spinal cord to communicate with muscles, organs and glands all over the body.

iii. Inter neurons (association neurons or relay):

- They are found exclusively in the central nervous system.
- It is the link between sensory and motor neurons
- They pass signals from sensory to motor neurons or integrate these function.

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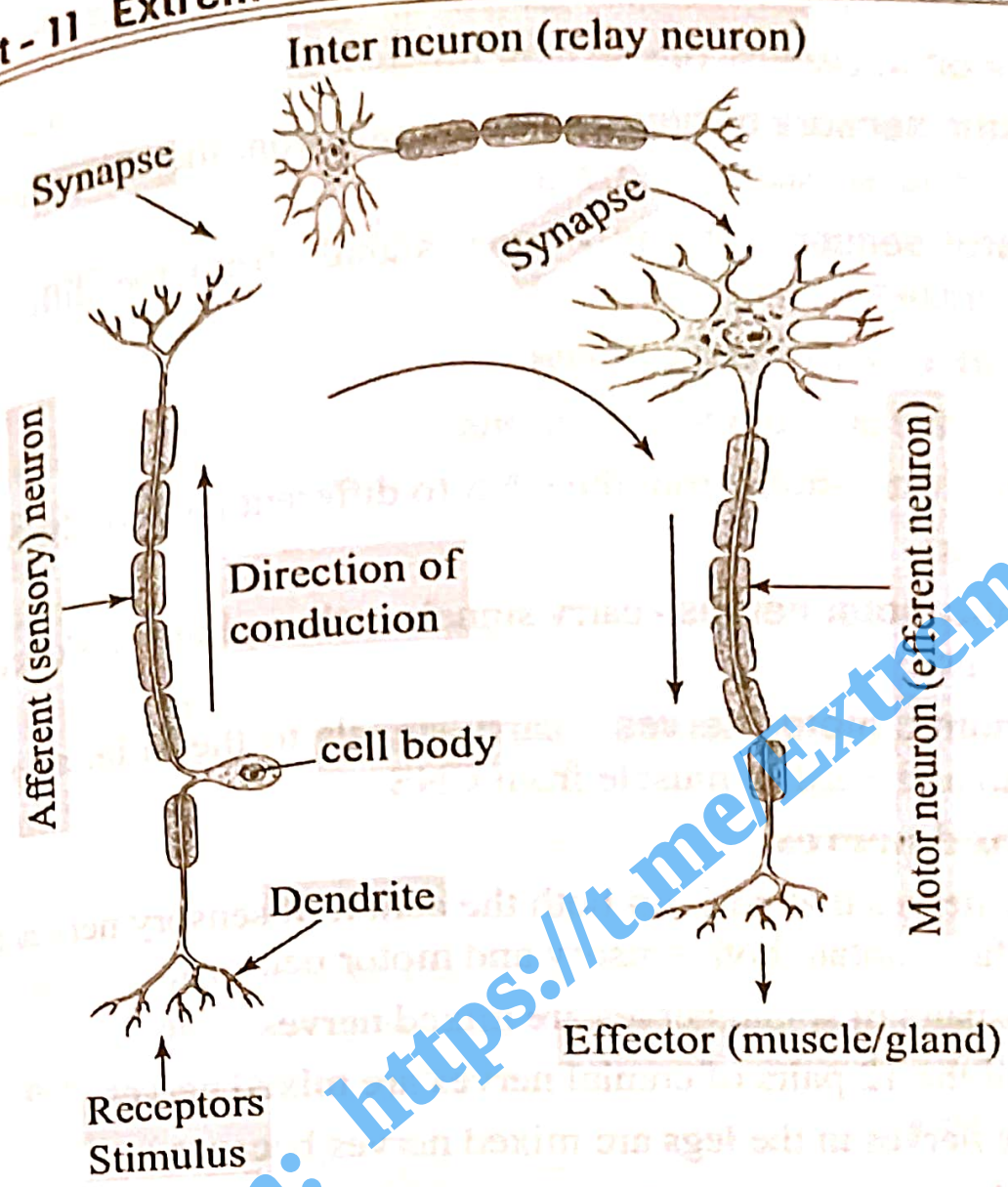

Summary of information flows through the nervous system

Stimulus → Reception by sensory receptor → transmission by afferent neuron → integration by **inter neurons in** CNS → transmission by efferent neuron → action by effector.

11.1.3 What is Nerve ?

It is a group of neurons or bundles of neurons with blood vessels and connective tissue.

Type of nerves

There are three major Types of nerves. These are:

I. Sensory or afferent nerves

They are made only of sensory neurons. The carry signals to the central neural nervous system.

The optic nerves for vision and olfactory nerves for smell etc, are examples

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of nerves with a purely sensory function.

Sensory or afferent nerves include:

- ☑ **Somatic sensory nerves** - carry signals from the skin, muscles, joints and Special senses to the CNS.
- ☑ **Visceral sensory nerves** - carry signals from the different internal body organs to CNS.

II. Motor or efferent nerves

They are made only of motor neurons.

They carry instructions from the CNS to different parts of the body.

It includes:

- ☑ **Somatic motor nerves** - carry signals to the voluntary, skeletal muscles from CNS.
- ☑ **Autonomic motor nerves** - carry signals to the different involuntary smooth and cardiac muscle from CNS.

III. Mixed nerves

These are nerves that perform both the action of sensory nerves and motor nerves. They contain both sensory and motor neurons.

- ➥ All 31 pairs of spinal nerves are mixed nerves
- ➥ Four of the 12 pairs of cranial nerves are mixed nerves
- ➥ Sciatic nerves in the legs are mixed nerves because :

They produce movement and sensation in the feet and legs.

What is nerve tract?

It is a bundle of axon connecting nuclei within the central nervous system.

It is also called white matter because the myelin sheaths of the neurons given a white color.

It is concerned with either sensory or motor activity. It includes:

- ☑ Ascending tract that extends from the spinal cord to the brain
- ☑ Descending tract extends from the brain to spinal cord.

Illustrative Example

1. What type of signals are used by the nervous system to communicate?

   A. Electrical and chemical B. Electrical and mechanical

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C. Mechanical and chemical D. Only chemical

Explanation: The nervous system communicates using Electrical and chemical signals.

Answer: A

2. Gaps in myelin sheath are called as _____

A. Synapse C. Nodes of ranvier

B. Axon D. Myelinoids

Explanation: Myelin is a fatty white substance that surrounds the axon of some nerve cells. It forms electrically insulating layer, it is essential for the proper functioning of the nervous system. Gaps in myelin sheath are called as Nodes of ranvier.

Answer: C

3. Suppose a person holds a very hot object and does not feel a burning pain, on his had. Which of the following neuron are probably damaged in his hand?

A. Association C. Sensory

B. Motor D. Inter neuron

Explanation: Sensory neurons receive an impulse from sensory cells or receptors. Thus, the burning sensation is not felt on the hand, probably due to a damage of sensory neurons.

Answer: C

4. The nucleus of neuron is located in the:

A. Synapse C. Cell body

B. Motor neurons D. Axon

Explanation: nucleus and mitochondria are located in the cell body of neuron.

Answer: C

5. Which of the following structures are effectors?

A. Muscles C. Synapses

B. Motor neurons D. Sensory neurons

Explanation: Effector are structures that receive an impulse from efferent/motor neurons. These are muscles and glands.

Answer: A

6. Which of the following systems are responsible for the regulation and

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coordination of body activities?

A. Nervous and muscular systems

B. Respiratory and circulatory systems

C. Nervous and endocrine systems

D. Nervous and circulatory systems

Explanation: The coordinating system of the body are nervous system and endocrine system. Both systems work differently towards coordinating body activities.

- Nervous coordination uses electrochemical impulses.
- Endocrine system uses hormones/ chemical messengers.

Answer: C

7. Cell bodies of motor neurons is located in the:

A. The white matter of spiral cord

B. Grey matter of spinal cord

C. Dorsal root of spinal nerves

D. Ventral root of spinal nerves

Explanation: The cell bodies of efferent (sensory) neurons is located in the spinal ganglia of the dorsal root of the central nervous system. The cell bodies of efferent (motor) neurons is located in the grey matter of the spinal cord.

Answer: B

8. The central nervous system consists of:

A. Brain and spinal cord

B. Anterior and posterior ends of the brain

C. Spinal cord and spinal nerves

D. Cranial and spinal nerves

Explanation: The central nervous system (CNS) consists of the brain and spinal cord.

Answer: A

9. After information is processed in the central nervous system, instructions are sent to the body by specific neurons, which are known as:

A. Affecter neurons

B. Afferent neurons

C. Efferent neurons

D. Sensory neurons

Explanation: Afferent neurons - carry impulse from the sensory

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receptor to the CNS.
- Efferent neurons - carry instructions from CNS to effectors
- Effectors - are glands and muscles of the body that receive instructions form the CNS.

Answer: C

10. Fatty insulting layer of the axon membrane
A. Synaptic cleft
B. Axon membrane
C. Myelin sheath
D. Fatty acid

Explanation: Mylen sheath is the outer faty layer covering the axon.

Answer: C

11. Afferent and effect neurons are connected by:
A. Relay neuron
B. Sensory neuron
C. Afferent neurons
D. Motor neuron

Explanation: relay or association neuron s is located in the CNS, which is a bridge between sensory and motor neuron.

Answer: A

12. The types of nerve that perform the action of both motor and sensory neuron is
A. Afferent nerve
B. Efferent nerve
C. Mixed nerve
D. Nerve tract

Explanation: mixed nerves are nerves that perform both the action of afferent neuron.

Answer: C

13. The dendrites of a neuron can carry an impulse only _____
A. away from cell body
B. toward the brain
C. toward the cell body
D. toward the brain

Explanation :-
- Dendrites brings nerve impulse to the cell body, but **axon** takes impulses away from the cell body.

Answer : C

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11.1.4 The Nerve Impulse and Transmission

The nervous system relies on nerve impulses travelling along a neuron.

Nerve impulse a minute is electrical event that works as a result of charge difference across the membrane of the axon.

Resting potential and polarization

Resting potential is an electrical potential difference across the membrane when the cell is non - excited state

- A neuron that is not conducing electrical signal, usually at about - 65mV
- When an axon is resting, its membrane is **polarized.**

This indicates that the inside of the axon axoplasm/ cytoplasm of axon is more negative (more K^+ ions and less Na^+ ions) than the outside. The interstitial fluid /outside/ (less K^+ ions and more Na^+ ions).

Which means the outside is positive and the inside is negative charge.

- Na^+ and K^+ are transported across the membrane against their concentration gradient by carrier protein which is called Na - K pump and energy is used through ATP.

Action Potential, depolarization and re polarization

Action potential is a rapid sequence of changes of charge (positive and negative) across a membrane of nerve cell.

- A rapid change in polarity across a portion of an axonal membrane as the nerve impulse occur
- A short term change in the electrical potential on the surface of a cell when it is stimulated
- It uses gated channel protein in axon membrane to exchange Na^+ and K^+ ions

When the Na^+ voltage gates open, Na^+ moves into the cytoplasm of the axon, then axon is depolarized.

When an action potential occurs the axon is said to be **depolarized.**

The depolarized membrane stimulates the adjacent voltage channel so the action potential passes as a wave along the length of neuron.

- Then the neuron electrical potential changes from -65mv to + 40 Mv

As soon as depolarization takes place the neuron membrane permeable to K^+ in which rushes out of the cell. This restores the positive charge outside

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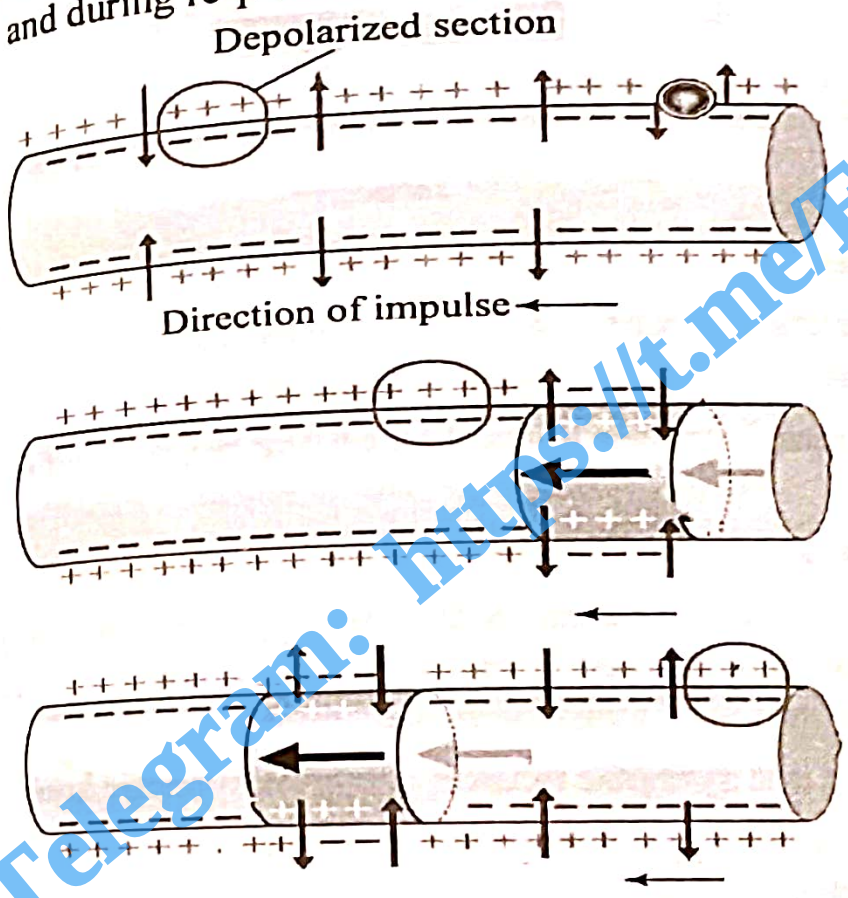
and the negative charge inside and the process is called **re polarization.**

- Then action potential changes from +40mv to - 65m V, re-polarization occur.
- In myelinated axons, the voltage gated ion channel are concentrated at the nodes of ranvier, therefore, the action potential jumps from node, to node at a speed of 200m/sec, this is called **salutatory conduction.**
- In non myelinated axons the impulse travels at 1m/sec.

Note: Threshold potential is a minimum voltage potential (depolarization) required for the initiation or generating an action potential, During depolarization, Na^+ ions move to the inside of the axon and during re-polarization, K+ ions move to the outside.



What is refractory period?

It is the time interval during which a nerve cell is incapable of responding to a stimulus.

- It is the time taken for a neuron to recover after an action potential.
- It is a very short period of time when another action potential cannot be generated.

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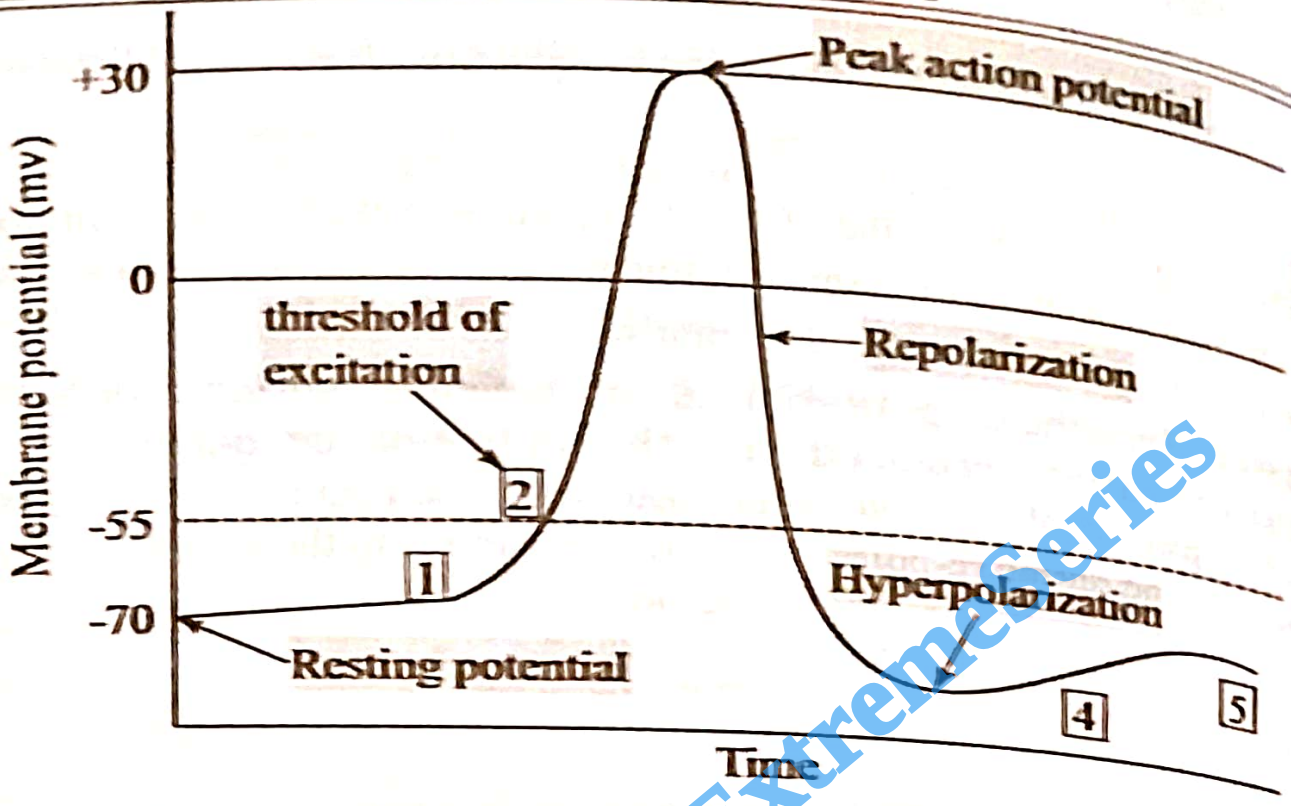

What is Synapse ?

Synapse are the junctions between neurons. Neurons do not touch each other. There is always a very small gap called the synaptic cleft.

Synapse is open space between the axons of one neuron and dendrite of the next neuron

Pre synaptic neuron is a type of neuron carries impulse towards synapse.

The membrane of the first neuron is the **pre synaptic membrane** and the membrane of the next neuron is called the **post synaptic membrane**, which transmit impulse away from synapse.

The axon terminal of pre synaptic neuron contains synaptic knob.

- ☑ Synaptic knob contains synaptic vesicles which store neurotransmitter made by presnaptic neuron.
- ☑ The synaptic knob comes close to the postsyncaptic neuron but does not touch it.
- ☑ The gap between presynaptic and postsynaptic neuron is the Synaptic cleft.

Chemical transmission across a synaptic cleft

It is carried out by means of neurotensmitters found in synaptic vesicle

- ➥ Nerve impulse reaches the axon terminal at synaptic knob.
- ➥ Gated chemical for Ca^{+} ion open and Ca^{+} enter the terminal.

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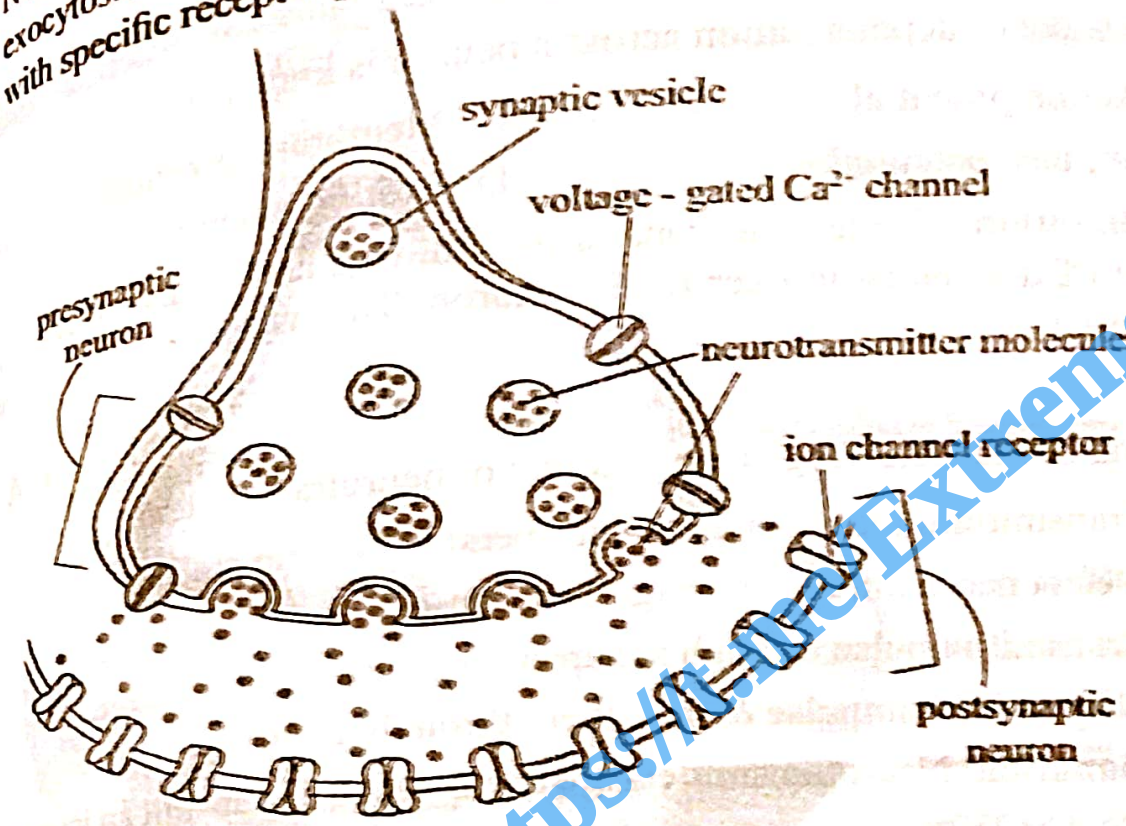
- The rise in Ca^{+} stimulates synaptic vessel to merge with the pre synaptic membrane to release neurotransmitter
- Neurotransmitter molecules are released into synaptic cleft by exocytosis and diffuse into post synaptic membrane where they bind with specific receptor proteins



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What are Neurotransmitter?

Neurotransmitters are chemical messengers in the body. The function is to transmit signals between nerve cells and target cells. These signals help to regulate various body function.

They are found in both the central nervous system (CNS) and the peripheral nervous system (PNS).

Neurotransmitters is excitatory or inhibitory depends on the receptor it binds.

Excitatory transmitter promotes the generation of an action potential in the receiving neuron, whereas an inhibitory transmitter prevents generation of an action potential.

For example glutamate is excitatory transmitter and Gamma amino butyricis inhibitory transmitter.

Acetylcholine /Ach/ is an abundant neurotransmitter in the human body. After the effect of neurotransmitters, they are digested by enzymes in the post synaptic membrane or retaken up by precipitant membrane.

Illustrative Example

14. The wave of depolarization across a neuron is known as:

A. Action potential
B. Impulse potential
C. Membrane potential
D. Resting potential

Explanation: While a neurons is transmitting an impulse, there is a wave of depolarization across the neuron. This wave is called action potential.

Answer: A

15. Which of the following is the function of neurotransmitters?

A. Transmit impulse through axon fibers
B. Detect nerve impulse through synapse
C. Transmit impulse through synapse
D. Detect nerve impulse & pass them through dendrite

Explanation: Neurotransmitters are chemicals that transmit an impulse across a synapse.

Answer: C

16. Voltage regulation means _____ ?

A. Na^+/K^+ pumps controls potential
B. Unless the voltage is regular gates will not respond
C. Oscilloscope only can control potential
D. With change in the membrane potential voltage gated ion channels open and close

Explanation: With changes in the membrane potential voltage gated ion channels open and close, this is regulation of voltage.

Answer: D

17. Depolarization and re polarization changes that occur during the action potential are produced by _____

A. Negative feedback loops
B. Simple diffusion of ions down their concentration gradient

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C. Carrier mediated transport
D. Active transport pumps along the neuron membrane

Explanation: Na^+ channels open at the beginning of action potential and moves the axon causing depolarization, when K^+ channels open re-polarization takes place out of the axon.

Answer: D

18. Diffusion of K^+ out of the cell makes the inside of it less positive, or more negative and acts to restore the original resting membrane potential. This process is called ____

A. Repolarization
B. Depolarization
C. Hyperpolarization
D. Overshoot

Explanation: Diffusion of K^+ out of the cell makes the inside of it less positive, or negative and acts to restore the original resting membrane potential. This process is called is called repolarization.

Answer: A

19. Which of the following statements about action potential is false?

A. Na^+/K^+ pumps are directly involved in creating the action potential
B. Na^+ and K^+ concentrations are not significantly change during an action potential
C. This includes both positive and negative feedback loops
D. Only a relative small number of Na^+ and K^+ actually diffuse across the membrane

Explanation: The concentration of K^+ is higher inside and concentration of Na+ more outside.

Answer: B

20. Which of the following does not act as a neurotransmitter?

A. Acetycholine
B. Epinephirne
C. Norepinephrine
D. Cortisone

Explanation: Cortisone is steroid hormone. It is released by the adrenal gland in response to the stress. Cortisone is not a neurotransmitter.

Answer: D

21. Transmission of the nurve impulse across a synapse is accomplished by the :

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A. movement of Na^+ and K^+

B. release of a neurotransmitter by a dendrite

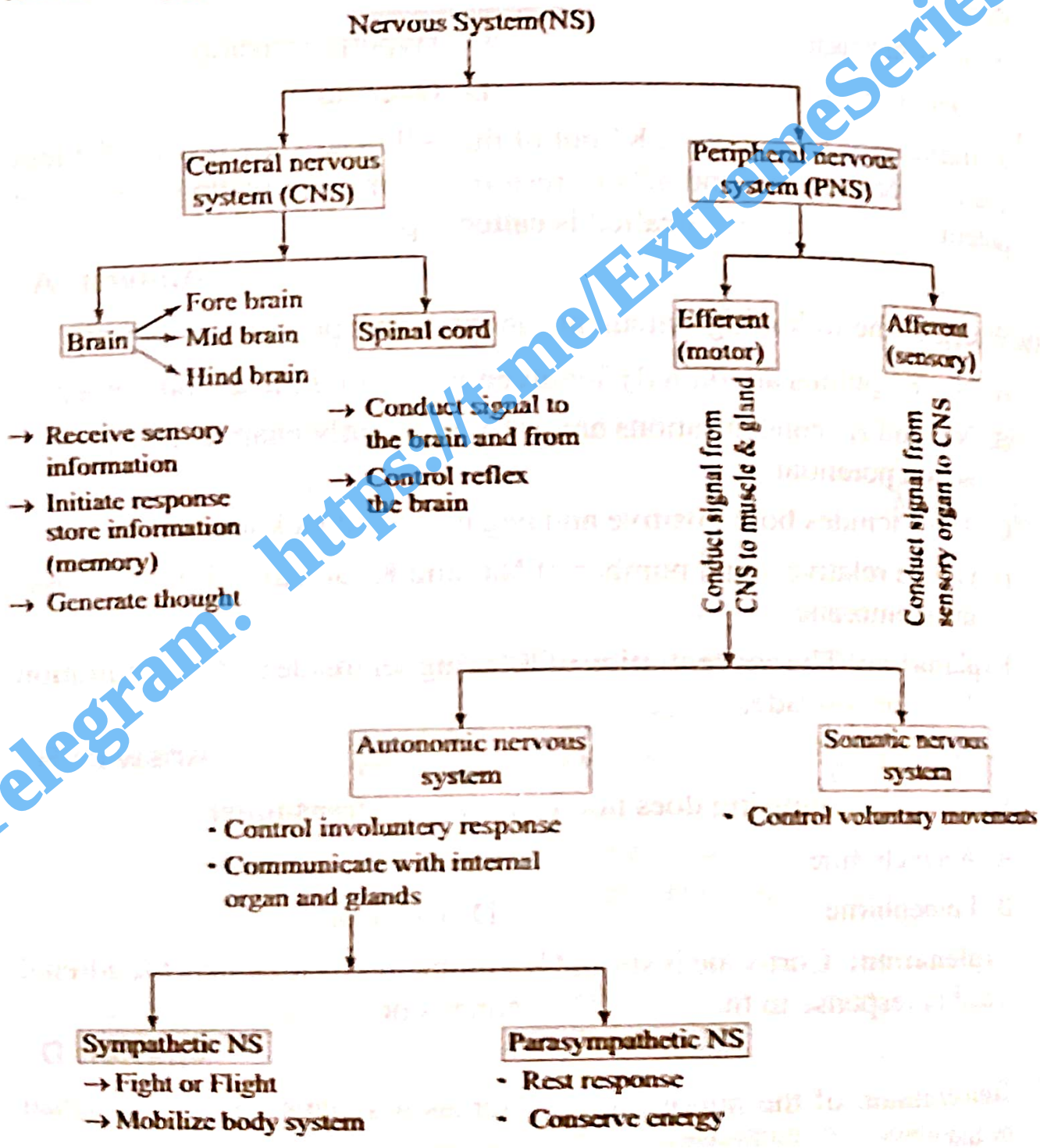
C. release of a neurotransmitter by an axon terminal

D. release of a neurotransmitter by a cell body

Explanation :- A nerve impulse causes the release of neurotransmitter by synaptic knob in the axon terminal into synaptic cleft.

Answer : C

11.2 Types of Human Nervous System



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Human nervous system has two main parts:

1. The central nervous system (CNS) which is made up of brain and spinal cord.

2. Peripheral nervous system (PNS) which consists of the nerves that branch out from the brain and spinal cord.

The peripheral nervous system (PNS) is divided into the somatic and automatic nervous systems. The somatic nervous system is responsible for controlling voluntary action such as walking and talking, while the autonomic nervous system keep the involuntary body function such as heart beat, breathing, and blook pressure.

The Human Brain (CNS)

It is delicate organ, consists of millions of nerve cells, which control the whole activities of the body.

Major structures that help to protect the brain are skull, meninges, and cerebrospinalfluid .

The brain is enclosed by hard bony structure is called **cranium or skull.**

Three membranes are called **meninges** the lie between the bony conversing and the nervous tissues.

The meninges consist of three connective tissue, layers, listed outside to inside: dura mater, arachniod mater and pia mater layer

i. Dura mater

- It is the outermost layer of the meninges
- It is lies closest to the bone of the skull.
- It consists of two layer of tough and fibrous connective tissue.
- Periosteal layer - lines the inner surface of the bone of the cranium
- Meningeal layer that lies closest to the brain, continuous with the dura mater.

ii. Arachniod mater

- It is the middle layer of the meneges, lines next to the dure mater.
- Between the dura mater and the arachnoid mater there is a space called the subdural spaces that contains serous fluid
- The space blow the arachnoid mater and above the pia is called **sub arachnoid.**

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- The subarachnoid space contains CFS/ cerebrospinal fluid/ and is also home to some of the larger blood vessels serving the brain.

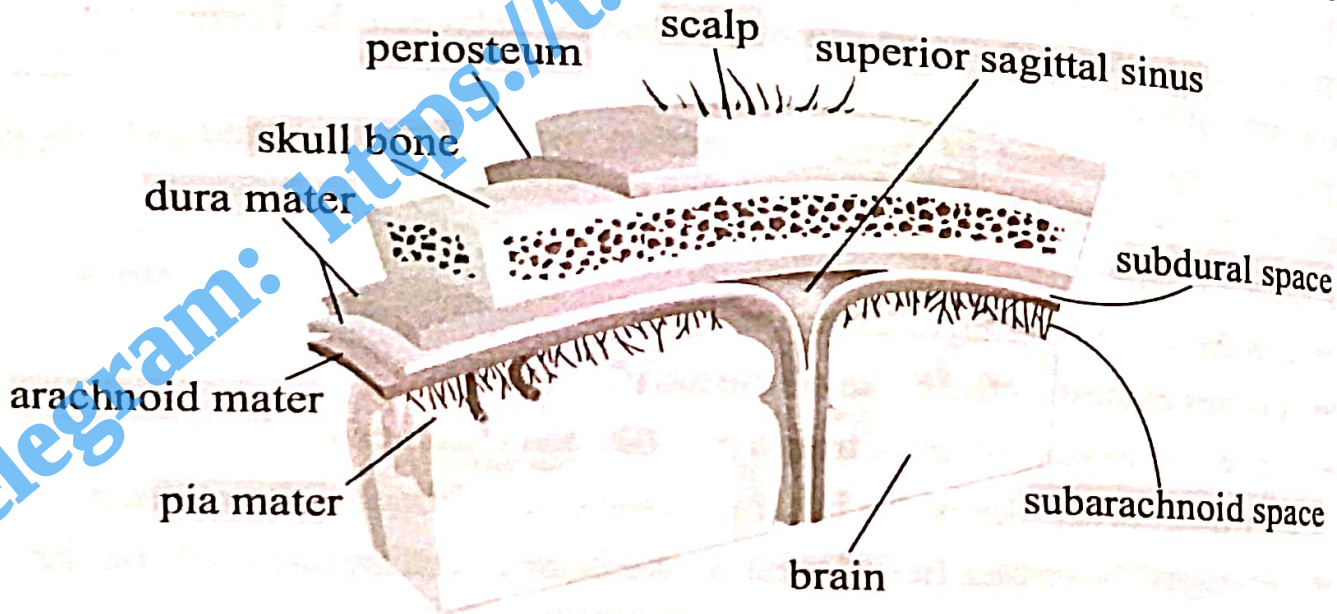
iii. Pia mater

- It is innermost layer, located under subarchnoid space
- Tightly adhered to the surface of brain.
- It contain many tiny blood vessels that serve the brain

Cerebrospinal fluid

It is a thin fluid similar to plasma and has several important functions:

- Acting as cushions, and shock - absorber.
- It supports the weight of the brain
- It protects the brain from damage
- It helps to maintain a uniform pressure around the brain.
- It helps the exchange of nutrients and waste products between neurons and CSE. /correction CSF/
- The following diagram shows that the different structures used to protect the brain.



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11.2.1 Parts of the Brain

The human brain is divided into three regions: **fore brain, midbrain** and **hind brain**

- The fore brain gives to the thalamus, hypothalamus, and cerebrum
- The hind brain develops into the medulla, pons, and cerebellum

- Mid brain acts as a relay station for tracts passing between the cerebrum and spinal cord or cerebellum

i. **Hind brain** : has three distict structure : Medulla, pons & cerebellum

☑ **Medulla:** it is located in Continuous with spinal cord below pons and above spinal cord.

It has the following functions:

- Contains vital centers (clusters of neuron cell bodies) that control heartbeat, respiration, and blood pressure.
- Contains centers that control swallowing, coughing, vomiting, hiccups, coughing sneezing.
- Contain vasomotor center to regulate blood pressure

☑ **Pons:-** It acts as **relay center** between parts of the brain and the spinal cord - thus important in **integration.**

functions of pons are :-

- Contains bundles of axons travelling between the cerebellum and the rest of CNS.
- Connects various parts of brain with one another; contains respiratory center and sleep center.

☑ **The Cerebellum** is mainly concerned with learned skills, such as balance, posture, movements.

ii. **Fore brain**: consist of thalamus hypothalamus and cerebrum

Thalamus:- it is located in at the top of brain stem.

It has the following functions :

- Receive information via nerves from various parts of body and send it the appropriate part of the cerebrum for integration.
- Main sensory relay center for conducting information between spinal cord and cerebrum.

Hypothalamus:- it is located just below thalamus, pituitary gland is connected to hypothalamus by stalk of neural tissue.

functions of hypothalamus are :-

- Help to maintain homoeostasis by regulating body temperature, water balance, hunger, thrust, appetite, and fat metabolism:

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- Regulates pituitary gland
- Important in emotional and sexual responses, and in sleep - wake cycle

Cerebrum:- it is the largest, most prominent part of human brain. Cerebrum is divided into right and left hemispheres, each divided into lobes:

- → **Frontal Lobe:-** it is the Center of motor control,memory, reasoning, intelligence, imagination, thinking, judgement. Initiate voluntary muscle movement
- → **Parietal lobe:-** it is the Center of taste, speech, somatosensory integration
- → **Temporal Lobe:-** it is the Center of hearing and balance
- → **Occipital lobes:-** it is the Center of Sight (Vision)

Major function of cerbrum is Center of intellect, memory, consciousness, and language: also sensation and motor functions.

Cerebrum has two parts; outer gray matter and white matter

Cerebral cortex (outer gray matter):- consists of the cell bodies of neuron, which is protected by three meninges:

- → Dura mater (outer),
- → Pia mater (inner),
- → Arachnoid (middle)

Functionally, cerebral cortex is divided into three areas :

- ☑ **Motor area:-** Control movement of voluntary muscles.
- ☑ **Sensory area:-** Receive incoming information from eyes, ears, pressure and receptors, etc.
- ☑ **Association area:-** Sites of intellect, memory, language, and emotion: interpret incoming sensory information.

Cerebra cortex (White matter):- It Consists of myelinated axons of neurons that connects various regions of brain. The axons are arranged into bundles (tracts)

White matter Connects the following:

- Neurons within same hemisphere of the brain
- Right and left hemisphere
- Cerebrum with other parts of brain and spinal cord

iii. **Mid brain** - It is location in the middle part of the brain just above the

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pons.
- It is the center of visual and auditory reflexes (blinking, pupil reflex, adjusting ear volume)

11.2.2 The Spinal cord

Spinal cord is elongated cylindrical structure, composed of nervous tissue. It extends from medulla obligation to the second lumbar vertebrae of the back bone. The spinal cord has a much simpler structure to the brain.

The spinal cord is encased and protected by:

- The vertebral column.
- Cerebrospinal fluid
- Meninges: are membranous around the spinal cord

The nerves that come out of the spinal cord are known as the spinal nerves.

- Spinal nerves are nerves stretch to the arms, legs, trunk/ head and neck/ and to the rest of the body.

The spinal nerve is divided in to the two roots. These are: dorsal and ventral root.

i. Dorsal root contains only sensory neurons and carry information towards the spinal cord.

ii. Ventral root contains only motor neurons and carry impulse from the spinal cord.

Spinal cord consists of grey matter and white matter

- The grey matter is located in the motor inner part whereas the white matter is found on the outside part.

Functions of the spinal cord.

- Transmission of impulse between brain and the rest of the body.
- It coordinates many reflex actions

The peripheral Nervous System (PNS)

PNS contains all the nerves that lie outside of the nervous system. It consists of 31 Pairs of cranial 12 pairs of spinal nerves

- Cranial nerves originate in the brain and mostly terminate in organs of the head and upper body
- Spinal nerves originate in the spinal cord and extend to parts of the

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body below the head

It is subdivided into the afferent & efferent division of the PNS

i. The afferent division of the PNS

- ➥ It carries signals to the PNS
- ➥ It includes all the neurons that transmit sensory information from their receptors.

ii. The efferent division of the PNS

- ➥ It carries signal from the CNS to the muscle and glands to bring the desired responses.

The efferent division of the PNS is further divided in to:

Somatic and autonomic nervous system

☑ The somatic (motor) Nervous System

It carries signal from CNS to skeletal muscle, who is voluntary.

☑ The Autonomic Nervous System

It is part of PNS that is responsible for regulating involuntary body functions, such as blood flow, heartbeat, digestion, and breathing.

The autonomic nervous system is further divided into two. These are sympathetic and parasympathetic

Sympathetic and parasympathetic divisions have antagonistic (opposite) effect on target organs.

Sympathetic Nervous System

The sympathetic division is active during time of stress.

It prepare the body for the flight - or - fight responses. It prepares the body to expenditure energy to respond to environmental threat. When action is needed it triggers a response by accelerating:

- ➥ The heart rate
- ➥ Breathing rate
- ➥ Boosting the blood flow to muscles
- ➥ Activating sweat secretion
- ➥ Dilating the pupils

Parasympathetic Nervous System

It helps to maintain the normal body functions and conserve physical resources. It allows returning the bodies to a normal resting state. Once a threat is recognized, this system will:

- ➥ Slow the heart rate
- ➥ Slow breathing

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➥ Reduce blood flow to muscles ➥ Constrict the pupils

What is a Reflex action ?

Reflex actions are rapid responses to internal or external stimuli which allow the body to maintain homeostasis. They may be **Somatic** or autonomic but all involve the sequence

Spinal cord is the control center for reflex action.

Receptor ⟶ Sensory neuron ⟶ CNS ⟶ Motor neuron ⟶ Effector

➥ When the body is in danger, it can responded to the situation without conscious thought.

➥ They are usually involved to avoid danger or damage.

➥ Reflex arc is the pathway that nerve impulses travel when a reflex is elicited or induces.

There are five essential parts in reflex arc. These are:

1. **Receptors** - They detects a change (the stimulus) and generates impulse
2. **Sensory neurons** - They transmit impulse from receptors to the CNS
   ➥ They enter the spinal cord through the dorsal root.
3. **Central nervous system**
   It contains spinal cord, one or more synapses and inter neuron.
4. **Motor neurons**
   They transmit impulses from the CNS to the effector .
   ➥ They leave the spinal cord by the ventral root.
5. **Effectors**
   They perform their characteristic actions by muscle glands.

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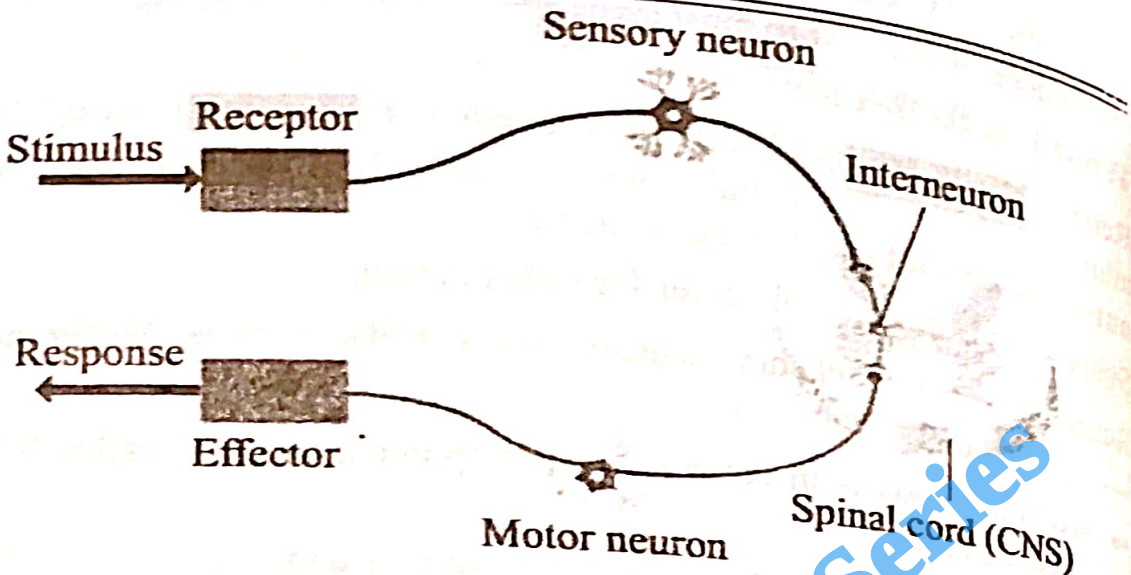

Illustrative Example

22. _____ response is the function of sympathetic nervous system.

A. Rest and restore
B. Fight or flight
C. Decrease respiratory rate
D. Increase respiratory rate

Explanation: Sympathetic nervous system "fight or flight" response.

Answer: B

23. Brain is situated in _____

A. Benign
B. Polonium
C. Meninges
D. Cranium

Explanation: The skull is composed of 2 part: The cranium and the *mandible*. Brain is situated in cranium and it protects the brain.

Answer: D

24. Hypothalamus is the basal part of which of the following

A. Fore brain
B. Mid brain
C. Hind brain
D. Mesencephalon

Explanation: The hypothalamus is the basal part of the diencephalon, fore brain ans it regulates a wide spectrum of body functions. It contains several groups of system is neurosecretory cells called nuclei which produce hormones.

Answer: A

25. Sympathetic nervous system is associated with _____

A. Fight and flight
B. Fear and rage

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C. Hormonal secretions D. Skeletal muscles

Explanation: Sympathetic nervous system is a part of the autonomic nervous system. It activates fight or flight response.

Answer: A

26. Somatic nervous system is made up of _____

A. Relay neurons C. Sensory neurons

B. Associate neurons D. Motor neurons

Explanation: Somatic nervous system is a part of the peripheral nervous system. The Somatic nervous system consist of afferent nerves and efferent nerves.

Answer: D

27. _____ controls the voluntary functions of the body.

A. PNS B. CNS C. Neurons D. ANS

Explanation: Peripheral nervous system controls the voluntary functions of the body, it extends between CNS and body parts

Answer: A

28. Which part of the brain is concerned with the regulation of body temperatures?

A. Medulla oblongata C. Cerebrum

B. Cerebellum D. Hypothalamus

Explanation: Hypothalamus contains small nuclei with a variety of functions. It links nervous system to the endocrine system via pituitary gland.

Answer: D

29. The decoding and interpretation of visual information are carried out by _____

A. Parietal lobe C. Occipital lobe

B. Frontal lobe D. Temporal lobe

Explanation: Occipital lobe is a visual processing centre of brain containing most of the anatomical region of the visual cortex.

Answer: C

30. Describe the major parts of Hind brain.

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Explanation: Hind brain contains three major parts. These are pons, cerebellum and medulla oblongata.

31. Which function is associated with pons?

A. Sleep, arousal and the relay of the information
B. Emotion memory and audition
C. Color perception, depth perception
D. Pain perception vision and audition

Explanation: Pons is the major part of the brain. Is involved in the control of breathing and communication between different parts of the brain.

Answer: A

32. Which of the following is the largest part of the brain?

A. Thalmus
B. Cerebelleum
C. Cerebrum
D. Hypothalamus

Explanation: Cerebrum is the largest parts of the brain. It consists of cerebral cortex and sub cortical structures. It is the uppermost region of the central nervous system.

Answer: C

33. Which part of the brain is present on the top portion of medulla in hindbrain?

A. Pons
B. Cortex
C. Frontal lobe
D. Parietal lobe

Explanation: Pons is present on the top portion of medulla in the hindbrain. It helps in breathing control

Answer: A

Conceptual Example

34. A women has a stroke and as a result is unable to move the left side of her body properly. Where did the stroke probably occur?

Explanation :- The left side of the body is controlled by the right motor area, and vice versa.

35. When we are worried about something such as exam results then which parts of the brain are involved ?

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A. Hind brain
B. Fore brain
C. Mid brain
D. Medulla

Explanation :- The fore brain interprets this as stress and brings about the release of adrenaline.

Answer: B

36. An inter neuron can relay information
A. from a sensory neuron to a motor neuron
B. only from motor neuron to another motor neuron
C. only from a sensory neuron to another sensory neuron
D. from neuron to another neuron.

Explanation :- Sensory neuron ⟶ Inter neuron ⟶ Motor neuron

Answer : A

37. A person under the increasing influence of alcohol doses, first his ability to think clearly; second his ability to walk straight ; third his ability to breath normally. Based on this evidence which is the correct order in which alcohol affects the central nervous system ?
A. Cerebellum ⟶ Medulla ⟶ Cerebrum
B. Medulla ⟶ Cerebellum ⟶ Cerebellum
C. Cerebrum ⟶ Cerebellum ⟶ Medulla
D. Cerebellum ⟶ Cerebrum ⟶ Medulla

Answer : C

38. Speech and intelligence are associated with the
A. Cerebral hemispher
B. Cerebellum
C. Medulla oblongata
D. Hypothalumus

Explanation :- The role of cerebrum is reasoning, learning and memory, language, and speech

Answer : A

39. The cell bodies of motor neurones and association neurones are located in the :
A. Spinal ganglia
B. Grey matter of spinal cord

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C. White matter of spinal cord

D. Dorsal root of spinal ganglia

Explanation :- Grey matter is largely composed of cell bodies of motor neurones and associate neurones

Answer : B

Drug abuse

Drug (or substance) abuse is when a person uses substance in the point of excess and/ or dependence. A drug is a chemical that alters body structures or biological function. Many drugs that affect the nervous system by interfering with or promoting the action of neurotransmitters.

Drugs can prevent, treat, or cure many illnesses. However, drugs, whether legal or illegal, can also be misused or abused.

- Legal drug are used for the mild pleasure they bring, to be sociable.

Psychoactive drugs alter the functioning of the CNS. Many psychoactive drugs produce physiological dependence and addiction when abused.

Drug dependence is when a person use a drug again and again and become addicted.

Addition is marked by a change in behaviour caused by the biochemical changes in the brain after continued use of substance.

Abuse of drugs can damage the body, in some cases, can result in death.

Several classes of psychoactive drugs are commonly abused:

A. **Depressants**, such as alcohol and barbiturates, generally decrease the activity of the central nervous system.

- Hight doses of depressants such as alcohol can depress respiratory functions enough to cause death.

B. **Stimulants**, such as cocaine and nicotine, generally increase the activity of the central nervous system.

- At high doses, stimulates can be fatal.
- Nicotine, which is found in tobacco products, increases heart and bathing rate, blood pressure, and overall activity level.

C. **Inhalants** such as some glues are also psychoactive drugs.

- Inhalants cause disorientation, confusion, and memory loss.
- They can also cause brain and other organ damage as well as respiratory failure.

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Effects of using drugs

☑ Lysergenicacid diethyl amide (LSD) and similar drugs cause sensory distortion, vivid waking called **hallucinations. For example**

- THC, found in cannabis (marijuana) and hashish, causes memory and judgment problems. The most potent is delta - 9 - tetrahydrocannabinoid
- Narcotics, such as heroin, have sedative effects that can be fatal at high doses.

☑ **Cigarette smoke contains nicotine, the highly addictive and very toxic stimulant found in the leaves of the tobacco plant, Nicotiana tabacum.**

- In the brain, nicotine mimics the action of an important neurotransmitter by binding to its receptors.
- Smoking causes permanent changes in the brain.
- After a while, the smoker is addicted.

☑ **Alcohol (ethanol) is a widely used and frequently abused psychoactive drug.**

- Alcohols a depressant that produces a sense of well - being when taken in small amounts.
- As more alcohol is consumed, reaction time increases, Coordination, judgment, and Speech become impaired.
- Alcohol consumption produces a state of intoxication known as being "drunk". Drunkenness occurs as the blood - alcohol affects on the normal functioning of the brain.
- Alcohol addiction is chronic and relapsing disorder associated with compulsive alcohol drinking.
- Alcohol use disorder is impaired ability to stop alcohol despite social, occupational, or health consequences.

☑ **Catha edulis (Khat)**

- The leaves of Khat are chewed by people for it is stimulant.
- Its young buds and tender leaves are chewed to attain a state of euphoria and stimulation.
- Khat contains a drug that affects the brain known as **cathinone**
- The drug is absorbed into blood stream through the membrane of the

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mouth and the stomach

- It acts quickly within 30 minutes of time before it is broken down and removed in the liver.
- Addicted people feel depressed, tired and unable to concentrate without getting the drug
- The plant has a stimulant effect similar to amphetamine, cocaine or very strong coffee depending on how much it is consumed.
- Khat is a plant grown commonly in the **horn of Africa.**

11.3 Sense organs

Sense organs are body parts that receive stimuli in such a way to excite nerve cells to send information to the brain. They are organs having nerve endings and sensory receptors to receive the external stimuli. There are five different sense organs: the skin, eye, tongue, nose and the ear.

11.3.1 The Skin

Skin is the layer of usually soft, flexible outer tissue covering the body of vertebrate animal. It is one of the largest organs in the human body in surface area and weight.

The skin contain a huge variety of sensory receptors. These are

1. Pressure receptors /Pacinian corpuscle/
2. Temperature receptors /thermo receptors/
3. Touch receptors /Meissner's corpuscles/
4. Pain receptors /Nociceptors/ etc.

Functions of the skin: protection, regulation and sensation

- It protects against the loss of water by evaporation
- It prevents gaining water by osmosis while swimming or washing
- It protects the body from the entry of bacteria and other pathogens.
- It protects the body from damage by UV light.
- It is an excretory organ (nitrogenous wastes are lost with the sweat).
- It is vital in controlling the body temperature.

Basic components of the skin

The skin has three main layers. These are: Epidermis, Dermis, and Hypodermis

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i. The epidermis

- It is outermost layer of skin
- It is made up of dead cells.
- It provides a waterproof barrier and creates the skin tone/condition of tension/
- It contains cells, such as: Keratinocytes, Melanocytes, and Langerhans cells.

☑ **Keratinoccytes**

They produce the protein known as **keratin**

Keratin is the main components of epidermis, which is used to:

- Makes up hair, nails and the surface layer of the skin.
- It forms the rigidity of the skin.
- It helps with the barrier protection that skin offers/produce antimicrobial peptides.

☑ **Melanocytes**

They are cells that produce skin pigment known as **melanin**

☑ **Langerhans cells**

- They prevent things from getting into the skin.
- They stop water loss and also protect against the entry of **pathogens**.
- They are responsible for antigen presentation during an immune response.
- Antigen presentation is the process of preparing foreign protein antigen for recognition by the immune system.

ii. The dermis

It is located beneath the epidermis, and the middle layer of the skin.

It contains:

→ Tough connective tissue

→ Hair follicles that contains the roots of hairs, Sweat glands, sensory receptors and blood vessels.

- It is involved in temperature control in homoeostasis
- It contains collagen and elastic proteins necessary for skin health.
- The proteins help the skin to go back to its original state after being str

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etched.

iii. The hypo-dermis

- It is the lower layer of the skin.
- It contains fatty tissue, which used as an energy store and insulation layer/protecting against heat loss.

Thickness of the skin:

The skin can be categorized as thick and thin.

- The thinnest skin covers the eyelids.
- The thickest skin is present on the soles of the feet and palms of the hands .
- Thick skin has no hair follicles or sebaceous glands whereas thin skin does.

11.3.2 The Tongue

The tongue is a muscular organ in the mouth covered with a moist, pink tissue called the **mucosa**, which is a membrane rich in mucus gland

- ☑ The tongues is anchored to the mouth by webs of tough tissue and mucosa/mucus secreting cells/.
- ☑ In the back of the mouth, the tongue is anchored into the **hyoid bone**
- ☑ It contains tiny bumps/projection/ called **papillae** given the its rough texture.
  - The taste buds cover the surface of the papillae.
- ☑ Taste buds are connected to nerves running into brain.
- ☑ The tongue is vital for tasting, chewing and swallowing of food and also for speech.
- ☑ The sensory receptors of the tongue are sensitive to **solutions of certain chemical substance**

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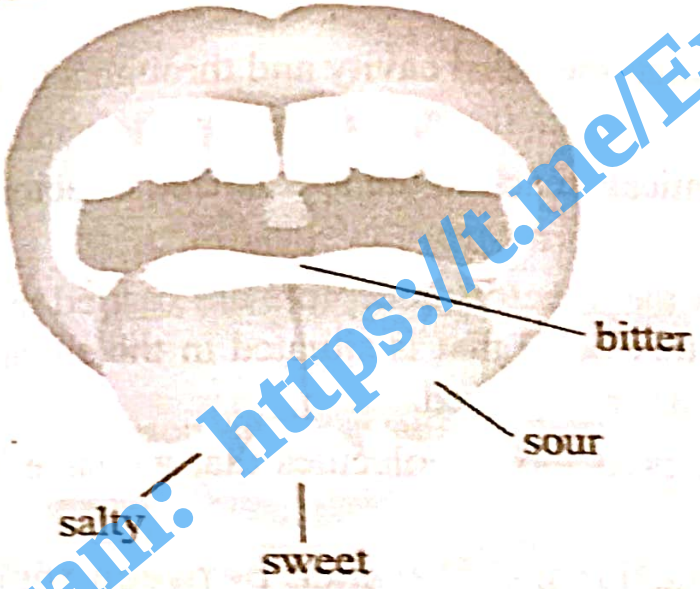
☑ The taste buds are located on the upper surface of the tongue and a lesser extent on the surface of the throat.

☑ There are **five basic taste sensations**. These are Sweet, Sour, Bitter, Salt and umami.

☑ The newly discovered tests is called **umami** that identifies the flavour found in foods such as meat, cheese, broth and mushroom.

For many year anatomists had thought that the receptors for the four known senses had their areas of different greats concentration on parts of the tongue:

- The tip part of the tongue is sensitive to both salt and sweet.
- The middle part of the tongue is sensitive to sweet.
- The side part of the tongue is sensitive to sour.
- The back side if the sensitive to bitter.



The current idea has clearly shown that in fact all of the five different taste organs are spread out all over the tongue, although some of them may seen to be a greater concentration in certain places.

11.3.3 The Nose

It is a sense organ of smell (olfaction).

Anatomy and physiology of nose:

☑ It contains nasal concha or turbinate bone/spongy bone/.

- Turbinate bone is a long, narrow and curled bone shelf protrudes into the breathing passage of the nose.
- The turbinate bones divide the nasal airway /cavity/ into four groove

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like air passages.

- Turbinate bones are responsible for inhaled air to flow in a steady and regular pattern.
- The two holes on the nose are called nostril.

☑ The olfactory receptors in the nose help to identify:

- Food, mates and predators
- Provides sensual pleasure in the odor of flowers and perfume.
- Warning of danger.

Another function of the nose is:

- Conditioning of inhaled air/ the nose makes it more humaid and warmer/.
- Hairs inside the nose prevent large particles from entering the lungs.

How do we smell?

- The air passes through the nasal cavity and through a thick layer of mucus.
- The dissolved chemicals reach to the specialized olfactory receptor cells.
- The impulses from the olfactory receptors are carried out by olfactory nerves to the olfactory build that is situated in the situated in the fore brain where smells are recognized.
- The brain then interprets those molecules that we have held up to the noses.

Note: the major functional difference between taste and smell receptors are:

The smell receptors are more specialized for detecting vapors coming to the organism from distant sources but test receptors are specialized for detection of chemicals present in the mouth itself.

Smell receptors are much more sensitive than testes receptors.

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Illustrative Examples

40. Organs involve in the sensation of the body is known as _____

A. Organ system
B. Muscular system
C. Nervous tissue
D. Sensory organ

Explanation: Organs which include hearing, seeing, taste and smell are

known as sensory organs. For example: eye, ear, noes, skin, and tongue.

Answer: D

41. The receptor which is absent in tongues is _____

A. Bitter receptors C. Photo receptors

B. Taste receptors D. Sweet receptor

Explanation: Photo receptors are sensitive to light. It is not seen in tongue. It is present in eyes.

Answer: C

42. What are the two holes on the noses called?

A. Septum C. Cartilage

B. Blow holes D. Nostrils

Explanation: Nostril is one of the two holes in the nose through which we breathe smell and sneeze.

Answer: D

43. Made out of a piece of that separates nasal air passages into four grooves _____

A. Nostrils B. Cartilage C. Bridge D. Turbinate

Explanation: The nasal turbinate bone separates the air passage.

Answer: D

44. Which receptor is present in the nose?

A. Photo receptors C. Olfacto receptors

B. Telo receptors D. Phono receptors

Explanation: Olfactory are present in the nose. It is narrow and spindle shaped. Life span is about 2 months ans is replaced by basal cells.

Answer: C

45. Which taste bud is present on the lateral sides of the tongue?

A. Salty B. Sweet C. Bitter D. Sour

Explanation: A man can perceive four basic modalities of taste i.e. Sweet, sour, salty and bitter. Other testes like pepper chocolate etc are experienced by a combination of two or more fundamental tastes.

Answer: D

46. Which of the following senses are detected by chemoreceptors?

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A. Smell and balance
B. Smell and taste
C. Taste and touch
D. Vision hearing

Explanation: Smell and test are detected by chemicals in the form of solution and vapor, respectively.

Answer: B

47. Receptors for the sense of taste are :

A. Utricullus
B. Taste buds
C. Sacclus
D. Olfactory cells

Explanation: the tongue contains taste buds, which contain sensory receptors for chemical testes.

Answer: B

48. The protective layer of the skin is:

A. Dermis
B. Hypodermis
C. Epidermis
D. None of these

Explanation: epidermis is the outer layer of the skin, which protects from pathogen, danger.

Answer: C

49. One of the following is the function of the skin, which one is it:

A. Protection from pathogens
B. Protect ultraviolet radiation
C. Regulation of body temperature
D. All are possible answer

Explanation: Functions of the skin: protection, regulation and sensation.

Answer: D

50. The fatty insulting layer of the skin is:

A. Epidermis
B. Dermis
C. Hypodermis
D. Hair Follicle

Explanation: Hypo dermis is the innermost layer of the skin, it contain fat layer, which used to insulate and store energy.

Answer: C

51. The layer of the skin which consists of blood vessels, lymph hair follicles and sweat gland is:

A. Hypodermis
B. Dermis
C. Epidermis
D. Hair follicle

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Explanation: dermis contains blood vessels, lymph vessel, hair follicle, and sweat glands.

Answer: B

52. The newly discovered sense of taste that is characterized by savoy taste is:

A. Bitter B. Sweet C. Salty D. Umami

Explanation: Umami is the newly discovered sensation of taster.

Answer: D

53. Which of the following senses are detected by the same organ?

A. Sight and hearing C. Sight and smell

B. Taste and smell D. Hearing and balance

Explanation: Ear is an organ that is used for the senses of hearing and balance.

Answer: D

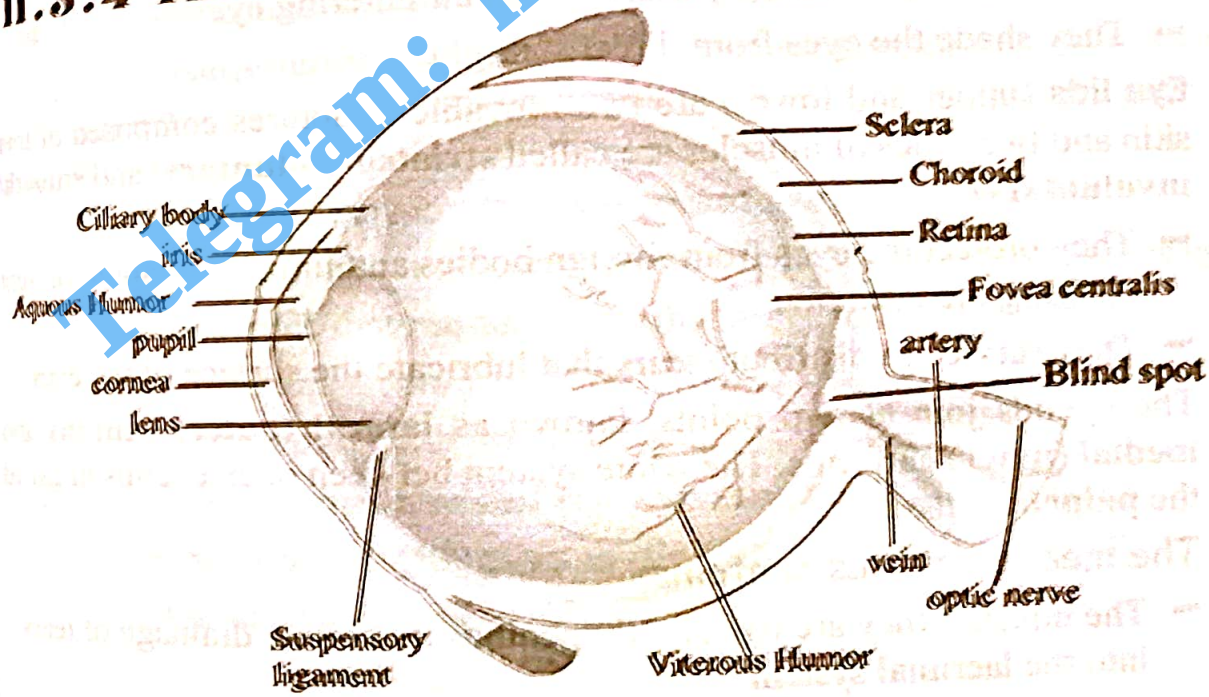
54. Which of the following sense is NOT located on the skin?

A. Touch B. Pain C. Pressure D. Smell

Explanation: Smell receptor is located in the nose.

Answer: D

11.3.4 The Human Eye



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The eye is the sensory organ of vision that responds to light, transmits stimuli to the brain for interpretation and forms image.

It is delicate organ, protected by:

i. Orbit or Socket

The eyeball is located in the cavity is called eye orbit or socket formed by different bones of the skull.

The eye ball is attached to the wall of the socket by extrinsic eye muscles/ extra ocular muscles/.

- There are six muscles attached to the outer surface of each eyeball.
- It controls six different directions of eye movement.
- **Four rectus muscles** are used for straight movement and the two oblique muscles are used for diagonal movement.
- In the orbit, the eye ball is surrounded by a cushion of fat.
- The **bony orbit** and **fat cushion** protects the eyeball.

ii. Eye brows, eye lashes, and eye lid

Eye brows: protect the eyes from entry of sweat (perspiration), which run down from forehead, and shade the eyes from direct sunlight

Eye lashes - it is projection of stiff or rigid hairs curving out ward along the margin of eyelids.

- They filter dust and dirty matter from air entering eye.
- They shade the eyes from direct sunlight

Eye lids (upper and lower) are two movable structures composed of thin skin and two types of muscles are called **striated/ voluntary/** and **smooth/ involuntary/.**

- They protect the eyes from foreign bodies and limit the amount of light entering the eye.
- They service to distribute tears that lubricate the surface of the eyes.

The eyelids join at two points, known as **lateral (outer) canthus** and **medial (inner) can thus.** The white system between open eyelids in called the **palpebral fissure**

The medial canthus contains:

- The puncta - they are two small openings that allow drainage of tears into the lacrimal system

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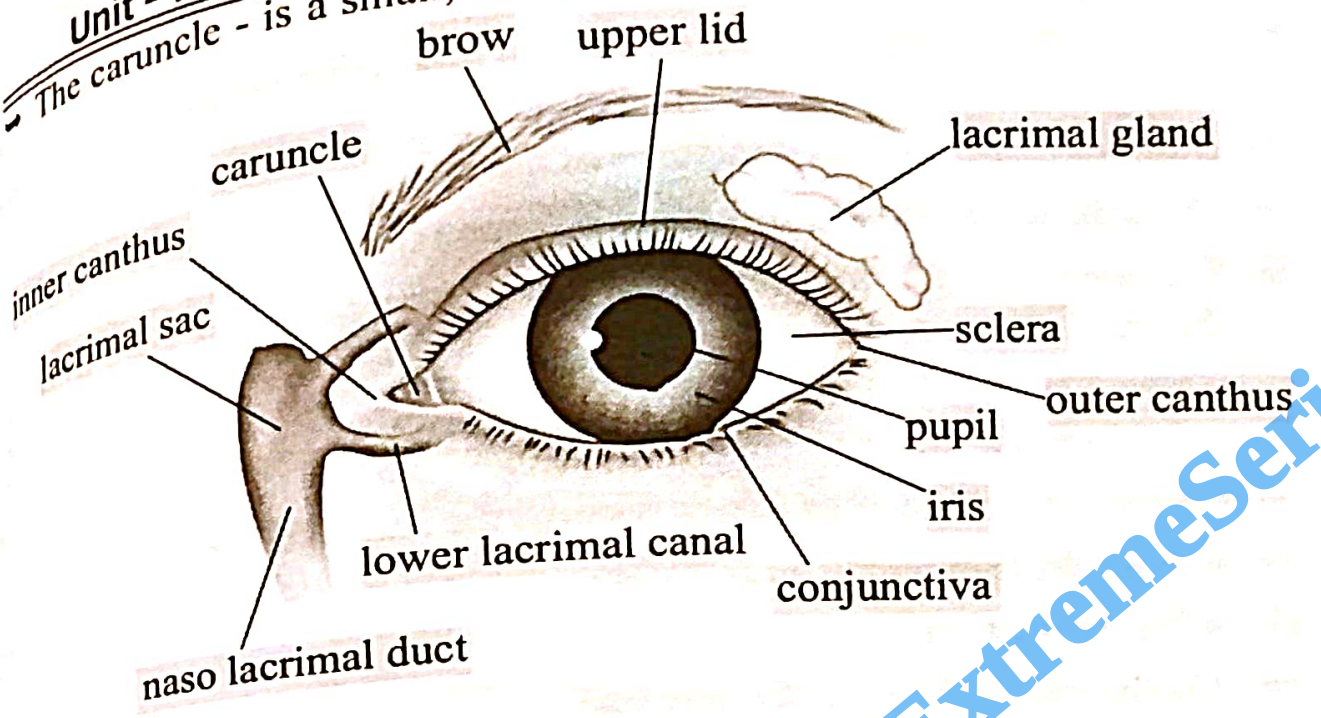
➥ The caruncle - is a small, fleshy mass that contains sebaceous glands.



iii. Conjunctiva

Conjunctiva is a thin and transparent continuous membrane, which allows for:

➥ Inspection of underlying tissue.

➥ Serves to protect the eye from foreign bodies.

Conjunctiva is divided into two portions:

A. **Palpebral conjunctiva** that lines the inside of the eyelids

B. **Bulbar conjunctiva** that converse most of the anterior eye and merging with the cornea at the limbus /forms the border b/n transparent cornea and opaque sclera.

iv. The lacrimal apparatus /lacrimal gland/

It is located in the upper outer corner of the orbital cavity just above the eye.

It consists of glands that produce tears that serve to lubricate the eye.

➥ The tear contains an enzyme is called **lysozyme** which kills bacteria.

➥ The tear is used to wash away dust particles.

The internal structures of the eye and their functions:

The internal wall of the eyes is divided in to three layers:

1. The sclerotic layer.
2. The choroid layer.
3. The retina

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i. The sclerotic layer (sclera)

- It is a white, external layer, dense and protective visible portion of an eye.
- It supports the internal structures of the eye.
- The anterior portion of this layer is cornea (the "window of the eye")

ii. Cornea:- It is the transparent anterior or front part of the eyes. It covers the pupil and iris. It is well supplied with nerve endings and making it responsive to pain and touch.

It is important to:

- Let the entry of light to passes through the lens to the retina.
- Refract the light along with the lens.

iii. Choroid layer

- It is the middle layer of the eye ball
- It has many blood vessels that provide food and oxygen to the retina.
- It is dark layer/bluish in color is used to prevents light from reflecting internally
- It is continuous anteriorly with the ciliary body and the iris.

iv. Ciliary body

- It is the thickened portion of the choroid layer on the region around the lens.
- It contains tissues, blood vessels and ciliary muscles.

v. Ciliary muscles

- They form bundles of circular and radial muscles.
- They change the shape of the lens by changing the tension of suspensory ligaments during accommodation

vi. Suspensory ligaments

- They are elastic like structures.
- They held the lens in position with ciliary muscles.

vii. Lens

- It is a biconvex, transparent, avascular/without blood vessels/ and encapsulated /covered with fluid/ structure.
- It is located immediately posterior /behind/ to the iris.

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- It is made up largely from proteins.
- It is held in place by suspensory ligaments and the ciliary muscles.

viii. Iris

- It is pigmented, colored portion of the eye because it contains the pigment chromatophores.
- It is visible externally.
- It controls the hole of the pupil according to the light source.
- It controls the amount of light entering the retina.

It contains two muscles: Circular muscle and radial muscles

In dim light, the radial muscle contract and the circular muscle relax

- As the result the pupil is dilates (open wide) and lots of light can enter the eyes.
- During parasympathetic intervention the pupil is open wider

In bright light, the circular muscles (sphincter pupillae) of the iris contract and the radial muscles relax.

- As the result the pupil becomes constricted (very small). This reduces the amount of light that enters into the eye.
- During sympathetic intervention the pupil is constricted.

ix. Pupil

- It is the small aperture /hole/ located in the center of the Iris.
- The size is changed in response to change in light intensity.
- It allows light to enter and focus on the retina.

x. Aqueous humour

It is colorles and transparent fluid that fills the space b/n cornea and lens/ aqueous chamber/

It is a clear liquid substance produced by the ciliary body.

Important for:

- It cleanses dirt and nourishes the cornea and lens.
- It maintains intraocular fluid pressure
- Maintains the shape of eye ball.
- It refracts or bends light that enters to eye to focus on the retina.

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xi. Vitreous humour

It is a jelly like substance that fills the space b/n lens and retina /vitreous chamber/.

Important for:

- ➥ Maintaining the shape of eye ball.
- ➥ It refracts or bends light that enters to the eye to focus on the retina.

xii. Retina

Retina is the inner most layer of the eye where image is formed. It extends only to the ciliary body anteriorly. It receives visual stimuli and sends it to the brain.

It consist of photoreceptor cell, known as rods and cones.

- ➥ Rods and cones contains light sensitive protein called photopigments
- ➥ Rods are highly sensitive to dim light.
- ➥ Cones function in bright light and are sensitive to color

☑ **Optic disc**

- ➥ It is a cream - colored and circular area located on the retina.
- ➥ It is the place where the optic nerve enters the eyeball.

☑ **Blind spot**

Blind spot is the place where the optic nerve stands or leaves. It does not contain light is not image formation here. Therefore, there is no image formation here.

☑ **Yellow spot of fovea**

- ➥ It is also called fovea centralis.
- ➥ It is a retinal depression where clear **image** is formed
- ➥ It is located adjacent to the optic disc in the temporal section of the fundus /part of the eye opposite the pupil/

☑ **Optic nerve**

It is the type of nerve that carries impulse from the retina to the visual center of the brain.

Focusing the light/image formation

Light is reflected from external objects and enters the eyes.

The curved surface of cornea, aqueous humour, lens and vitreous humour

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refract light and focus it on the ratina. Thus the image is formed on the retina /fovea/.

- When an image is produced on the retina, the light - sensitive cells are stimulated.
- They send impulses to the brain along affecter (senssory) neurons in the optic nerve.
- When the brain receives these massages it interprets the information and enable the person to see.
- The image formed on the retina is real, inverted and smaller than the object.
- The inversion and the size of an image are corrected in the visual center of the brain.

Common eye defects and their corrections

In some people the normal method of accommodation in the eye that enable them to see close - up and distant objects with equal clarity doesn't work properly. There are a umber of common eye defects that can affect many people of all ages. These are: Short sight (myopia), Long sight (Hypermetropia), Astigmatism, Bllindness, and Colorblindness

1. Short sigh (myopia)

A short sighted person can focus objects that are close to him/her but cannot focus distant objects.

Short sighted is caused by: rigid and too curved lens, long eye ball, and weak ciliary body.

- Thus image of distance object is formed (focused) in front of the retina.

Short sight can be corrected by wearing concave or diverging lens.

2. Long sight (Hypermetropia)

- A long - sighted person can focus clearly on things that are at a distance but can not focus near objects clearly.
- Long sighted is caused by: weak lens, smaller eye ball, strong ciliary muscles.
- Thus image of near object is formed behind the retina.
- The problem is corrected by wearing **converging or convex lens**

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Illustrative Examples

55. Which of the following helps in maintaining the shape of the eye?

A. Conjuctiva
B. Iris
C. Vitreous humor
D. Retie

Explanation: Vitreous humor is a thick fluid mass that helps in maintaining the shape of the eye.

Answer: C

56. Make the INCORRECT statement about the lens?

A. Protein found in the lens are α, β and γ crystalline
B. Composed of water and proteins
C. Mainly made up of water
D. It has blood supply

Explanation: Though lens does not have a blood supply, It is active, Lens is basically made up of water and protein and many verebrates have lens protein α, β and γ crystalline.

Answer: D

57. What is Sclera?

A. Cornea
B. White part of the eye
C. Red part of the eye
D. Lens

Explanation: Sclera is the white part of the eye white cover the entire eye except for cornea and is made up of collagen fibroblast.

Answer: B

58. Name the part of the eye, which is involved in focusing light on the retina?

A. Sclera
B. Iris
C. Cornea
D. Ciliary muscle

Explanation: Fibrous tunic is composed of anterior cornea and posterior scaler. It is a surface coating of an eyeball. Transparent cornea helps in focussing light on the retina.

Answer: C

59. Among the following which is one is more vascularized and

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pigmented?
A. Cornea B. Iris C. Ciliary body D. Choroid

Explanation: Choroid is highly pigmented with melanin and vascularized structure which forms the posterior region of vascular tunic. It absorbs scatterd light and also supplies blood.

Answer: D

60. Protein found in the retina is _____
A. Crystalline C. Collagen
B. Opsin D. Rhodopsin

Explanation: Rhodopsin is a biological pigment found in rods of the retina. It is G - protein coupled receptor. It is extremely sensitive to light.

Answer: D

61. What helps to see color?
A. Rods B. Cons C. Lens D. Retina

Explanation: Rods and cones are two types of photoreceptor in human retina. Rods are responsible for vision at low light levels and cones helps to see color.

Answer: B

62. Which part is a colorful parts of the eye?
A. Pupil B. Iris C. Eyeballs D. Cornea

Explanation: Iris is a thin, colorful circular structure of the eye. It responsible for controlling the diameter and size of the pupil.

Answer: B

63. Aqueous and vitreous humours are divided by _____
A. Lens B. Iris C. Retina D. Optic nerve

Explanation: Aqueous humours fill both anterior and posterior chambers of the eye. Vitreous humours are a clear gelatinous mass found in back of eyeball. Aqueous and vitreous humours are divided by lens.

Answer: A

64. The inner most layer of human eye is _____
A. Sclera B. Cornea C. Choroid D. Retina

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Explanation: retina is the inner most layer of human eye. It is light sensitive layer. The retina converts the light rays into impulses that travel through the optic nerve to brain where they are interpreted as the image we see.

65. Color blindness is due to defect in _____ **Answer: D**

A. Cones
B. Rods
C. Rods and cons
D. Rhodopsin

Explanation: Color blindness is reduced ability to distinguish between certain colors. It is a very common defect. Color blindness is due to defect in cones.

66. Color perception in human due to _____ **Answer: A**

A. Rhodopsin present in rod cells
B. Rhodopsin present in cone cells
C. Iodopsin present in cone cells
D. Iodopsin present in rod cells

Explanation: Iodopsin is present in cone cells. Iodopsin consists of the protein and a bound chromophore, retinal.

Answer: B

11.3.5 The human Ear

Ears are the auditory sense organ of our body, specialized for receiving sounds for hearing, balance and position of the body.

- The human ear has two types of mechano - receptors. One for hearing, the other is for balance
- The ear is divided into three sections, namely: the out ear, the middle ear, and the inner ear

1. The outer ear /external ear/

- It is the external part of the ear, which collects sound wave and direct them into ear
- It is a tube that passes inward from the side of the head to the ear drum.

It consists of: Pinna (auricle), ear canal and eardrum or tympanum.

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Pinna (auricle) - is the funnel - shaped external part of the ear, made up of cartilage and skin.

- It is the only visible part of the ear

It is used to:

- Collect and direct sound vibration in to ear canal.
- Judge the direction from which the vibrations are coming.
- Overcome the difference in air pressure inside and outside the ear

ii. **The ear canal /auditory/**

It is the tube that directs sound waves in to the tympani membrane or ear drum.

It consists of gland that produce waxes

- The wax is used to lubricate the ear drum and trap dusts.
- Contain small hairs that filter out dust particles entering the ear canal.

iii. **The eardrum or tympanic membrane**

- It is a sheet of very thin membrane at the end of the ear canal.
- It receives sound waves from the ear canal and directs the vibration in to the middle ear
- It is protected by the curve of ear canal, and the wax by trapping dust and insects

2. The middle ear

The middle ear is a cavity filled with air behind the eardrum. It consists of: Ossicles, Eustachian tube, oval and round window

i. **Ossicles -** are three tiny bones in the middle ear, namely:

- Hammer (malleus), anvil (incus), and stapes (stirrup).
- They are the smallest bones in the human body.
- The malleus attached to the eardrum, and the stapes to the oval window
- Anvil bones is between malleus and stapes.

ii. **Eustachian tube -** is a tube that connects the middle ear to the throat.

It equalizes the pressure on both sides of the ear drum to prevent the eardrum from bursting.

iii. **Oval window and round windows** - are openings and covered by very thin membrane, found at the end of the middle ear.

- Oval window directs sound waves to the inner ear.

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➥ Round window helps the movement of fluids in the cochlea.

3. The inner ear

It is fluid filled cavity of the ear, consists of:

i. The sacculus and utriculus

➥ They are two sac - like structures, filled with fluid and lined with hair cells.

➥ They are right angles to each other.

➥ They are concerned with the sense of balance and posture.

ii. The semicircular canals

➥ They are three in numbers

➥ They are called semicircular canals because each forms a half circle.

➥ The sacs are filled with fluid and lined with hair cells.

➥ Signals are sent to cerebellum of the brain.

➥ They are concerned with orientation, position of the head, and maintain balance

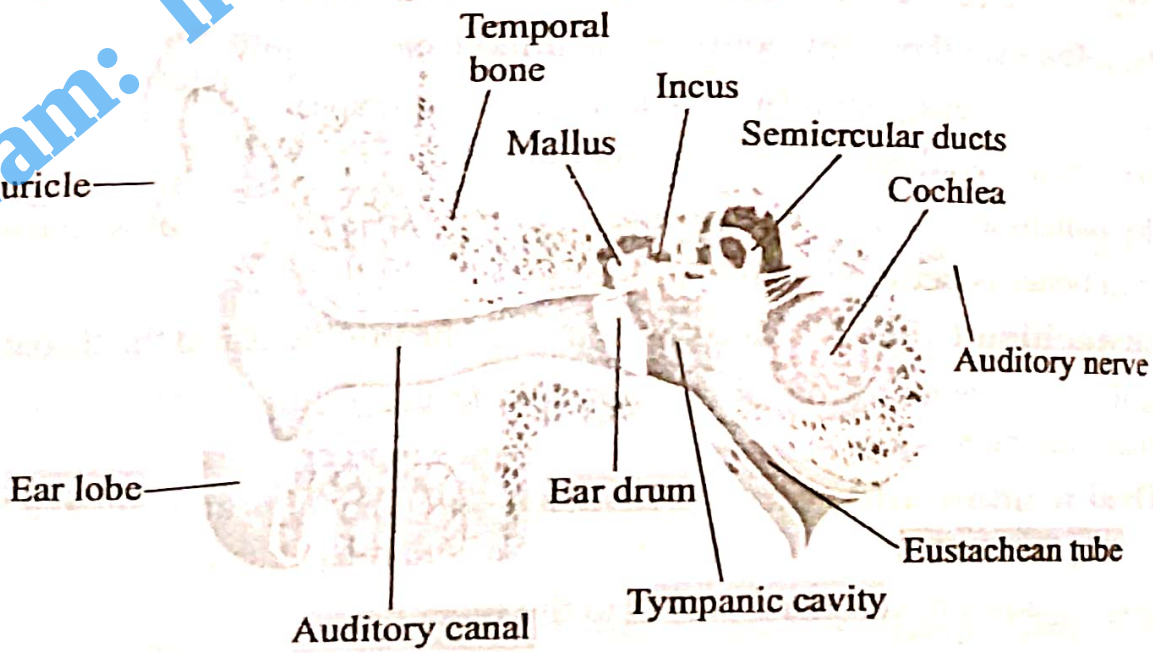
iii. Cochlea

➥ It is the coiled inner part of the ear.

➥ It is filled with fluid and lined with tiny hair cells that are pushed back and forth by sound waves.

➥ It is a place where hearing is generating.

➥ It contains auditory nerves that carry impulse to the brain



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11.3.6 The Mechanism of Hearing

The pinna collects sound waves and directs them to the eardrum through the ear canal.

When sound waves hit the eardrum, it vibrates.

The vibration of the ear drum magnifies the sound, which are then transmitted to the ear ossicles

- The ear ossicles /hammer (malles), anvil (incus) and stapes (stirrup)/ also amplify the vibrations.
- From the stapes the vibration of sound passes to the oval window.

Vibration of oval window transmits the sound wave to cochlea.

- Vibrations of the fluid in cochlea cause the hair - like sensory cells to move.
- The movement of hair like sensory cells causes production of nerve impulses
- The impulses are transmitted by auditory nerve to the brain for interpretation.

The movement of sound wave into ear from outside to inside is:

Sound wave → Pinna → ear canal → hammer (malleus) → anvil (incus) → stapes (stirrup) → oval window → cochlea → auditory nerve → Brain

11.3.7 The Sense of Balance and Movement

The sense of balance and movement or motion is achieved by semicircular canal of the ear.

The swellings on each of the semicircular canal are called ampullae

- Contain sensory cells, which are hair like cells, attached to the sensory nerve endings
- These cells are enclosed in a core of jelly substance are called **cupula**.

When the head moves, the fluid in the semicircular canals also move in the opposite direction.

- Causes the copula to tilt and pressing the hairs of sensory cells.
- These create nerve impulses in the sensory nerve endings.
- Then the nerve impulse are transmitted by auditory nerve to the brain.
- The brain then interprets the direction and speed of motion of the body

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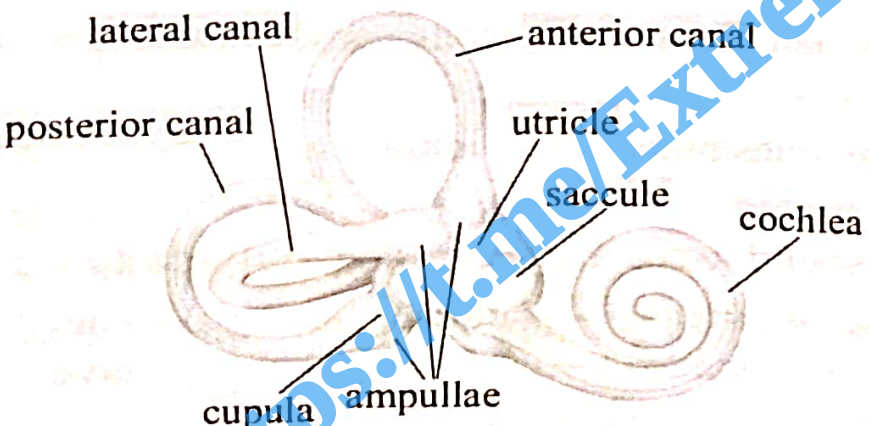
or head.

Note: fast spinning of the body followed by instant interruption causes dizziness. This is because the fluid in the semicircular canals keeps moving after the spinning of balance has stopped. The utriculus and sacculus are concerned with your sense of balance and posture.

- It contain sensory cells with protruding hairs embedded in a jelly - like substance containing tiny particles of chalk are called otoliths.
- When the head is tilted on one side, the otoliths move in the opposite direction
- The brain detects the angle of tilt and return the body to its normal position and posture.



Illustrative Example

67. Name the tiny bone of the middle ear, which convert sound into vibration?

A. Stapes B. Malleus C. Incus D. Anvil

Explanation: The space of the middle ear is filled with air with the typmpanic membrane separates the inner and external ear. Eardrum transfer sound energy to middle ear. Which in turn convert it to vibrate.

Answer: A

68. In human ear, secretion of wax is done by _____

A. Cochlea C. Basilar membrane

B. Vestibule D. Ear canal

Explanation: Human ear has the number of sweet gland called ceruminous glands. These glands are present on the ear canal from

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exterior to interior part. The secretion of these glands called cerement or wax.

Answer: D

69. Internal ear is filled with _____
A. Perilymph
B. Endolymph
C. Lymph
D. mesolymph

Explanation: Endolymph is the fluid contained in the membranous labyrinth of the inner ear. It is also called as scarpa's fluid.

Answer: B

70. Sequence of ossicles from outside to inside is _____
A. Malleus - incus - stapes
B. Incus - stapes - Malleus
C. Stapes - incus - malleus
D. Malleus - stapes - incus

Explanation: Ear ossicles are three bones in middle ear. They are the smallest bones in humans. The term ossicle means tiny bones. From outside to inside, Malleus - incus - stapes.

Answer: A

71. Which of the following is balancing organ?
A. Organ of corti
B. Cochles
C. Vestibular region
D. Eardrum

Explanation: Vestibule is the central art of inner ear and is situated medial to eardrum behind cochlea. It contains organ of hearing (cochlea) and organ of balance (labyrinth).

Answer: C

72. What separates the outer ear from the middle ear?
A. Auricle B. Ear wall C. Cochlea D. Ear drum

Explanation: Eardrum is a thin cone - shaped membrane. It separates the external ear from the middle ear. It is also called a tympanic membrane.

Answer: D

73. Which part of the ear takes sound waves and turns them into vibrations?
A. Middle war
B. Ear wall
C. Ear canal
D. Inner ear

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Explanation: Middle ear contain 3 bones. These 3 bones form a connection from the eardrum to the inner ear. As sound waves hit the eardrum, the eardrum moves back and forth causing the ossicles to moves

74. Eustachian tube connects _____

Answer: A

A. Pharynx with middle ear
B. Middle ear with external ear
C. Middle ear with external ear
D. External ear with internal ear

Explanation: Eustachian tube connects Pharynx with middle ear. It controls pressure within the middle ear making it equal with he air pressure outside the body.

75. What are the two bones of the middle ear that fit to the eardrum and the oval window respectively?

Answer: A

A. Stapes and anvil
B. Malleus and anvil
C. Stapes and incus
D. Malleus and stapes

Explanation: Malleus (hummer) fits into the ear drum and stapes (stirrup) fits into the oval window.

76. What do we call the swelling on each semicircular canal of the human ear?

Answer: D

A. Eardrum
B. Capula
C. Eustachian tube
D. Ampullae

Explanation: Ampullae is the swollen portion at the base of semicircular canals. It detects the movement of fluids.

77. Which of the following structures of human ear collects sound waves?

Answer: D

A. Eustachian tube
B. Ear canal
C. Pinna
D. Cochlea

Explanation: Pinna is a cartilaginous flap found at the most outer part of the ear. It is used to collect the sound waves.

Answer: C

78. Which of the following structures is found at the most outer part of the

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ear. It is used to collect the sound

A. Ossicles
B. Tympanic membrane
C. Semicircular canal
D. Pinna

Explanation: Pinna - Collects sound & detect direction of sound

Answer: D

79. Explain the reason why fats spinning of the body followed by instant interruption cause **dizziness.**

Explanation: this is because the fluid in the semicircular canals keeps moving after the spinning has stopped, and send nerve impulse to the brain.

11.4 The Endocrine System

It is system that contains a collection of glands that produce hormones which regulate:

→ Metabolisms → Tissue function
→ Growth and development → Biological process

- It is a diverse collection of Cells, Tissues, and Organs
- Endocrine glands produce and secrete hormones which are chemical messengers that regulate many physiological processes.
- The endocrine system has two main components that are glands and hormones.

Hormones - a substance that is made in one cell or tissue and that causes a change in another cell or tissue locater in a different part of the body

- The blood carries the hormone to every cell in body, but only the target cells for a given hormone can respond to it. The hypothalamus and the pituitary gland work to control the function of the endocrine system.
- The major endocrine glands are: hypothalamus, pituitary, thyroid, adrenal, ovaries, testes, pineal, pancreas, and gastrointestinal tract

Gland: is an organ that produces and release substances. There are two types of glands: Exocrine and Endocrine glands

1. Exocrine gland

- They are glands which release their cellular secretions (enzymes, sweat, etc.)

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- Have ducts or tubes.
- Secretary products are: Sweat, enzymes, mucus, sebum
- Route of secretion: secretary products are released to an internal organ or the external surface through a duct

Exocrine glands in includes:

- Sweat glands
- Salivary glands
- Sebaceous glands, pancreatic and liver gland, and mammary glands.

2. Endocrine gland

- They are glands that produces hormones
- Have no ducts, so they are sometimes known as ductless glands.
- Route of secretion: The endocrine gland secret hormones directly into the blood stream, eventually reaching the target organ or tissue
- They secrete hormones directly onto the blood, and the chemicals are carried from glands to the body parts throughout the bloodstream.

Endocrine glands includes :

- Pitutary gland
- thyroid gland
- parathyroid glands
- A drenal glands

I. Pituitary Gland

It is a small size gland found in the middle of the lower surface of the brain. This gland is about 1 c.m in diameter. Pituitary gland consists of two main lobes. **Anterior lobe** (anterior pituitary) and **posterior lobe** (posterior pituitary).

A. Anterior lobe or Anterior pituitary

The Anterior lobe of the pituitary gland produces a number of hormones which control the activities of all other endocrine glands. Anterior pituitary gland - secretes six hormone.

i. Growth Hormone (GH) / somatotropic hormone.

This is also known as somatotropic hormone. This hormone is necessary for the normal growth and development of the body.

ii. Thyrotrophic Hormone / Thyroid stimulating hormone (TSH).

This hormone is also known as Thyrotrophic Hormone. It controls the activities and growth of thyroid gland.

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iii. Adrenocorticotropic Hormone (ACTH).
This hormone stimulates the adrenal cortex to synthesize its hormones. It also regulates the activities of Adrenal glands.

iv. Follicle - stimulating Hormone (FSH)
It stimulates the ovary in females to synthesize estrogen. It also stimulates testes in male to provide spermatozoa.

v. Luteinising Hormone (LH)
It controls the secretion of estrogens and progesterone in the ovary and testoterone in the testis.

vi. Luteotrophin Hormone / prolactin.
The other name of this hormone is prolactin. It stimulates milk production and secretion in mammals. It also maintains the uterine growth and development during pregnancy.

B. Posterior lobe or posterior pituitary

The posterior lobe of the pituitary gland secrete two hormones : oxytocin and vasopressin / Antidiuretic Hormone (ADH)

i. **Oxytocin** - this hormone has two functions :
   - ➥ It contracts the uterus during delivery or birth of the child
   - ➥ It also helps in the ejection of milk from the mammary gland

ii. **Vasopressin** - this hormone is also known as Antidiuretic hormone (ADH). Its functions are :
   - ➥ It helps in decreasing urine out put (conserve water)
   - ➥ Increasing blood pressure

II. Thyroid Gland

It is situated in the lower part of the neck and extended on both side of the larynx. There are two lobes in thyroid gland, one on each side of the trachea. These two lobes are connected by a strip of thyroid tissue called the "isthumus" which lies in front of the trachea.

The thyroid gland synthesizes and secretes two hormones :

1. Thyroxine (T. 4) 2. Trilodothyronine (T. 3)

The major function of thyroid hormones are : Regulation of metabolic rate, such as :

- ➥ Increase in the absorption and utilization of glucose

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- Helps in physical and mental growth and development of the body
- Increase in the rate of cholesterol synthesis in liver
- Helps in protein synthesis

The secretion of thyroid hormones is controlled by Thyroid stimulating Hormone (TSH) of anterior pituitary Gland. The thyroid hormones influence growth and metabolism .

Disorders of thyroid Function

A. Hypothyroidism (decreased secretion)

i. Deficiency of the thyroid secretion at birth produces a cretinism. It produces mental retardation and slow physical development

ii. **Myxoedema** occurring after birth. Slow metabolic process, retardation of physical growth, weight gain, slowness of mind and speech, skin thickened, dry and hair falls.

iii. **Endemic goiter** occurs due to deficiency iodine in food. It produces enlargement of the thyroid gland.

B. Hyperthyroidism (Increased secretion)

i. The metabolic rate is raised and the body temperature may be higher than normal.

ii. Loss of weight, nervousness, high pulse rate, cardiovascular symptoms may arise which can lead to heart failure.

iii. **Grave's disease** or exophthalmic goiter is due to excessive production of thyroid hormone which produces protrusion of eyeballs.

III. Parathyroid Gland

They are four small glands situated two on each side of the thyroid gland in the neck. They are embedded on the posterior surface of the thyroid gland.

The parathyroid glands are composed of mass of epithelial cells, which secrete the **parathyroid Hormone (PTH)** or parathormone. The secretion of PTH is not under nervous or hormonal control. Stimulate osteoblasts to release calcium into blood to increase **blood calcium** level

Disorders of Parathyroid Function :

i. **Hypoparathyroidism : (Less secretion). It is also known as "Hypocalcaemia" which leads to Tetany.**

- There is deficiency of the calcium content in blood

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- It is characterized by muscular contraction, convulsions of the hands and feet etc

ii. **Hyperparathyroidism : (Excess secretion)**

- Enlargement of the glands
- Calcium balance is disturbed. It produces "Osteitis fibrosa Cystica."
- The calcium may be deposited in the kidney, causing renal stones and kidney failure.

IV. Adrenal Glands

The adrenal glands are two in number, Which are situated just on the top of each kidney like a cap. Each adrenal weighs about 5 gms. This gland is called the gland of emergency.

The adrenal glands can be divided into two parts. These are : outer cortex and inner Medulla

☑ Outer Adrenal Cortex :

This part has three distinct layers of cells. They are

i. **Zona glomerulose : it is the outer layer which secretes the hormone mineralocorticoids**

It is aldosterone, which help to maintain electrolyte and water balance of the body

ii. **Zona fasciculate : this is the middle layer of the cortex which secretes glucocorticoid hormones. It is cortisol, cortisone and corticosterone, which influences carbohydrate metabolism**

iii. **Zona Reticularis : it is the inner layer secreting sex steroids (androgen and estrogen).**

They are androgens in males and estrogen females. They influence growth and sex development. The secretion of the hormones of the adrenal cortex is controlled by Adrenocorticotropic Hormone (ACTH).

Hypo - secretion of Adrenal cortex leads to **Addison's disease** which characterized by loss of weight, hypotension, pigmentation of skin etc.

Hyper secretion produces **Cushing's Syndrome**, which is called as Moon Face. It is characterized by deposition of fat on face, neck, diabetes and hypertension.

☑ Adrenal Medulla

This is the inner part of Adrenal glands. It secrets adrenaline, which is used

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for :

- Increasing the rate of heart beat
- This hormone acts as **fight and escape** from an enemy
- It helps constriction of skin and blood capillaries
- The energy level of the body is increased by this hormone as it raises blood glucose.
- Helps in dilation of the pupil of the eyes
- Relaxation of the intestine can be possible by this hormone

Adrenal medulla also produces non - adrenaline hormone, which is used for :

- Helps in constriction of small arteries
- Increase the Blood pressure

Note : In times of emergency like shock, fear, anger, danger, the sympathetic and parasympathetic nerves stimulate the adrenal medulla, so that large amount of adrenalin and non - adrenaline hormones are secreted which make immediate physiological adjustment.

V. Pancreas

The pancreas lies under the stomach in front of the abdomen. It performs both **exocrine and endocrine** functions. As an exocrine gland, pancreatic juice is secreted from the secretory cells **"acini"** which helps in the digestion of food.

The endocrine part is called **" Islets of Langerhans".** The Islets are present mostly in the tail portion of pancreas.

There are about two million islets in the pancreas.

The islets contain Three major types of cells, the alpha, beta and delta cells.

i. **Alpha cells** from 25 % of the islets which secrete the hormone glucagon.

Glucagon : It increases the blood sugar level by breaking down glycogen in the liver to glucose. It also helps in mobilization of stored fat.

ii. **Beta cells** constitute 60 % and secrete insulin hormone.

Insulin : it decreases the level of glucose. It promotes formation of glycogen from glucose in the liver and muscles. It prevents fresh synthesis of glucose and stimulates the utilization of glucose in the skeletel muscle.

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- Decrease in the synthesis of insulin or any type of disorder gives rise to the disease **diabetes.**
- **Type 1 diabetes** (when pancreas secretes no insulin), **type 2 diabetes** (due to obesity, overeating, and lack of exercise)

iii. **Delta cells** form 10% of endocrine tissues and secrete **somatostatin** hormone.

Somatostatin : it prohibits secretion of insulin and glucagon from the islets ; slows down the assimilation of food from intestine.

VI. Sex Glands (ovary and testes)

- Ovaries in the female secrete estrogen and progesterone
- Testes in the male secrete testosterone

i. **Estrogen** : it is the female sex hormones secreted by the ovaries. Its functions are regulation of menstrual cycle, and development of Secondary Sexual Characteristics :

- Development of mammary Glands
- High pitched voice
- Deposition of subcutaneous tissue
- Growth of hairs in public and axillary region
- Influences the growth and development of fallopian tubes
- It influences behavioral and psychic patterns in females

Note : The secretion of estrogen is controlled by follicle stimulating hormone of Anterior pituitary gland.

ii. **Progesterone** : it is a female sex hormones and its functions are as follows :

- Menstruation and development of uterus and breast
- Preparation of the uterus to receive the fertilized ovum
- During pregnancy, this hormone stimulates growth of mammary glands

iii. **Testosterone** : this hormone is secreted from the male sex organ (testes). The important functions are :

- Stimulation of spermatogenesis
- Promotes growth and activity of prostate, penis and scrotum.
- Development of **secondary sexual characteristics** like moustache, deep voice, skeletal and muscular growth.

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Note : The secretion of testosterone is controlled by luteinizing hormone of Anterior pituitary.

VII. Thymus

It is present in the upper chest cavity on the trachea. It is partly an endocrine gland and partly lymphoid structure. It lies behind the sternum but in front of the heart. It is consisting of two lobes in which a number of lobules are there. The thymus is bigger in size at birth; it grows in size until puberty, then the size gradually decreases.

The hormone secreted is thymosin or thymine. Its functions are as follows

- It controls the production of lymphocytes (body defense)
- Regulates transmission of minerals in bones
- It plays some role in sexual development

VIII. Pineal Gland

It is a small gland situated in the roof of the brain. It is covered by corpus collosum and cerebral hemispheres. The shape of this gland is like a pine cone. It weighs about 150 mg.

It produces a hormone called **melatonin,** which is involved in the sleep / wake cycle, regulate circadian rhythms. It affects sleep, wakefulness, reproduction and mood

X. Hypothalamus

It connects the nervous system to endocrine system. It controls the other endocrine glands and regulates all the major internal function.

For example :

It stimulate the secretion of ACTH, inhibit prolactin secretion, stimulate the secretion of GH, stimulate the secretion of TSH

Illustrative Example

80. Low secretion of the growth hormone causes which of the following diseases

A. Acromegaly
B. Gigantism
C. Dwarfism
D. Thyroid

Explanation: Low secretion of growth hormone results in stunted growth and which known as dwarfism while the over secretion of the

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mone cause the abnormal growth of the body leading to

Answer: C

the following hormones regulate the growth of mammary

. Prolactin B. TSH C. FSH D. ACTH

Explanation: Prolactin regulates the growth of the mammary glands and formation of milk in them. ACTH stimulate the synthesis and steroid hormones called glucocorticoids from the adrenal cortex.

Answer: A

82. Which of the following hormones are known as glucocorticoids from the adrenal cortex.

A. LH and Oxytocin C. LH and FSH

B. LH and vasopressin D. FSH and Oxytocin

Explanation: LH and FSH stimulate the gonadal activity and are therefore referred to as gonadotrophins. In males, LH stimulates the syntheses and secretion of hormone called androgens from the testis.

Answer: C

83. What do delta cells secrete?

A. Cortisol C. Pancreatic enzyme

B. Glucose D. Somatostatin

Explanation: Somatostatin is also known as growth hormone. It is a peptide hormone. Somatostation is secreted by delta cells.

Answer: D

84. The pineal gland secretes _____

A. Thyroxin C. Melnin

B. Melatonin D. Prolactin

Explanation: The pineal gland secretes Melatonin. It regulates sleep and wakefulness.

Answer: B

85. Which of the following is incorrect regarding the endocrine gland?

A. They are known as ductless glands

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B. Hormones are secreted through ducts in the bloodstream
C. Their secretions are called hormones
D. Theses glands lack ducts

Explanation: Endocrine glands lack ducts and the hence, known as ductless glands. Their secretion are called hormones. As these glands lack ducts, therefore they release their secretions directly into bloodstream.

Answer: B

86. Which of the following is not an endocrine gland?

A. Pituitary gland
B. Sebaceous gland
C. Pineal gland
D. Adrenal gland

Explanation: Pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus, and gonads are the organised endocrine bodies in our body. The endocrine glands and their hormones constitute the endocrine system.

Answer: B

87. Hypothalamus is a parts of _____

A. Midbrain
B. Forebrain
C. Hind brain
D. Future brain

Explanation: Hypothalamus is the pat of fore brain that contains a number of small nuclei with a variety of function. It links the nervous system to endocrine system.

Answer: B

88. Hormones produced in hypothalamus are _____

A. Oxytocin and estrogen
B. Oxytocin and testoterone
C. Oxytocin and Antidiuretic hormone
D. Oxytocin and thyroxine

Explanation: Hormones produced in hypothalamus are Oxytocin and Antidiuretic hormone. They have their specific function and specific target site.

Answer: C

89. Which of the following events could be a result of damage to hypothalamus portal system?

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A. Decreased secretion of ADH
B. Decreased and secretion oxytocin
C. Decreased and secretion TSH
D. Decreased and secretion PTH

Explanation: Secretion of Thyroid stimulation hormone will be decreased if there is any damage in hypothalamus region.

Answer: C

90. What conned hypothalamus to the pituitary?

A. Infundibulam
B. Corpus callosum
C. Anterior
D. Cerebral cortex

Explanation: Infundibulam is a funnel shaped structure. It is the connection between hypothalamus and posterior pituitary.

Answer: A

91. Deficiency of vasopressin causes _____

A. Goitre
B. Diabetes mellitus
C. Myxoedema
D. Diabetes insipidus

Explanation: Vasopressin is an antidiuretic hormone. It is synthesized by hypothalamus. Deficiency of vasopressin causes Diabetes insipid us.

Answer: D

92. Which of the following is responsible for sleep cycle movement?

A. Melatonin
B. Dopamine
C. Progesterone
D. Testosterone

Explanation: Melatonin is produced by a pineal gland and it regulates sleep ad wakefulness.

Answer: A

93. Which of the following hormone is responsible for ovulation?

A. LH
B. FSH
C. Progesterone
D. Testosterone

Explanation: LH is a hormone produced by gonadotrophic cells in the anterior pituitary gland. In females, an acute rise in LH triggers ovulation and development of corpus luteum.

Answer: A

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94. Stress hormone is _____

A. Oxytocin
B. Adrenaline
C. Vasopressin
D. Sex hormone

Explanation: Adrenaline is a stress hormone. It is produced by glands and certain neurons. It also known as epinephrine.

Answer: B

95. Enlargement of the thyroid is called _____

A. Diabetes
B. Goiter
C. Cretinism
D. Myxoedema

Explanation: Enlargement of the thyroid is called as Goiter. It is a non cancerous enlargement. It is caused due to iodine deficiency.

Answer: B

96. Hypothyroidism causes in an adult _____

A. Obesity
B. Diabetes
C. Cretinism
D. Myxoedema

Explanation: Hypothyroidism is also called as under active thyroid. A condition in which the thyroid gland does not produce enough thyroid hormone.

Answer: D

97. Grave's disease is due to _____

A. Hyperactivity of Islets of Langerhans
B. Hyperactivity of adrenal cortex
C. Hyperactivity of thyroid gland
D. Hyperactivity if adrenal medulla

Explanation: Grave's disease is also called as base dowse disease. It results in the overproduction of thyroid hormones.

Answer: C

98. Which of the following is the deficiency of parathyroid hormone

A. Glucose level lowered in the blood
B. Glucose level raised in the blood
C. Calcium ions in the body unbalanced
D. Rate of heart beat lowered

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Explanation: Parathyroid hormon controls calcium ion concentration in the blood.

Answer : C

11.5 Homoeostasis in the Human Body

What is Homoeostasis?

'Homoios' means like or the same and 'stasis' means state. Therefore homeostasis is the process of maintaining of normal internal conditions in a cell or in organism by means of self - regulating mechanisms.

The structure and function of the human kidneys

Kidneys are a pair of reddish - brown bean - shaped organs just above the waist. They are divided in three regions:

- The out cortex
- The inner medulla
- The most inner renal pelvis

They have hilums /depressions/ from where three tubes are passed out. The tubes are:

- The renal artery
- The renal vein
- The ureters

The functional units of kidneys are called nephrons.

The functions of kidneys.

- The kidney removes waste and extra water from the blood
- They filter blood and producing urine
- Help keep chemicals balanced in the body
- They reabsorbing mineral ions
- Regulation of pH
- They produce hormones like erythropoietin

11.5.1 Homeostsasis and Kidneys

- Blood flows into the kidney along the renal artery.
- The blood is filtered, so fluid containing water, salt, urea, glucose and

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many other substance are forced out into the kidney tubules are called **nephrons.**

- Then everything the body needs are taken back (reabsorbed), including all of the sugar and the mineral ions needed by the body.
- The amount of water reabsorbed depends on the needs of the body.
- The waste products like urea, excess ions and unwanted water of the body are lost as urine.

Parts of nephrons and their functions

- Nephrones are functional unit of the kidneys.
- Each kidney is made up of millions of nephrons.
- They are tiny microscopic tubules where all filtration and resbsorption takes place.

A nephron contains:

- Bowman's capsule
- First coiled (convolued) tubule
- Loop of Henle
- Second coiled (convoluted) tubule
- Collecting duct

i. Bowman's capsule

- It is a cup like thin membranous double walled capsule.
- It contains **glomerulus, which is a** collection of the blood vessels /capillaries/ where the pressure builds up so that ultrafiltration occurs.
- It is the site of the ultrafiltration of the blood /filtration through the medium such as capillary wall/.
- Blood in the capillsries is under high pressure because vessels feeding into capsule /afferent artioles/ are wider than the vessels leaving the capsule /efferent arteioles/.

The wall of the blood capillaries and wall of the capsule have small gaps which act as a filter.

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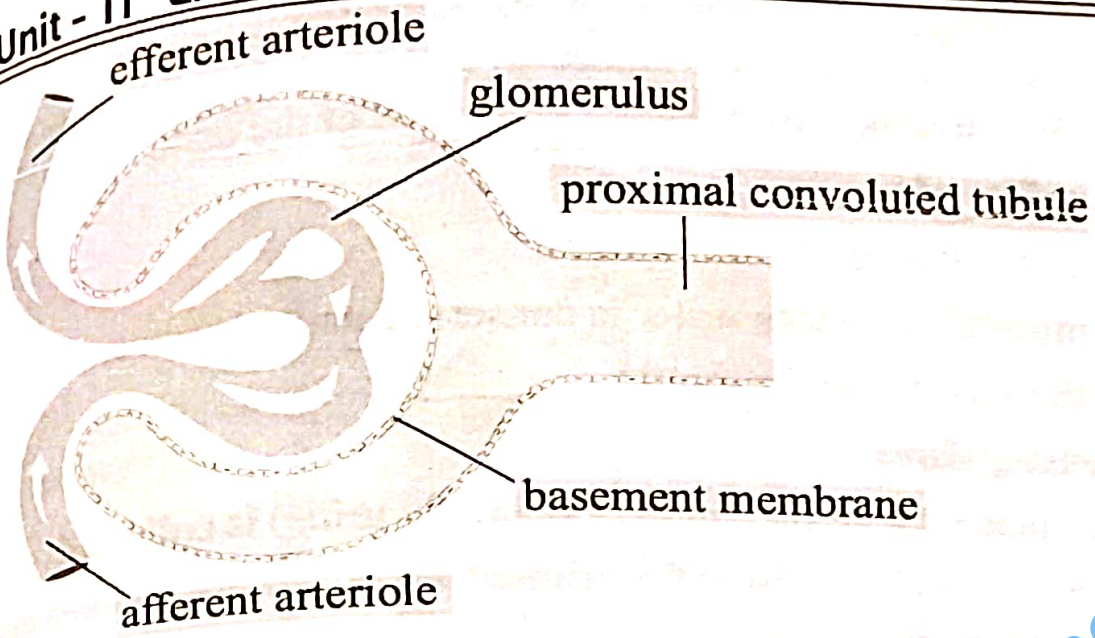

Larger molecules such as the blood cells and large blood protein can't pass the glomerulus and are passed to efferent arteriole

Smaller molecules such as the water, glucose, amino acid, urea and many other substance are forced out into the first coiled tubule.

ii. First coiled (convoluted) tubule

- It is a tubule where much of reabsorption takes place.
- It has micro villi to increase the surface area for re - absorption.
- Liquid which enters the first tubule is known as the glomerular filtrate.
- All glucoses, 67% of sodium ion and 80% of water in the glomerular are back to the blood in first tubule.

iii. Second coiled (convoluted) tubule

It is place where the water balancing is done.

If the body is in shortage of water, more water is reabsorbed into the blood under the influence of anti - diuretic hormone or ADH.

If the water concentration of blood becomes too low that is high blood salt concentration, the osmoreceptors in the brain detect change and stimulate the pituitary to release more ADH.

- ADH makes the second coiled tube more permeable to water as a result more water is reabsorbed back to the blood.
- Diuresis means passing urine, so anti - diuretic means preventing or reducing urine flow.
- Ammonium ions and some drugs are added into the second tubules from the blood to get rid or remove.

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- At the end of second soiled tubule all salt needed by the body has been reabsorbed and the excess is lost with most of the urea.

iv. Loop of henle

- It is the U - shaped part of nephron
- It is a place where more water is conserved and urine is concentrated.
- pH of the blood is maintained

v. Collecting duct

- It is the place where the liquid (essentially urine) is collected
- Urine contains about 1% of the original water, most of the urea and with no glucose at all.
- If the body seriously needs more water, water can be reabsorbed from collecting duct under the influence of ADH.

Note: If the water concentration of blood falls, high amount of ADH is released and if water concentration of blood rises, less ADH is released

Urinary bladder

- It is large storage sac that collects urine from both kidneys.
- Urine is formed in each kidney and passes out into the **ureters** which carry it to the **urinary bladder.**
- Urine is temporarily stored here until it is released via the **urethra**
- The following diagram shows that the different parts of nephron

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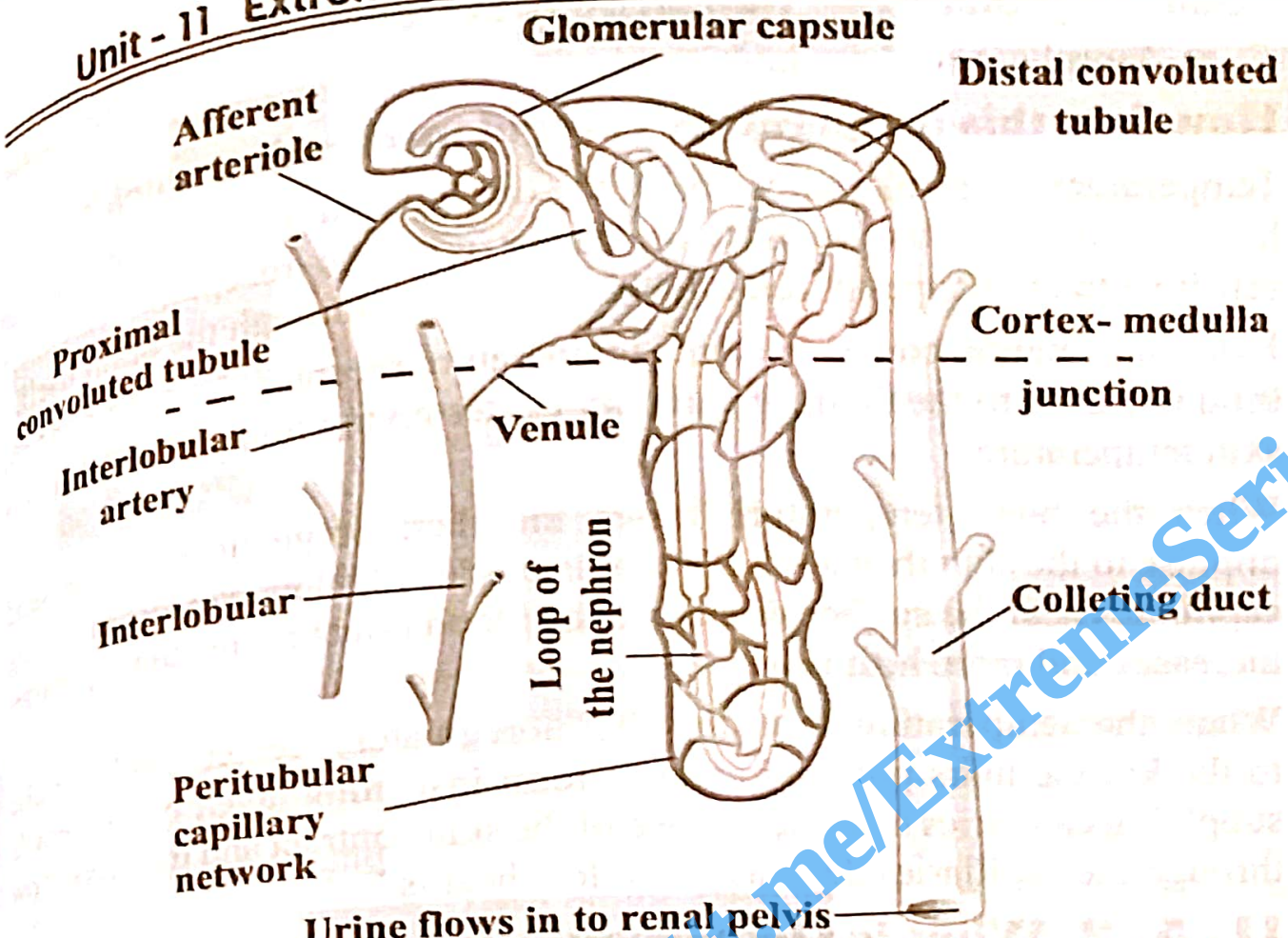

Controlling temperature/thermoregulation/

One of the most import factors which animals need to control is the internal or core body temperature. (Detail information is available in unit five)

With regard to temperature regulation animals can be divided into two groups:

1. Poikilotherms - are animals whose body temperature increases or decreases as the temperature of their surroundings raise or fall down
2. Homoiotherms - are animals whose body temperature is independent of the temperature of their external environment.

Temperature Regulation in homothermic animals

In homothermic animals there is temperature controlling center in their brain called the hypothalamus.

They have mechanisms to regulate their body temperature through:

- Physiological methods
- Behavioral methods

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➥ Morphological methods

How does this temperature control mechanism work?

Temperature control relies on the thromoregulatory center in the hypothalamus of the brain. The center contains receptors which are sensitive to the temperature of the blood flowing through the brain itself.

Extra information comes from the temperature receptors in the skin which send impulses to the thermoregulatory center giving information about the skin temperature.

When the body temperature raises the thermoregulatory center send impulse to the skin then a lot of sweat is produced, blood vessels supplying capillaries near the surface dilate and the blood flow through the capillaries increases and more heat is lost.

When the temperature falls the thermoregulatory center send impulse to the kin the hairs pull and erect to form insulating layer, blood vessels supplying capilaries near the surface of the skin contract and the blood flow through the capillaries decreases and less heat is lost.

11.5.2 What is Osmoregulation

It is process of controlling water and electrolyte (Ca, Na, K etc.) balance in the body. It is maintained by the kidneys

The balance of water and salts in the body is very important because the concentration of the body fluids changes, water will move into or out of the cells by osomosis and they could be damaged or destroyed

If we are short of water we produce very little urine and most water is saved for use in the body but if we have too much water then our kidneys produce lost of urine to get rid of the excess water.

The liver and chemical regulation

It is the largest internal glandular organ in the body that makes up around 5% of the body mass.

All the blood leaving the stomach and intestines passes through the liver.

It has a vary special blood supply that is called hepatic portal vein.

➥ Hepatic portal vein carries digestion products from gut to the liver to filter toxins

The liver carries out a wide range of functions:

➥ It regulates most chemical levels in the blood.

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- It controls the sugar level in the body through stored glycogen in the liver itself.
- It controls and balances the fat that we eat and cholesterol levels in the blood.
- It breaks down of worn out RBCs (breaking down of haemoglobin and storing iron).
- It forms bile and it is stored in the gall bladder before it is released into the digestive gut.
- It is used to control temperature due to abut 500 different chemical reactions producing heat.
- It controls toxins (detoxification), breaks down poisons including alcohol. Too much alcohol can cause cirrhosis (liver tissue is destroyed)

It is an important organ where protein metabolism take place:

- The liver breaks down excess amino acids and forms urea.
- The process of removing the **amino group** from excess amino acids is know as **deamination.** Then amino parts is converted into ammonia and then urea in the liver.
- The rest of the amino acid is converted into carbohydrate and can be used in cellular respiration or converted into fat to store in the liver.

11.5.3 Chemical Regulation in the Blood

Chemoreceptors are receptors that sense changes in the chemical composition of the blood.

Chemoreceptors are stimulated by change in the chemical composition of their immediate environment and send information to the brain to regulate cerdiovascular and respiratory function.

The respiration chemoreceptors work by sensing the PH levels of their environment thought the concentration of hydrogen ions because most carbon dioxide is covered to carbonic acid in the bloodstream.

Chemoreceptors are able to use blood PH as a way to measure the carbon dioxide levels of the bloodstream.

There are two major types of chemoreceptors: **Peripheral chemoreceptors** and **Central chemoreceptors.**

1. Peripheral chemoreceptors

They are extensions of the peripheral nervous system.

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They are located in the carotid and aortic bodies

- Carotid body/main artery in the neck/, which detects blood oxygen, carbon dioxide and PH.
- Aortic body/mass of chromaffin cells in a capillary plexus/ detects change in blood oxygen and carbon dioxide but not PH.

How do peripheral chemoreceptors work?

- They detect large changes in the partial pressure of oxygen (pO_2) as the arterial blood supply leaves the heart.
- When low levels of oxygen are detected afferent impulses travel via the glossopharygeal and vagus nerves/ carnial nerves/ to the medulla oblongata and the pons in the brainstem.

A number of responses are then coordinated to restore PO_2:

- ☑ The respiratory rate and tidal volume are increased to allow more oxygen to enter the lungs and subsequently diffuse into the blood.
- ☑ Cardiac output is increased to maintain blood flow and oxygen supply to the body's tissues.
- ☑ Change in the levels of O_2, CO_2 and H^+ in the blood cause compensatory changes in the level of ventilation.

2. Central chemoreceptors

They are located on the ventrolateral surface of **medulla oblongata**

- Detect changes in the P^H of cerebrospinal fluid.
- They also detect change in the arterial partial pressure of carbon dioxide (pCO_2).
- They can be desensitized over time from chronic hypoxia (oxygen deficiency) and increased carbon dioxide.

How do central chemoreceptors work?

- When changes in carbon dioxide are detected receptors send impulses to the respiratory center in the brain stem that initiate changes in ventilation to restore normal pCO_2.
- Detection of an increase in pCO_2 leads to an increase in ventilation that is more CO_2 is exhaled, then p CO_2 decreases and returns to normal.
- Detection of a decrease in pCO_2 leads to a decrease in ventilation that is less CO_2 is retained in the lungs, the pCO_2 increases and returns to

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normal

As arterial pCO_2 rises, so does the CO_2 content of cerebrospinal fluid (CFS) because the blood - brain barrier /BBB/ is permeable to CO_2 but not to hydrogen ions.

The pH of the CSF in inversely proportional to the arterial pCO_2:

- A small decrease in pCO_2 leads to an increase in the of the CSF which stimulates the respiratory centers to decease ventilation.
- A small increase in pCO_2 leads to a decrease in the pH of the CSF which stimulates the respiratory centers to increase in ventilation.

Illustrative Example

99. Part of nephron impermeable to salt is _____

A. DCT
B. Collecting ducts
C. Ascending limb of loop of Henle
D. Descending limb of loop of Henle

Explanation: Henle's loop is made up of the descending loop and an ascending loop. Descending limb of loop of Henle is impermeable to salt and doesn't allow any salt to pass through it

Answer: D

100. Vasopressin is called as _____

A. Synovial fluid
B. Neurotransmitter
C. Antidiuretic hormone
D. Growth regulating substance

Explanation: Vasopressin is called an Antidiuretic hormone (ADH). Vasopressin stimulates reabsorption of water and reduction of urine output.

Answer: C

101. Mark the mismatch part.

A. Henle's loop - concentration of urine
B. DCT - Absorption of glucose
C. PCT - Absorption of sodium and potassium ions
D. Bowman's capsule - Glomeruler filtration

Explanation: Glucose absorption is occur in the bowman's capsule

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DCT is distal convoluted tubule. It is the portion of kidney nephron between the loop of Henle and collecting tubule.

Answer: B

102. Urea sythesis occurs in

A. Kidney B. Liver C. Pancras D. Muscle

Explanation: Urea synthesis occurs in the liver. Since there is an arginase enzyme present in the liver the synthesis of urea occurs in liver.

Answer: B

103. Main function of Henle's loop is _____

A. Passage of urine
B. Filtration of blood
C. Formation of urine
D. Conservation of urine

Explanation: Loop of Henle conducts urine with each nephron. It helps in the recovery of water and sodium chloride from urine. It helps in the conservation of urine.

Answer: C

104. Osmoreceptors in hypothalamus are involved in _____

A. Secretion of GH
B. Secretion of ADH
C. Secretion of ACTH
D. Secretion of Aldosterone

Explanation: Osmoreceptors are sensory receptor found in hypothalamus. It detects changes in osmotic pressure.

Answer: B

105. Which of the following is NOT the function of the kidney?

A. Excretion
B. Deamination
C. Water balance
D. Osmoregulation

Explanation: Osmoregulation (water balance) and excretion are the functions of the kidney. While, deamination (removal of excess omino acids from the body) is the function of the liver.

Answer: B

106. Why do we produce less urine during hot and dry weather than during cold weather for the same amount of water of drink?

A. Because we produce less amount of waste.

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